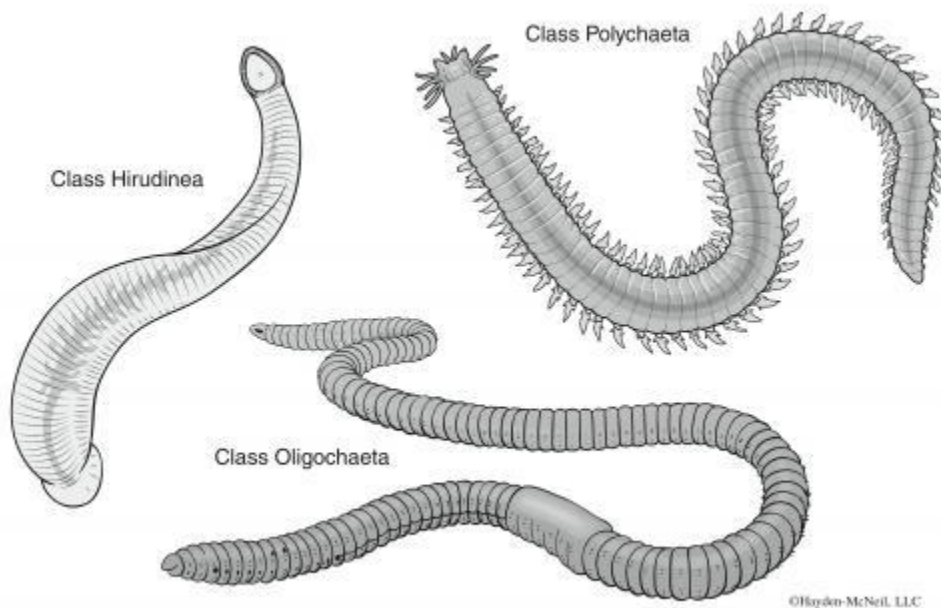


Welcome to BIPN 145

This is a course about how we study the nervous system.
As a neuroscientist, you'll be asked to think, plan, and share your findings.

This lab manual describes the set of experiments
you'll be conducting over the course of the quarter.



This course is for everyone.

Neuroscience is a diverse discipline, with folks from many different academic and personal backgrounds. We'll be working together to create an equitable and inclusive environment of mutual respect, in which we all feel comfortable to share our moments of confusion, ask questions, and challenge our understanding.

Everyone should be able to succeed in this course.
If you do not feel that is the case, *please let us know*.

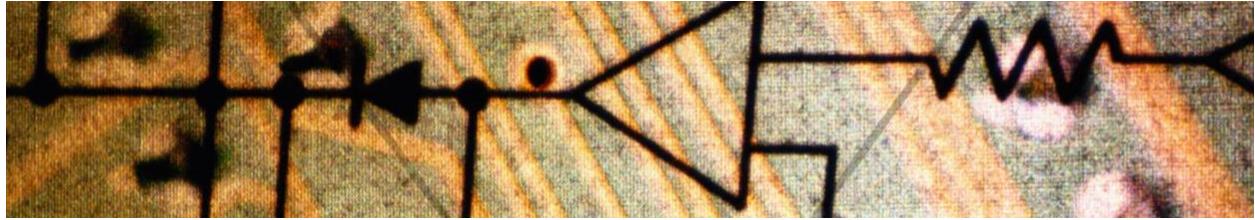
**This lab manual was written and edited by
Ashley Juavinett, Isabella Maita, and Marc Marino
and includes adaptations of AD Instruments protocols.**

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Experiment #1: The neuron as an RC circuit



Background & Goals

The brain as a circuit board

We often describe the brain in terms of circuits, which is a useful and pretty accurate metaphor. This lab focuses on RC circuits. RC stands for **resistance** & **capacitance**. Building on these fundamental concepts we will build and manipulate a basic working circuit representation of the biophysical properties of a neuron.

Modeling resistance

A resistor converts electrical energy into heat when current flows through it. A current flowing through a resistance creates a voltage drop across that resistor, described by Ohm's law:

$$V = IR$$

V = voltage (Volts)

I = current (Amps)

R = resistance (Ohms, Ω)

In a neuron, the **resistance** is determined by the opening and closing of ion channels, which allow current to flow in and out of the cell. When more ion channels are open, more ions are able to flow. Ion channels can be represented with resistors in an RC circuit, since they resist the flow of electricity.

It's important to note the units in the equation above: **1 Amp** flowing through a **1 Ohm** resistor produces **1 Volt** across the resistor. In electrophysiology, we typically work with different scales of numbers:

millivolts (mV) = 10^{-3} V

nanoamps (nA) = 10^{-9} A

megaohms (M Ω) = 10^6 Ω

Sometimes you'll see Ohm's Law with **conductance** instead of resistance. Conductance (measured in siemens and often denoted by *g*) is the *inverse* of resistance. In other words, it's equal to $1/R$.

Kirchhoff's circuit laws

Kirchhoff's laws state two relatively intuitive facts:

1. The voltage at a given point in a circuit is uniquely defined. Thus:
 - a. If there are two (or more) paths to go from point A in a circuit to point B, the voltage drop along each of those paths is the same.
 - b. The voltage drops around a closed loop must sum to zero.
2. The sum of all currents going into a point equals the sum of all currents going out of it.

Modeling capacitance

Capacitors are defined by two conducting substances separated by a non-conducting substance in the middle. Capacitors accumulate electric charge on the insulator. The accumulated charge creates an electric field between the conductors and stores electrical energy. Salty fluids that make up the intra- and extracellular solutions of the cell are conductive, the lipid bilayer is not. Therefore, the neural membrane acts as a capacitor.

When one conducting region of the capacitor accumulates a charge, an electric field is created, which pushes the charge off of the next conducting region. This phenomenon only lasts a short amount of time, producing a brief current which is expressed as:

$$I = C * dV/dt$$

I = current (amperes)

C = capacitance (farads)

dv/dt = rate of voltage change with time

To summarize and add a couple useful symbols:

Circuit component	Symbol	A current step (I) like this...	... produces this voltage (V) response.	Equation
Resistor				$V = IR$ R is measured in ohms (Ω)
Capacitor				$I = C dV/dt$ C is measured in farads (F)

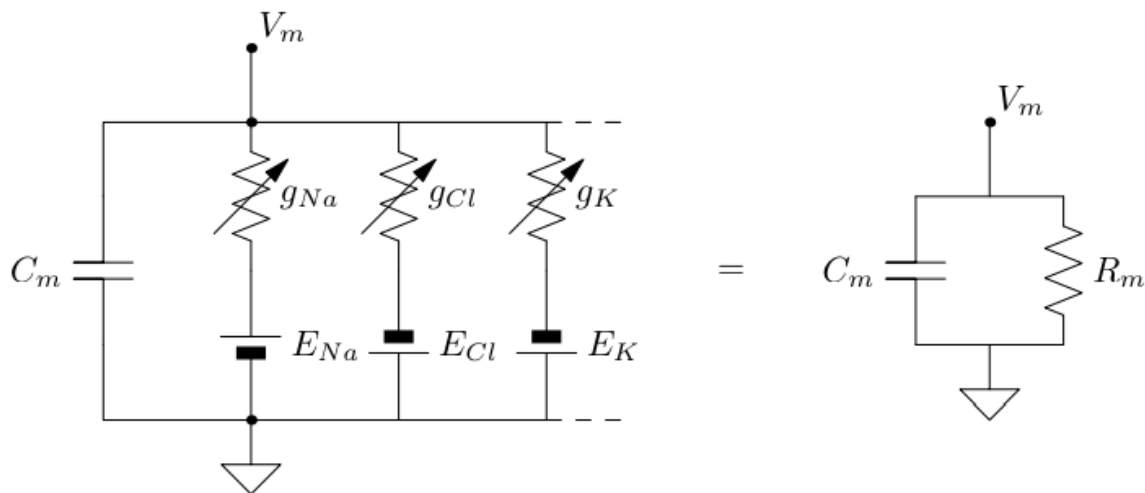
Modeling concentration gradients

In a neuron, there's also a steep concentration gradient due to the different concentrations of ion on either side of its membrane. This is the basis of the neuron's **resting potential**. The inside of the cell is more negative than the extracellular cell space -- typically by about -60 mV, depending on the cell type. We'll use batteries to model this offset of voltage.

Putting it all together

With these components – resistors, capacitors, and a battery – current flow in a neuron can be modeled. In reality, there are many different ion channels, and multiple different ion conductances (left in the figure below). We'll simplify each of these into one membrane resistance, and we'll model the membrane of the cell with one capacitor (right, below).

When we send current into the circuit on the right, the capacitor will charge. In both a neuron and an electrical circuit, voltage rises to a steady state level asymptotically in the response to an external current (e.g., from an action potential or passively passing charge). Once the external current is shut off, the voltage drops in a similar asymptotic way. As you'll discover today, this charging time course depends on a few different factors.



From Barbour, 2014, "[Electronics for Electrophysiologists](#)"

These characteristics of the membrane voltage can be measured with a simple voltmeter in the RC circuit. Opening more ion channels (or resistors, as in this RC circuit), which decreases resistance and increases conductance, will express a more rapid increase in voltage on the voltmeter, visually demonstrating the properties of resistance in a neuron.

We can calculate a **time constant**, which represents the time it takes to reach a steady state after a change in voltage across the membrane:

$$\tau = RC$$

τ = time constant (seconds)

R = resistance (ohms)

C = capacitance (farads)

Breadboards & electronics are useful tools for neuroscience research

Lastly, these types of circuits are really useful for designing neuroscience experiments. Many labs rely on breadboards and microcontrollers (e.g. Arduino) to create custom setups. Once you know how to wire up a circuit and program an Arduino, you can do pretty much anything.

In this lab you will:

- Learn how to interpret circuit diagrams and model them on breadboards
- Demonstrate an understanding of Ohm's Law and its applications to resistance in a neuron
- Describe the properties of resistance and capacitance in terms of both an electrical circuit and a neuron
- Show how changing the conductance of ions creates a resting membrane potential
- Observe how the capacitor generates an asymptotic rise in voltage rise
- Describe how the membrane resistance affects the time constant

This lab is largely based on "[Electric Circuits for Electrophysiology](#)", adapted by Daniel Wagenaar, originally by Kevin Briggman et al. as well as [Dabrowski et al., JUNE, 2013](#).

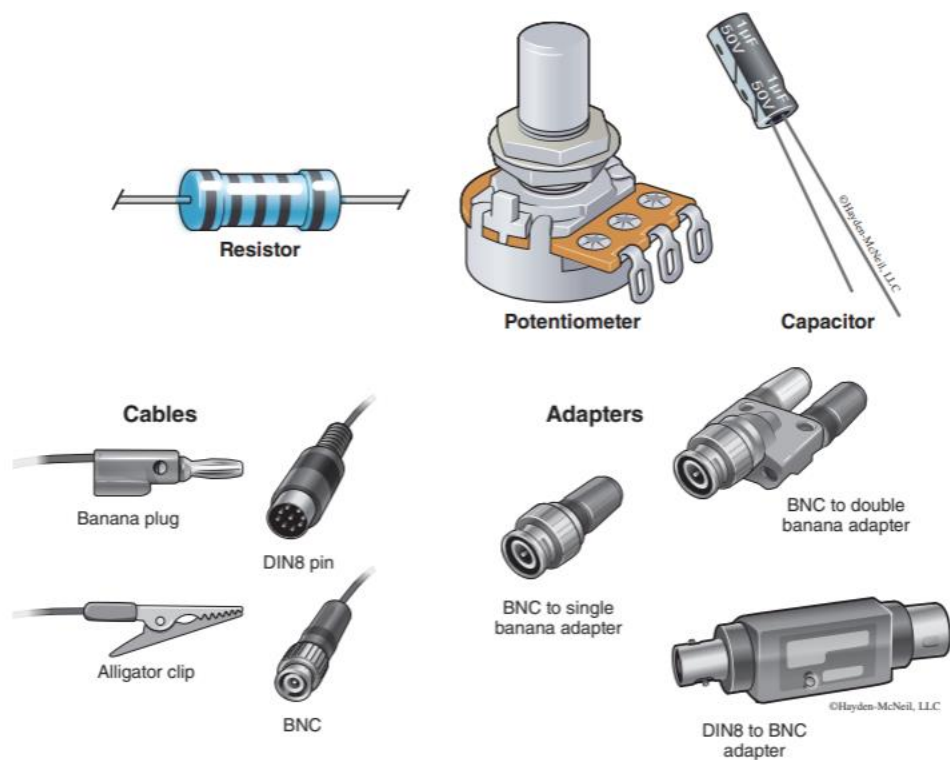
Lab Protocol

Parts

- PowerLab & Computer
- 1.5 V(x2) battery & holder (2)
- Resistors: 3 k Ω , 15 k Ω , 33 k Ω , 100 k Ω (3), 220 k Ω , 470 k Ω
- 1 μ F capacitor
- 1-100 k Ω potentiometer
- Jumper wires (small colored wires)

Cables & Adapters

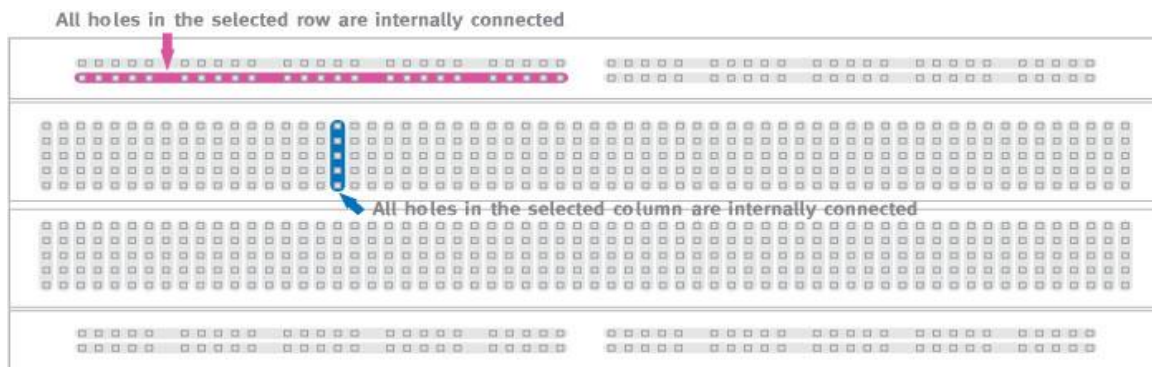
- DIN8-BNC adapter (1)
- BNC to double banana adapter (2)
- Banana plug cable (4)
- Alligator clips (4)



Getting familiar with your breadboard

Breadboards are really useful ways to prototype and model electrical circuits (and in our case, neural circuits). Some holes on the breadboard are connected together, so putting wires into connected holes provides an electrical contact between them. These connections are organized with two main rails (rows) on the sides, typically used for power, and many columns. Knowing how these components are connected is crucial to understanding how to use a breadboard.

Identify the rows and columns (as indicated below) on your breadboard:



From <http://wiring.org.co/learning/tutorials/breadboard/>

Meet your PowerLab

First, it helps to get familiar with our equipment. Look for the following on your PowerLab. I've highlighted items that are particularly important for our experiments — make sure you know where these are, and what they do/indicate:

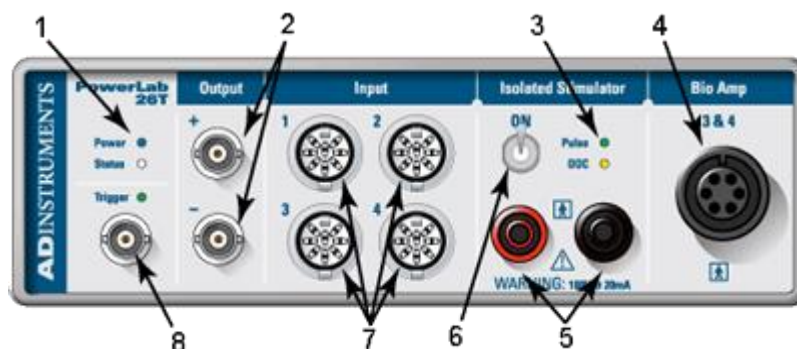


Image courtesy of ADInstruments ©

1. Power indicator light: illuminates when the PowerLab is turned on
2. **Analog output connections: provide a voltage output in the 10 V range. Note:** This is NOT safe for direct connection to humans.
3. Isolated Stimulator status light: indicates if the Isolated Stimulator is working properly (green) or out of compliance (yellow)
4. Bio Amp input: connects a Bio Amp Cable to the PowerLab; **reads as inputs 3 and 4 in LabChart8**
5. Isolated Stimulator outputs: for connecting stimulating electrodes to the Isolated Stimulator
6. Isolated Stimulator switch: turns on/off the Isolated Stimulator
7. **Inputs: 8-pin DIN connectors for attaching electrodes and any input you'd like to measure.**
8. Trigger input: can be used to start or stop a recording event

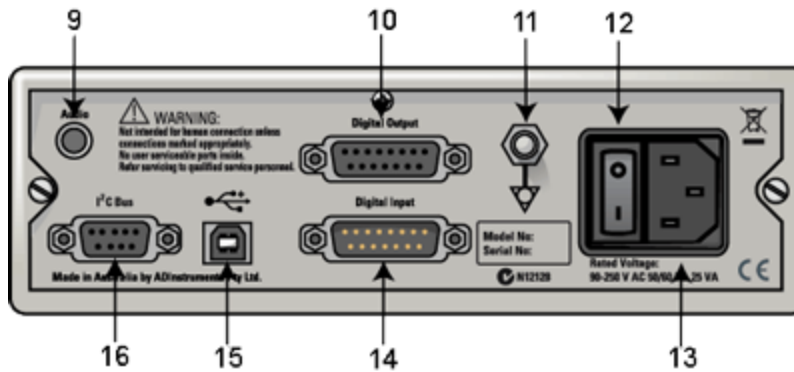


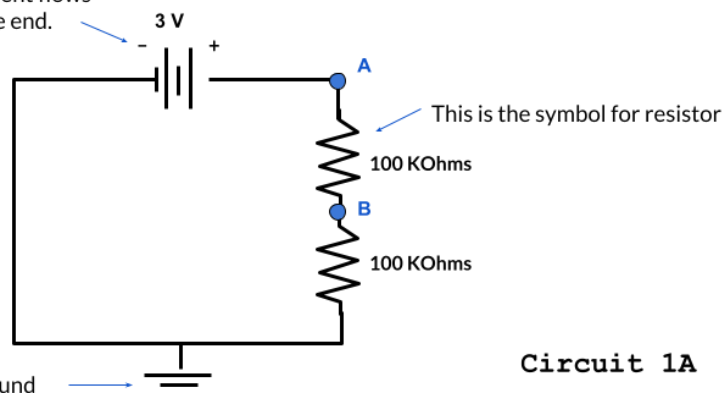
Image courtesy of ADInstruments ©

- 9. Audio output connector: standard phono jack for sound output
- 10. Digital Output Connector
- 11. **Earthing post**: used to ground the PowerLab, if grounded power supply is unavailable
- 12. **Power switch**
- 13. Power cord connector
- 14. Digital Input Connector
- 15. **USB connector**: connects computer to the PowerLab

Circuit #1: A model of ion channels in a membrane

First, we'll set up a simple circuit with a battery (representing the concentration gradient) and a resistor (representing an ion channel). We'll put two 100 K Ω resistors in **series** (head-to-tail):

This is the symbol for a battery. Current flows from the negative end to the positive end.



Circuit 1A

1. Build the circuit above on your breadboard. It helps to start with the battery ends in the rails, and work your way around. The **red** wire of the battery is the positive terminal.

Note: Don't worry if it takes some time to figure out how this translates to a breadboard.

CAUTION: Do not connect the + and - ends of the battery. This will short the circuit, heat, and melt the plastic.

2. We'll use the PowerLab as our **voltmeter** to record the voltage in our circuit. Plug a **BNC-DIN8 adapter** & double banana adapter into **Input 1** of the PowerLab.
3. Connect the ground/reference side (**black**, -) of the double banana adapter to the circuit ground with alligator clips.

Note: The "ground" in the circuit here is really going to be our reference electrode. It should be connected to the black side of Input 1.
4. Connect the "hot" pin (**red**, +) of the double banana adapter to point A on the diagram above.
5. Turn on the PowerLab 26T.
6. Open LabChart 8. You should see that the PowerLab is connected (don't worry if it tells you other devices are *not* connected):

PowerLabs:

✓ PowerLab 26T

7. Open a file with the **Circuit Lab** settings, downloaded from <http://bit.ly/labchart>. LabChart settings files have ".adiset" as their extension. When you download this file, you can open it with LabChart directly. Or, you can open LabChart on the desktop and then open the downloaded .adiset file.
8. In LabChart, go to **Setup > Channel Settings**.
9. Click on the Input Amplifier button to open the **Input Amplifier** for Channel 1. This gives you a livestream of the voltage on that channel and will allow us to measure the voltage in our circuit today, effectively acting as an oscilloscope.

Note: If you are not observing a voltage > 0 on the Input Amplifier, refer to the TROUBLESHOOTING table at the end of this protocol.

What is the voltage at point A (relative to ground)? _____

10. Move your recording pin so that you're recording from point B.

What is the voltage at point B (relative to ground)? _____

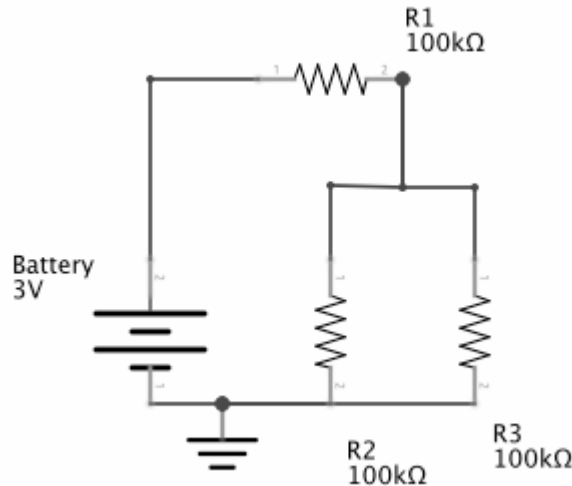
Resistance in series can be calculated as $R_{\text{total}} = R_1 + R_2$. What is the total resistance?

Total resistance = _____

Using $V = IR$, calculate the total current (in μA) in this circuit. (You can treat your battery as if it is exactly 3V).

Current = _____

8. Next, add another resistor in **parallel** (side-by-side):



Circuit 1B

9. Repeat the measurements, measuring the voltage across R1. Is it the same as with the two resistors in series (point B above)? If not, why is it different?

Resistance in parallel can be calculated with the equation $1/R_{\text{total}} = 1/R_1 + 1/R_2$. What is the total resistance in this circuit (Hint: figure out the parallel branch, and then add in series)?

Total resistance = _____

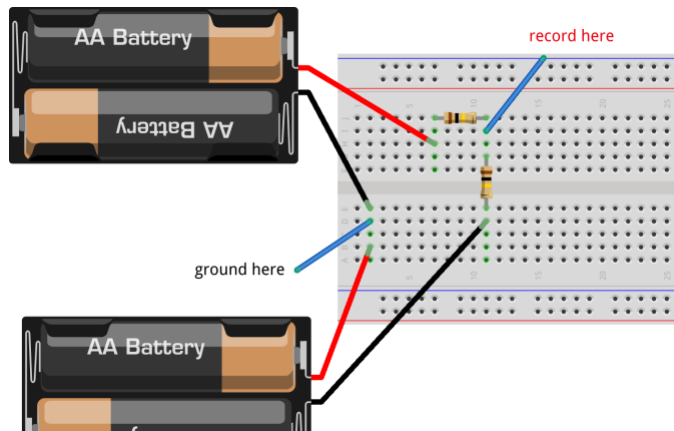
What is the total current in this circuit? Current = _____

Circuit #2: The resting membrane potential

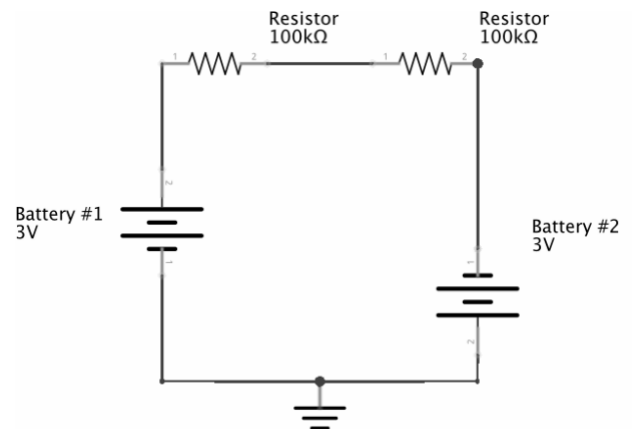
Now, we'll take our circuit up a notch with two batteries that represent the gradient of different ions. We'll also define our circuit in terms of conductances, since that makes more sense when we're talking about ion channels. The balance between the two concentration gradients (batteries) and conductances (resistors) is what leads to a resting membrane potential.

A circuit with two concentration gradients

Build the following circuit on your breadboard. Note the orientation of the two batteries.



Circuit 2A



What is the voltage between your electrode and ground? _____

Does that mean no current is flowing?

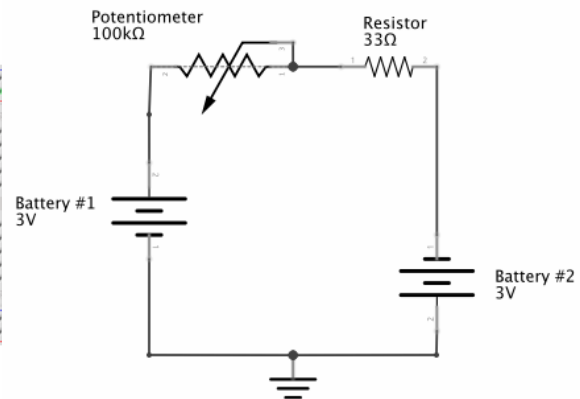
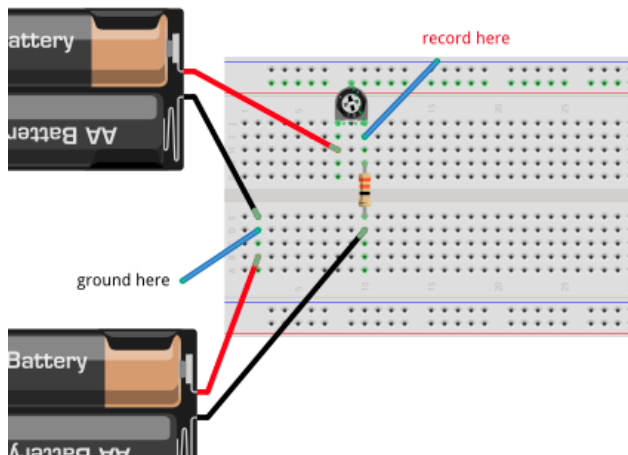
What is the current in this circuit (you can add the *absolute* voltage of the two batteries)?

Current = _____

Adjusting the resistance to create a non-zero membrane potential

How then can we get a non-zero resting membrane potential? There are two ways: We could either have non-equal batteries (i.e. different concentration gradients) or non-equal conductances (i.e. different permeabilities for different ions). In the next exercise, you'll test the effect of changing conductances.

Swap out one of the resistors for a **potentiometer**, so that your circuit looks like the one below. Potentiometers (often called pots) have 3 middle pins -- connect to the middle pin and one of the outer pins. The 3rd pin can be left unconnected. You can insert the pins of the potentiometer directly into the breadboard.

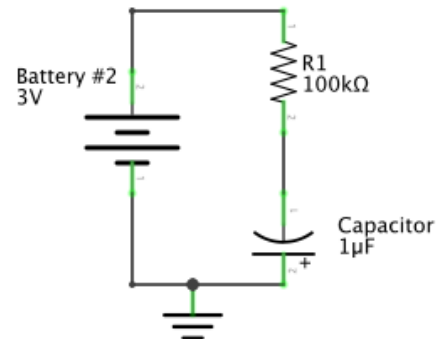
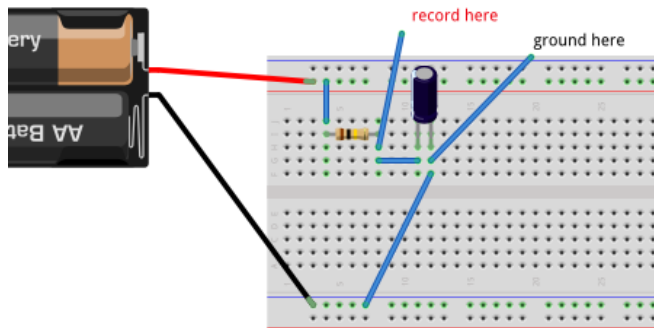


Circuit 2B

You can change the resistance of the potentiometer by turning the round bar sticking out of it. Watch what happens to the voltage as you change the resistance. Can you get it to cancel out the other branch exactly so that the voltage between A and B becomes zero again?

Circuit #3: A resistor-capacitor (RC) circuit

Finally, what we came here for: an RC circuit. So far, all the measurements have ignored how fast the voltage changes take place. Indeed, if membranes and electrodes were purely resistors, all voltage changes would take place instantaneously. However, the changes always have a delay because the membranes also have properties of **capacitors**.



Circuit 3

1. Build the circuit above on your breadboard.

Note: The voltage is depicted as a battery here but is actually coming from the **Output** of the PowerLab, using the BNC to double banana plug adapter. Use two banana plug cables and a couple of jumper wires to connect it to your breadboard.

2. The voltmeter goes into Input 1 of the PowerLab (similar to Circuit #2).
3. Go to **Setup > Stimulator** to open the Stimulator window. This is the window where you can modify the shape, timing, and amplitude of the stimulus output.
4. Make sure that the Waveform Name is set to **Pulse**.
5. In LabChart, set up the Stimulator to stimulate with 1 V Pulse Height for 0.5 Pulse Width.
6. Change your view to **Scope view**, which looks like the sweeps of an oscilloscope. The advantage of Scope View is that it allows the overlay and averaging of your data. You can tell which view you're in from the menu that looks like this (Scope is highlighted in red):



7. In Scope view, with your Stimulator ON, press **Start**. You should see a waveform on your screen. Make sure the range of your recording channel is 1 V. The range determines the range of values the PowerLab can possibly record,
Note: you may need to Autoscale the channel (right click on it) to be able to see the entire trace. See visualization tips below.

LabChart Visualization Tips

Vertical scaling

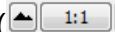
The Scaling buttons are on the left side of each channel's Amplitude axis. The up button will double the vertical scale shown, and the button will halve the vertical scale shown

Right-clicking in the channel will show several options for displaying data:

1. **Auto Scale Channel** will automatically adjust the amplitude axis so the maximum value is just larger than the maximum value of visible data in this channel.
2. **Show Points as Dots** and **Show All Channels with Dots** will show you the individual points the PowerLab is sampling.
3. If you changed the size of the data channel, **Equalize Channel Heights** will make all the data channels the same size again.
4. **Split View** inserts a divider vertically in the data creating two separate scrollable regions. This allows you to view recorded data while still monitoring new data. Note: This only works in Chart View.

Horizontal Scaling

The horizontal scroll bar provides the simplest way of moving backwards and forwards through your file and works the same as it would in any other computer program. You can think of your recording as a large strip of paper of which only one part can be seen at any one time.

By using the View Buttons at the bottom right corner () you can compress or expand the Time axis to see more or less of a waveform. The left button will compress your data; the right button will expand it. If you select the ratio button, a pop-up menu appears in which you can choose the new compression directly.

The time constant should equal $\tau = RC$. What *should* the time constant be (in seconds), given the resistors and capacitor you're using? Mind your units.

$\tau =$ _____

We can also measure the time constant by measuring how long it takes the curve to rise or decay. One time constant (τ) is defined by when the circuit rises to **63.2%** of its total charge (alternatively, we could see how long it takes to decay $1/e$, about 37%). In this case, it's $1\text{ V} * 0.632 = 0.632\text{ V}$.

To *empirically* measure your time constant, follow these steps:

2. Make sure your voltage is very close to zero at baseline.
3. Using the Marker Tool (right click anywhere in the window and choose "Clamp to trace") find the point on the trace that is equal to 63% of the maximum voltage (this may or may not be close to $\sim 0.632\text{ V}$ depending on whether your capacitor is fully charging).
4. Move your cursor to the start of the stimulus. In the upper right corner of the window, you'll see something that says Δt . This is the change in time since the mark on the curve, and your time constant.

What is your recorded time constant with 1 V stimulation, a 100 k Ω resistor, and a 1 μ F capacitor?

τ = _____

How close is your recorded time constant to the value you calculated with $\tau = RC$?

In engineering, 5τ is considered the time it takes for a capacitor to fully charge (technically, it's at ~99% of its full charge). The time constant is symmetrical for charging as well as losing charge.

How long would it take this patch of membrane to return to baseline, then? What does that mean for any computation happening in this membrane?

Record & plot your time constant with different resistors

Next, we'll see how the time constant of the circuit changes in response to the resistance strength. Use the following resistors in your circuit, and determine the time constant for each one.

Resistance	Time Constant
3 k Ω	
15 k Ω	
33 k Ω	
100 k Ω	
220 k Ω	
470 k Ω	

Check your time constants against the literature

Let's circle back to biology. Are the time constants you measured similar to what you'd find in a real neuron? Check this website to look at a collection of examples from the literature:

https://www.neuroelectro.org/ephys_prop/4/

Choose a paper from the example above and look for the reported time constant. Is the time constant in your circuit similar? If not, why might it be different?

Troubleshooting

Observation	Likely issue(s)	Possible solution
There isn't any voltage in your circuit	The power isn't connected properly; your battery is off or dead	Make sure everything is connected correctly on your breadboard. Consult http://wiring.org.co/learning/tutorials/breadboard/ . Directly check the battery voltage.
Your signal is really noisy	You're not grounded; your circuit isn't complete	Double check your circuit is connected correctly
You've got "rabbit ears" on your waveform	Your voltage is reversed somewhere	Check the battery is connected in the right direction.

Experiment #2: String Nervous Systems

Background & Goals

Why are we recording from a string?

The purpose of this lab is to get familiar with the recording equipment and software that we'll be using in this course.

First, we'll learn how you can configure LabChart 8 to acquire data and send out pulses of electricity. Then, you'll record the stimulus artifact (the passively conducted signal from the stimulus) from a saline soaked piece of string. The string mimics some of the properties of the earthworm that you'll observe in a future experiment. This will give you the opportunity to learn how to use the stimulating and recording devices on the PowerLab, as well as reduce noise and stimulus artifacts from your recordings, before you actually have to do all of this in an experiment.

Asking questions during this lab is a great idea, and will help you succeed in future experiments.

Today you will:

- Configure a recording experiment in LabChart 8.
- Determine how to record signals from a string.
- Identify sources of electrical noise in your rig.

This lab was adapted from Experiment #1 from the BIPN 105 & BIPN 145 Lab Manuals, as well as AD Instruments.

Lab Protocol

Supplies

- String
- Dissection tray
- PowerLab 26T, connected to computer
- Bio Amp
- Sewing pins (2)
- Faraday Cage
- Large Petri Dish
- Eyedropper (1)

Cables & connectors

- BNC to single banana adapter (2)
- BNC to double banana adapter
- BNC cord
- BNC-DIN8 adapter
- Banana plug cords (2)
- Alligator clip adapters (2)
- Needle electrodes

Solutions

- Saline

Setting up LabChart 8

LabChart 8 is the software we'll be using to record electrophysiology data. Before performing our experiment, we need to set up LabChart to **acquire** and **display** the data as we want.

1. Turn on the PowerLab 26T.
2. Open LabChart 8. You should see that the PowerLab is connected (don't worry if it tells you other devices are *not* connected):

PowerLabs:

✓ PowerLab 26T

3. Open a new experiment by choosing the **New** button (bottom left hand corner).
4. First, we need to decide how many channels to record. Go to **Setup > Channel Settings**.
 - a. At the bottom of the window, change the number of channels to 3, and make sure that they are all on (box on the left is checked).
 - b. Rename the first channel "Stimulus" and the second channel "Raw Recording."
 - c. Set the range for Channel 1 (your "Stimulus" channel) to 5 V.
Note: The range determines the range of values the PowerLab will record, and when using the BioAmp (later in this lab), it will actually change the amplification.
 - d. Set the range for Channel 2 (your "Raw Recording" channel) to 100 mV).
 - e. Make the sampling rate for Channel 1 "40 k/s" and ensure that "Same sampling rate for all channels is checked."
 - f. Make sure that the far right channel says "No calculation" for all three channels.

Finding sources of electrical noise

One of the biggest issues in electrophysiology is noise from overhead lights and other electronics in the room. It'll help to get an understanding of where noise in your rig is coming from.

1. Connect a **BNC-DIN8 adapter** and a **double 3 way-BNC adapter** to **Input 2** of the PowerLab. Plug in a red banana cable to the positive (red) side of the connector.
2. Hold the lead (the end of the banana cable) in your hand.
3. In the **Channel Settings** window, click on **Input Amplifier** for Channel 2. The Input Amplifier window will show the voltage that you're recording. This allows for precise setting of the input range for a recording channel and provides filtering options.
4. The signal at your rig may be variable. To adjust the sensitivity of the channel, choose an appropriate range setting from the Range drop-down list in the Input Amplifier dialog. **Note:** The number displayed in the range menu indicates the **maximum input voltage** currently selected. Notice how as you decrease the range the vertical scale changes and the small rhythmic deflections that appear on the signal trace increase in amplitude.
5. Choose a range such that your signal fills about $\frac{2}{3}$ of the window.
6. Observe the change in the recorded voltage with the electrode lead held overhead (near the lights) and then near the ground (away from the lights).
7. Try the lead inside of your Faraday cage. Is the signal more or less noisy? (Tip: is your Faraday cage actually grounded? How could you ground it?)

Questions for reflection

How does the signal change as you move the electrode lead around?

What do these sources of noise mean for our recording configuration, and our attempts to record electrical activity?

Recording stimulator outputs

We'll use our PowerLab as a stimulator that can send voltage pulses into our specimen. We can change the timing, shape, amplitude, and repetition of these stimuli via the Stimulator window in Lab Chart. We'll directly record these stimuli by sending the output of the PowerLab into an input.

1. Change your view to **Scope view**.
2. In Scope view, set the Duration to 20 ms. This changes how long each sweep of data is.

Duration: 20.0 ms

3. Remove the **BNC-DIN8 adapter** and BNC cable from **Input 2**.
4. On the PowerLab, connect the **(+) Output** to **Input 1** using the **BNC-DIN8 adapter and BNC cable**.
5. Go to **Setup > Stimulator** to open the Stimulator window. This is the window where you can modify the shape, timing, and amplitude of the stimulus output.
6. Make sure that the Waveform Name is set to **Pulse**.
7. Change the range of the Pulse Height to -5 to 5 V by clicking the ruler button next to the Pulse Height option ().
8. Configure the following by clicking & then editing the values in the window:

Start Delay	1 ms
Pulse Width	0.1 ms
Pulse Height	0.150 V
9. Close the Stimulator window.
10. Go to **Setup > Stimulator Panel** to open up a smaller window where you can directly modify the stimulus. This window can stay open while you're recording.
11. Press **>Start** in the main LabChart GUI.
12. Increase the stimulator voltage (Pulse Height) and watch the level change in the window.
13. Try a negative stimulus voltage.
14. Open the Stimulator window again and vary the delay, pulse width, and waveform shape, and see what happens.

Questions for reflection:

Why would you want to record the stimulus output?

Why would you want to delay the stimulus output?

Why would you want to send different shapes of waveforms into the string (and ultimately a nervous system)?

II. Recording the stimulus artifact

Lastly, we'll add an amplifier (the BioAmp) to our circuit, and use this record from a string soaked in saline. The string doesn't have a nervous system, but we'll add some voltage with our stimulator and observe the change in voltage from another point on the string.

Set up your string stimulation experiment

Your setup for the string stimulation will look like this:

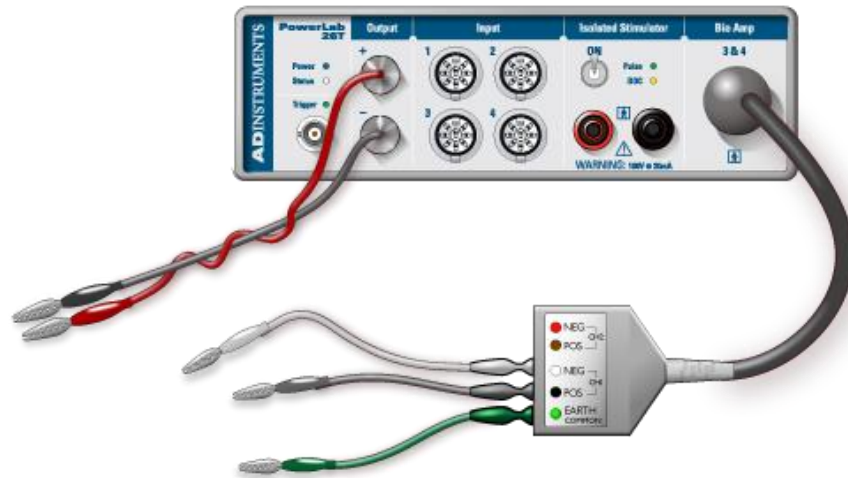
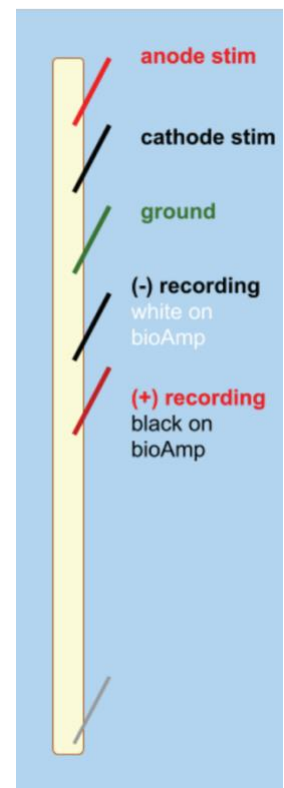


Image courtesy of ADInstruments ©

1. Remove all of the cords from the PowerLab.
2. Connect your two single BNC to banana cable adapters into the (+) and (-) **Outputs** of the PowerLab.
Note: The color of these *does not matter* but it can help to be consistent, where **red** is the anode (+ **Output**) and **black** is the cathode (- **Output**).
3. Using banana cables, connect the (+) **Output** to your **anode** stimulating electrode (the sewing pin) and the (-) **Output** to the **cathode** stimulating electrode.
4. Connect the BioAmp to the PowerLab.
5. Connect the (+) **recording**, (-) **recording**, and **ground needle electrodes** to your string and into the corresponding spots on the BioAmp on the same side as the stickers that label each connection.
Note: The colors on the BioAmp defy convention — both colors are listed on the diagram at right for clarity.
6. Change the configuration of your recording so that you are now using the BioAmp.
 - a. Go into **Setup > Channel Settings**.



- b. Make sure that BioAmp is listed under the Input Amplifier column for Channel 3.
 - c. Uncheck Channel 2 — we don't need to record from it anymore.
 - d. Change the name of channel three to "BioAmp Recording"
7. We've removed our ability to directly record the stimulus output, so we need another way to know when the stimulator sends outputs. Open the **Stimulator Settings** and set the "Marker Channel" to Ch1 (Your "Stimulus" channel). With this setting, LabChart will put a small tick on Channel 1 to show you when the stimulus was sent.

Stimulate your string!

1. Place your string in saline in the small petri dish, and give it a few minutes to soak. If it's not completely soaked, you will not get a stimulus artifact.
Note: If your string dries out during the experiment, you can use an eyedropper to add saline to it.
2. Set the stimulator voltage to 1 V, and press **Start**. You should see your stimulus marker, like you did before, but you should also now have a recording on Channel 3, the BioAmp channel.
3. Change the scaling if you cannot see the entire deflection.
Note: This recorded deflection is called the **stimulus artifact**. Recovery from the artifact should be rapid, however, so that the baseline is reached within one millisecond or so after the stimulus pulse.
4. If you do not see anything, increase the voltage (no more than 5 V).

Is the amplitude of the stimulus artifact the same as the amplitude of the stimulus? Why or why not?

5. Increase the stimulator voltage and stimulate again. What happens?
Note: Change the name of Scope pages, or make comments on the recording, in order to indicate what you've done during that run of the stimulus. You need to know the **metadata** for the experiment once you're ready to export.
6. Change the stimulus to a negative voltage and stimulate again. Does the stimulus artifact change?
7. Choose one of your pages (separate recordings) on the left, where you have a stimulus artifact with a clear peak. **Using the Marker tool (see box below), measure the latency from when the stimulus outputted from the software to the peak of your stimulus artifact.** How much of a delay is there? How much of a delay should there be? (How fast does an electrical current flow?) Note that you can clamp the Marker to one channel and look at the change in time in a separate channel.

8. Change the location of the stimulating electrodes. How does this change the shape of the artifact?
9. Remove the ground electrode. How does this change your recording?

Using the Marker Tool

The Marker () is located at the bottom left of the Chart View and can be dropped on any part of the waveform to allow relative measurements to be made. If you move the Marker to the data trace in Chart View, it will be placed on the same spot in the Scope View, and vice versa. Drag the Marker from the Marker Box to a location on the trace and release. The Marker does not have to be placed exactly on the waveform; it will attach itself to the waveform at the time position you dropped it.

When the Marker is in use, the amplitude and time values displayed are relative to the marked reference point. This means the time and amplitude values are now displayed as differences (Δ) between the Waveform Cursor and the Marker. This is very useful for measuring the time between events or measuring the relative amplitudes of parts of a waveform.

III. Save your file & export your data

LabChart buffers files to disk in case of a power failure or computer crash but it is wise to save work frequently when working with any computer. Saving files in LabChart is the same as saving any file on your personal computer.

Please also save LabChart files in your Google Drive folder or another cloud source that you can easily access.

For many of the labs in this class, you'll also want to export your data into another program, so that you can re-plot it and clean it up. We'll do that for this lab so that you're ready to do it for future lab reports:

1. Save the settings file for your experiment, by going to **File > Save As Settings** file in Lab Chart. Save it in a safe place (not locally on the computer). **You will need this settings file for the earthworm experiment.**
2. Save the data file for your experiment by going to **File > Save As** and save as a LabChart Data File (*.adicht).
3. Go to the Scope page for two of your favorite string stimulation experiments that you can compare and contrast. For example, two different voltages, or with two different distances between the stimulator & recording electrodes.
4. Follow the directions in Appendix A for exporting your data.

Troubleshooting

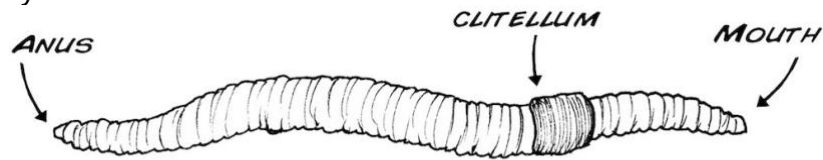
Observation	Likely issue(s)	Possible solution
You can see the stimulus marker on Channel 1, but no stimulus artifact on Channel 3	Your string may not be soaked in saline well enough	Add saline to the string
The recording trace is cut off	Your visualization needs to be re-scaled or your range is too small	Right click on channel and choose autoscale (see Visualization tips above), change the range on your recording channel
Your string recording trace has clear 60 Hz noise	You're not grounded, or there are ground loops	Make sure your Faraday cage & ground electrode are both grounded to the back of the PowerLab. If this doesn't help, consult an instructor.

Experiment #3: Earthworm Electrophysiology

Background & Goals

Earthworm essentials

Earthworms are a member of the Annelida phylum, which includes segmented worms such as the ragworm and our soon-to-be-friend, the leech. We'll be working with earthworms in the *Lumbricidae* family.



TOP (DORSAL) SIDE OF WORM DARKER THAN BOTTOM (VENTRAL)

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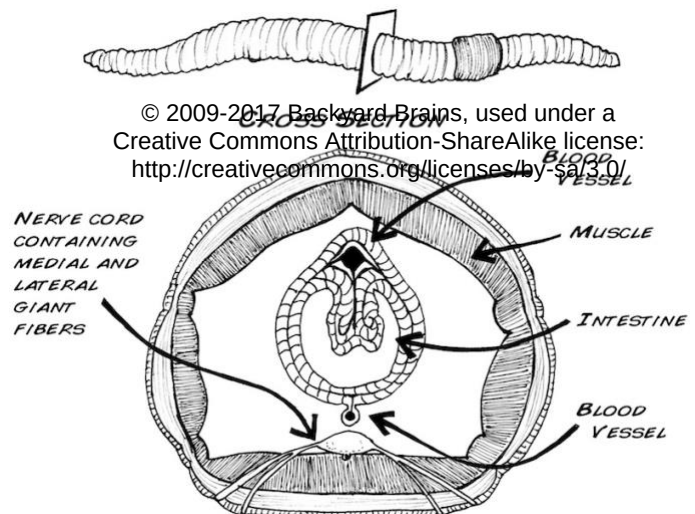
Earthworms are laid out with a mouth at one end, and an anus at the other. In the middle is the **clitellum**, a reproductive gland that can secrete a material that helps them form a cocoon. The clitellum is only visible in older worms that are ready to reproduce.

Earthworms have a closed nervous system, with one main blood vessel running down the length of their body. They breathe through their skin, so it is important to keep it moist. They crawl using muscles that are located just underneath the epidermis.

Earthworm nervous system

Earthworms have a relatively simple brain: a fused pair of nerve ganglia, predominantly located in their third segment. The earthworm has one main **nerve cord** on its **ventral** (bottom) side with three fibers running through it: one **medial giant fiber**, and two **lateral giant fibers**. These fibers are big bundles of axons. They serve as high-speed motor pathways that enable the worm to send movement signals throughout its body for quick, stereotyped reflexes.

Each giant fiber is made up of large individual cells (one per segment) that are electrically coupled to each other through gap junctions. This results in the rapid conduction of action potentials from cell to cell so that each giant fiber behaves as though it were a single axon. There are cross-connections between the medial and lateral fibers, so they often function as one unit.



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These fibers differ in important ways. The medial giant fiber transmits information from skin cells in the **anterior** (mouth) end of the worm, and the lateral giant fibers transmit information from the skin cells of the **posterior** end of the worm.

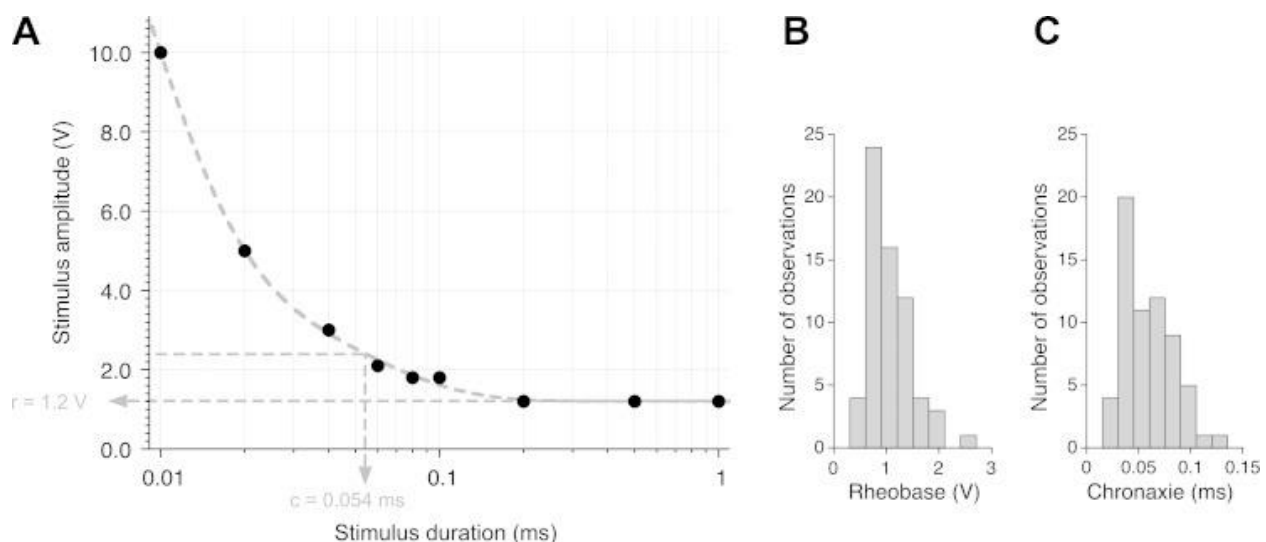
It's quite easy to listen in to the activity of these large fibers, even from the skin of the earthworm. Importantly, these fibers fire action potentials very rarely. This type of nervous system activity is often called **sparse coding**. That means that we should be able to elicit one action potential and cleanly record it.

Properties of action potentials

In this lab, we'll determine the **threshold**, **rheobase & chronaxie**, **refractory period**, and **conduction velocity** for the earthworm giant fiber system. We're recording **compound action potentials** (CAPs) from many axons in these fiber bundles, but the underlying principles are the same.

The **threshold** is defined as the minimum stimulus voltage that elicits an action potential 50% of the time. We'll determine this with short stimuli. Stimulus voltages that don't elicit an action potential are called **subthreshold**, stimulus voltages that *do* elicit an action potential are called **suprathreshold**.

In some neurons, we can elicit an action potential with a lower stimulus voltage if the stimulus length is longer. A longer stimulus length enables charge to build up that can ultimately elicit an action potential. However, there is a minimum stimulus voltage needed, even at infinite stimulus durations. The **rheobase** is defined as the minimum stimulus voltage that can elicit an action potential with very long durations of the stimulus. The stimulus duration at 2x the rheobase is called the **chronaxie**. Chronaxie is a useful measure of the excitability of a nerve — the most excitable nerves have the smallest chronaxie. We can plot this data in a strength-duration curve:



(A) Sample strength-duration curve for an earthworm, with the rheobase (r) and chronaxie (c) marked
(B & C) histograms showing the number of observations for rheobase & chronaxie, respectively. From Bähring & Bauer (2014).

The **absolute refractory period** is the minimum amount of time needed for a neuron to fire a second action potential. This is caused by the inactivation of sodium channels after an action potential — it takes time for them to close, to then be reopened by a depolarizing stimulus. Since we're recording from many axons, your measured refractory period will be more variable than for a single axon. We'll define the absolute refractory period as the time when the amplitude of the second CAP is 30% of the first.

After an action potential, the membrane of the axon is also hyperpolarized, due to the slowness of K⁺ channels closing. So, there is a period of time where the neuron requires more voltage to fire an action potential. This period is called the **relative refractory period**. In this lab, we'll identify it as the period where the amplitude of the second CAP is 30-90% of the first.

In these experiments you will:

- **Stimulate** the ventral nerve cord of the earthworm to elicit an action potential
- **Determine** the voltage that is needed to recruit the lateral giant fibers
- **Estimate** the absolute and relative refractory period between action potentials (optional)
- **Calculate** the speed of an action potential
- **Conduct** an experiment to test whether temperature impacts conduction velocity

References

Bähring, R. & Bauer C.K. (2014) Easy method to examine single nerve fiber excitability and conduction parameters using intact nonanesthetized earthworms. *Advances in Physiology Education* 38(3): 253–264.

Bullock, T.H. (1945) Functional organization of the giant fibre system of *Lumbricus*. *J Neurophysiol* 8:54–71.

Drewes CD, Landa KB, McFall JL (1978) Giant nerve fibre activity in intact, freely moving earthworms. *J Exp Biol* 72:217-227.

Kladt et al. (2010) Teaching basic neurophysiology in intact earthworms. *JUNE* 9(1):20-35.

Rudell & Fox (1991) The Propagation potential: An Axonal response with implications for scalp-recorded EEG. *Biophysical Journal* 60: 556-567.

Shannon et al. (2014) Portable conduction velocity experiments using earthworms for the college and high school neuroscience teaching laboratory. *Advances in Physiology Education* 38(1): 62-70.

Images from Backyard Brains: <https://backyardbrains.com/experiments/speed> & <https://backyardbrains.com/experiments/comparingnervespeed>

Lab Protocol

There are multiple days dedicated to these earthworm experiments. You are encouraged to complete as many of the experiments below while you have a functioning earthworm.

Supplies

- Worm (>3.5 g or >60 mm)
- Dissection tray
- PowerLab 26T, connected to computer
- BioAmp
- Sewing pins (3)
- Timer (on phone/computer)
- Ruler
- Thermometer
- Faraday Cage
- Large Petri Dish

- Eyedroppers (1)

Cables & connectors

- BNC to single banana adapter (2)
- Banana plug cords (2)
- Alligator clip adapters (2)
- Needle electrodes

Solutions

- Earthworm anesthetic (10% ethanol in Ringers)

I. Preparation

Setting up the PowerLab 26T

Our first step is making sure our PowerLab & Amplifier are wired up correctly. See the image on the next page for an overview.

1. Make sure the PowerLab is turned off.
2. Connect a single BNC adapter to the **(+) Output** of the PowerLab. Plug a **red banana plug cable** into the adapter.
3. Connect a single BNC to banana adapter to the **(-) Output**. Plug a **black banana plug cable** into the adapter.
4. Make sure that the BioAmp is connected to the PowerLab, with three different needle electrodes plugged into it, as in the figure below.
5. Once you're confident everything is connected correctly, turn on the PowerLab.

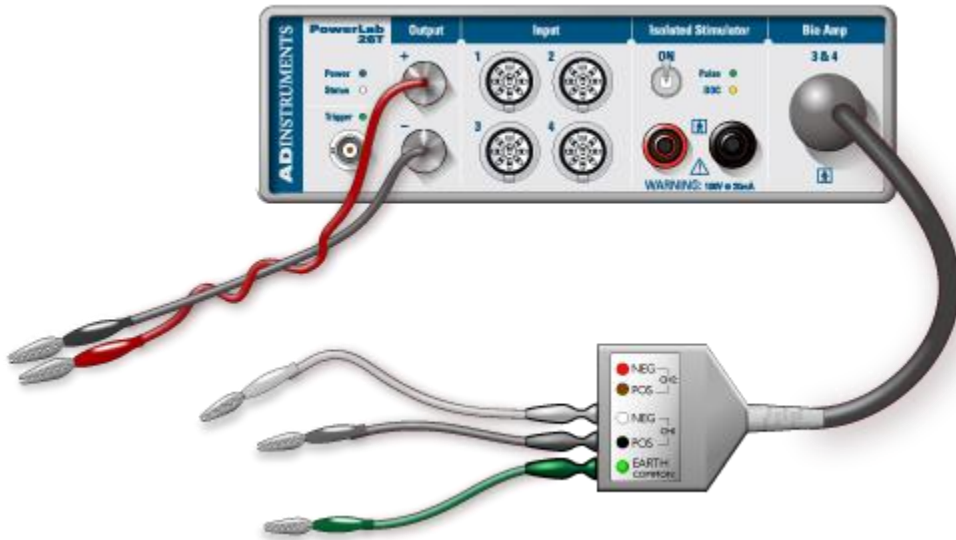


Image courtesy of ADInstruments ©

Setting up LabChart 8

1. Open LabChart 8 with the settings file you saved during the string experiment. Most settings are saved within this settings file. However, it's always good to ensure that you have the correct settings before beginning.
Note: If for some reason you cannot access your settings file, or have issues with it, you can find an appropriate settings file here: <http://bit.ly/labchart>. Use the "Earthworm bioAmp Settings"
2. Go to **Setup > Channel Settings** and confirm that you have three recording channels, and the third ("BioAmp Recording" should be connected to the BioAmp).
3. Go to **Setup > Stimulator** and change the Stimulator settings to be as follows. You'll need to open this window later to modify these settings. Make sure your Marker Channel is Channel 1.

Start Delay	0.1 ms
Repeats	1
Max Repeat Rate	1 Hz
Pulse Height:	0.2 V
Pulse Width:	0.2 ms

Setting up your earthworm

Note: Throughout every step of this procedure, **be careful not to stretch the earthworm**. This will damage the nerve and your experiment will not work.

1. Obtain a large, lively earthworm.

2. Anesthetize the earthworm by placing it in the petri dish with **10% ethanol (in Earthworm Ringer's solution)** for **5 minutes**.

Note: Do not leave your earthworm unattended in anesthesia. As with human anesthesia, the delicate balance between too little anesthesia and too much is tricky. With too little anesthesia, the earthworm will move around during the experiment, and the resulting muscle electrical activity (electromyogram) will drown out the small neural electrical signals you are interested in. Too much anesthesia and the nerves will not fire.

Note #2: Your earthworm may still react to pressure after 5 minutes — you can proceed with the experiment as long as you are able to handle the worm and pin it down.

3. When your earthworm is anesthetized enough to pin it down, remove the earthworm from the ethanol and place it on the dissection tray **with the dorsal (darker side) up**.
4. Pin the earthworm to the tray using one sewing pin on each end of the worm. These pins should not be placed through the middle of the worm, but slightly off to the side to avoid the main blood vessel.
5. Place the ruler down next to your worm.
6. Insert a third sewing pin into the anterior end of the earthworm, about 1 cm from the first pin. Connect the **anode & cathode** stimulating electrodes to the first two pins as shown in the image. Note that the distance matters more than the location relative to the clitellum.

Note: The anterior end is the end nearest the clitellum, a smooth, non-segmented band.

7. To keep the worm anesthetized, apply a few drops of the earthworm anesthetic along the length of the earthworm with the eyedropper. Soak up any excess fluid using a kimwipe.

Note: Make sure there isn't a large puddle of solution, and too much may interfere with your recording.

8. Insert the three recording needle electrodes into the worm as indicated at right (same as the string!)

Note: Make sure no wire is touching any other wire. Be careful when positioning the wires to avoid damaging the worm.

II. Determining the threshold of your fibers

In this experiment, you will send a stimulus into the earthworm while recording from it to determine the threshold voltage of the medial and lateral giant fibers.

Determine the threshold to activate the Medial Giant Fiber

1. Open your **Input Amplifier (BioAmp)** for your Action Potential channel to make sure that your signal looks reasonable. You shouldn't see 60 Hz noise, but it should look like a physiological signal. (If you have noise, see [Troubleshooting](#)).
2. Open the **Stimulator Panel** by going to the **Setup** menu. Your stimulator should start with:

Max Repeat Rate	1 Hz
Pulse Height	0.2 V
Pulse Width	0.2 ms

Note: Leave the **Stimulator Panel** open, we'll need it throughout the experiment.

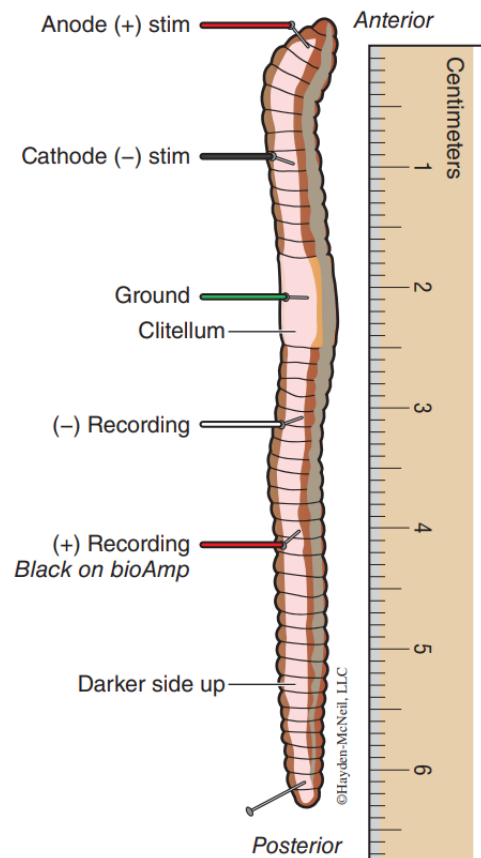
3. Press **Start**. LabChart should be in Scope View and should display one 20 ms sweep every time you click on Start. For each 20 ms sweep, a stimulus pulse is generated through the PowerLab Stimulator outputs.
4. Examine the recording — you should see a quick deflection very shortly after your stimulus pulse. The deflection just after the start of the sweep is your **stimulus artifact**, and is caused by spread of the stimulus voltage to the recording electrodes.

Note: The stimulus artifact can be very large in amplitude, and can exceed the range of the recording amplifier. It is okay if it is cut off.

5. Change the Pulse Height to **1 V**.

Each time you change the stimulus, click on Start and a new page will appear to the left of the **Scope View** Window. This is the **Page Explorer** and allows for easy navigation between 'pages' or sweeps of a file. Label the individual pages of your recording as you go along. This will make it much easier if you need to go back later.

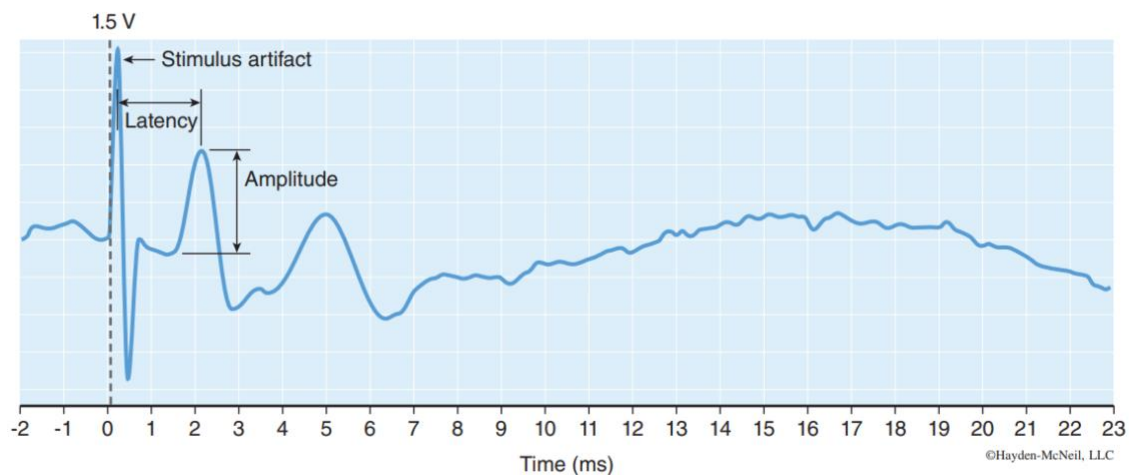
6. Increase the Pulse Height in **0.1 V** steps until you see an inflection *after* the stimulus artifact. Congrats, this means you've successfully elicited a response from the median giant fiber! When the response first appears, note the stimulus voltage as an estimate of the threshold stimulus strength.



Note: The visualization of our recording is key here. If you can't see anything, right click on the channel to change the scaling (or use the - and + buttons on the bottom left of the channel).

Note #2: If no response is seen with a stimulus of 3.5 V or greater, contact your instructor for assistance.

7. Once you've successfully confirmed that you can elicit a CAP from your earthworm, it's time to determine a more precise threshold. Find a more precise threshold by first **reducing** the stimulus by 0.2 V (or more) until you no longer see a CAP. To find the minimum voltage needed to induce a CAP, you will begin at a **subthreshold** voltage and then increase by 0.05 V until a CAP is once again elicited. Using these smaller steps will enable you to find a more accurate threshold. You should report your threshold (Table 1) to the hundredth decimal point (e.g. 1.25 V).
8. Save your data, but do not close the file.
9. When you are done collecting data, you will use the Marker Tool to measure the latencies and amplitudes, as indicated below:



Determine the threshold to recruit the Lateral Giant Fibers

In the next stage, you will attempt to elicit an action potential in the lateral giant fibers.

Note: Remember to keep the earthworm anesthetized!

1. Set the Pulse Height to your threshold for the MGF.
2. Keep increasing the stimulus in 0.1 V steps until you see a second inflection in your recorded trace.
3. Once you see a second inflection, follow the same procedure as above to determine the threshold where you can elicit an action potential from the lateral giant fiber about 50% of the time. Record the stimulus threshold in Table 1 below.
4. Save your data, but do not close the file.

Note: In some worms, the median and lateral fiber responses may not be separated – they will appear to have the same threshold. Other worms may not show the second response, possibly because the lateral giant fibers are damaged.

6. Save your LabChart file.

Table 1. MGF and LGF compound action potential (CAP) properties.

	Threshold	Amplitude	Latency
MGF			
LGF			

Questions for reflection

Did the shape or amplitude of the CAPs change with more stimulation voltage? What does this tell us about the action potential that is firing?

Which fiber fires an action potential sooner (with a shorter latency)?

Why would the MGF and LGF latencies differ?

III. Calculate the rheobase

In Part III, we'll change the stimulus amplitude and width to generate a strength-duration curve calculate the **rheobase** and **chronaxie** of the earthworm medial giant fiber (read the Background if you forget what these are). Note that "Duration" in LabChart is the length of the recording window in Scope view – the "duration" in strength-duration refers to the Pulse Width.

In the previous experiment, we gave a stimulus for 0.2 ms. Now, we'll modify the stimulus width and amplitude to determine what the minimum stimulus requirement is, regardless of the pulse width. The goal here is to determine the combination of stimulus duration and amplitude that works for your earthworm, but you should get data that is similar to the chart in the Background.

1. Write your MGF threshold in the column for 0.20 ms stimulus duration in Table 2.
2. Change your Pulse Height to the threshold for your MGF.
3. Systematically change the length of your stimulus (Pulse Width) to the values in the table below. As necessary, change the stimulus voltage to successfully get a CAP from your MGF. Given that these nerves may exhibit synaptic depression, you may want to start with the longest pulse widths (the right side of the table) rather than the shorter pulse widths.

- For each Pulse Width, record the stimulus voltage (Pulse Height) values that successfully elicit a CAP in Table 2.

Note: Be sure to label your pages in Scope view accordingly. You don't need to save the pages that *do not* elicit an action potential.

Table 2. Data for strength-duration curve.

Pulse Width (ms)	0.06	0.08	0.10	0.20	0.40	0.60	1	1.4	1.8
Stimulus Amplitude (V)									

Questions for reflection

Why does the required stimulus voltage for a CAP decrease as you increase the width of the stimulus?

IV. Compute the conduction velocity

(If you're doing this on a different day, follow the same steps for preparing LabChart & your earthworm as you did in the previous earthworm experiment.)

How far is your first recording pin (-) from your cathode (-) stimulation pin? Be sure to include units:

- In PowerLab, press **Start**. Confirm that you have two recording channels.
- If you are doing this experiment with a new worm, follow the same protocol as in the previous experiments to determine the threshold required to elicit an action potential from your median giant fiber.

What is the stimulus voltage needed to elicit an MGF action potential from this worm (if it's a different worm)?

In the cold condition, you may need more voltage (a *suprathreshold* voltage) to elicit an action potential. Add ~0.5 mV to your threshold voltage to determine an appropriate stimulus voltage for both the cold and room temperature conditions in your conduction velocity experiments.

Determine the MGF latency to spike

1. Using the Marker tool, measure the latency (in seconds) from the peak of the stimulus artifact to the peak of your elicited compound action potential.
2. Repeat this 10 times and record your data in the “MGF Baseline AP Latency” column in the table below. (You’ll fill out “Cold Latency” in the next step of the experiment.)
3. Since we know when the AP occurred, and we know the distance between your electrodes, we can calculate the conduction velocity of the action potential (reminder: $\text{speed} = \text{distance}/\text{time}$.) Record these values in the table below.

Test the impact of temperature on MGF conduction velocity

For this final part, you will design your own experiment to test the impact of temperature on conduction velocity. We will provide ice and a thermometer, but the rest is up to you!

Work with your group to determine a protocol for running your experiment — you’ll need to include this in the Methods section of your lab report. Think about what someone needs to know in order to replicate this experiment *exactly* as you did.

Note: Make sure that you have sufficiently cooled your worm (by at least 5 degrees Celsius) before measuring your Cold conduction velocity.

Once you have a plan, conduct your experiment. Record your results in the MGF Cold Latency column of the table below.

Trial	MGF Baseline Latency (s)	Conduction Velocity (m/s)	MGF Cold Latency (s)	Conduction Velocity (m/s)
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Questions for reflection

How fast, on average, is your action potential traveling down the MGF? _____

Why might temperature impact conduction velocity?

V. Determine the refractory period (optional)

In this experiment, you'll determine the shortest time between stimuli where you can elicit two different CAPs from the MGF. In other words, we'll determine the **relative** and **absolute refractory period** of these fibers.

1. Set your pulse height so that you can comfortably elicit a response from the MGF.
2. In order to play multiple pulses, we'll modify the number of repeats and speed of these repeats. In **Setup > Stimulator**, change Repeats to 2.
3. We also need to change the time between each stimulus. PowerLab thinks of this interval as Hz. You'll determine at what frequency (in Hz) to stimulate your worm. To start, we want 15 ms between each stimulus.

Note: Remember that Hertz (Hz) means events per second: 1 Hz = 1 event per second, and 10 Hz = 10 events per second. At 1 Hz, each stimulus is 1 second apart. The time between stimuli (the **interval**) can be measured by taking 1 over the frequency. As an example:

$$\text{Interval (s)} = \frac{1}{\text{Frequency (Hz)}}$$

If the frequency is 100 Hz:

$$\text{Interval (s)} = \frac{1}{100 \text{ Hz}} = \frac{1}{100 \text{ events/s}} = \frac{1}{100 \text{ s}} = 0.01 \text{ s} = 10 \text{ ms}$$

Instead, if we *know* that we want a 10 ms interval between each stimulus and would like to know what that translates to in frequency:

$$4. \text{ s} = \frac{1}{\text{Frequency (Hz)}} = \frac{1}{0.01 \text{ s}} = 100 \text{ Hz}$$

How many *milliseconds* would there be between each stimulus at 10 Hz stimulation?

For 15 ms between each stimulus, what should the frequency of stimulation (in Hz) be?

Fill out this chart with the corresponding Hz values for each interval length:

Interval (ms)	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1
Freq. (Hz)															

4. Change your **Max Repeat Rate** to the equivalent frequency for 15 ms intervals.
5. Change your Pulse Width to 0.2 ms and your Pulse Height to the threshold for your MGF.
6. Press **> Start** to record a trial with two stimuli, separated by 15 ms.
7. Right click and choose **Add Marker > Clamp to Trace** to place a marker at the moment the stimulus occurs. Move your cursor to the peak of the action potential. Look at the ΔV on the right hand side to see the change in voltage from that point.
8. Measure the amplitude of your first and second action potentials. Then, determine how strong the second was relative to the first. In other words, if AP amplitude #2 was half as high as #1, it would be 50%. Record your response in row one of **Table 3**.
9. Decrease the time between stimuli by 1 ms by changing the Repeat Rate to the Hz for a 14 ms interval.
10. Record the amplitudes and corresponding %s for each stimulus interval in **Table 3**.
11. At some point, AP #2 will diminish below 30% of AP #1. We can consider this our **absolute** refractory period.

Note: Depending of the latency of your MGF CAP, it may be difficult to see a CAP with a stimulus interval of 1 or 2 ms. Reduce the stimulus interval to as short as you can before the MGF CAP is cut off by the second stimulus. If the shortest stimulus interval is > 3 ms, you should move your pins closer together. If it is ≤ 3 ms, you should explain in your lab report discussion why you were unable to get data for very short intervals.

12. Save your LabChart file.

Note: If you have data that doesn't seem quite right, you should repeat your experiment. **You must be upfront about your number of repeats** in your lab report, comprehensively reporting all of the data that you collected.

Table 3. Amplitudes of CAPs elicited by stimuli with decreasing stimulus intervals.

Stimulus Interval (ms)	#1 CAP Amplitude	#2 CAP Amplitude	% #2/#1
15			

Questions for reflection

What was the shortest time between stimuli where the second action potential was less than 90% but more than 30% of the original CAP amplitude?

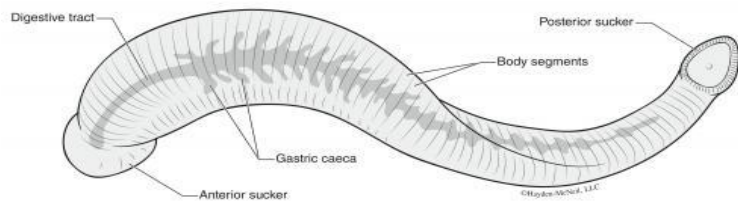
What was the shortest time between stimuli where the second CAP was diminished below 30%?

Which of these values is your absolute refractory period, and which of these is the relative refractory period?

Troubleshooting

Observation	Likely issue(s)	Possible solution
The recording trace is flat	Something is disconnected	Check all of your connections
I can't see a stimulus trace in the Stimulus channel	You don't have a marker channel set up	Go to Setup > Stimulator to set up the Marker Channel.
I can't see a stimulus artifact in my action potential channel	Your electrodes may not be in the worm, or your stimulus amplitude is too weak	Ensure all electrodes are in the worm (but not too medial), and try stimulating at 0.5 V
My worm is bleeding	The electrode is through the blood vessel	Move the electrode more lateral; if you don't get a CAP after this, get a new worm.
There is a stimulus artifact, but no visible response to stimuli between 1-3.5 V	Your electrodes may not be placed well, or your worm may be broken, your BNC cord or connection is broken.	Try to reposition the electrodes on a different place in the worm. If this doesn't work, get a new worm. Get a new BNC cord. Try stimulating via the negative output on your PowerLab.
My recording just looks like a big up and down squiggle (like a sine wave)	You're picking up 60 Hz noise	Ground everything (Faraday cage, ground recording pin, amplifier) to the ground pin on the back of the PowerLab. Averaging multiple trials can also get rid of noise. If you still can't get rid of the noise, ask an instructor for help.

Experiment #4: Leech Electrophysiology



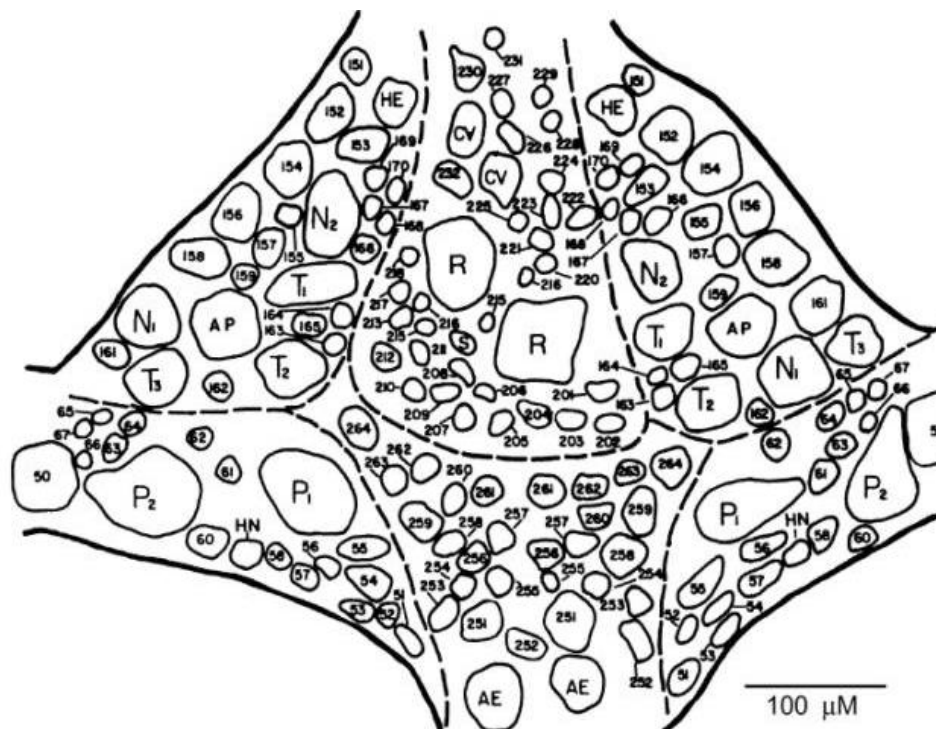
Background

It might seem strange, but leeches are fantastic models for neuroscience experiments. They have a rich, well-defined repertoire of behaviors and an easily accessible, stereotyped nervous system. In this lab, we'll spend two days recording from the medicinal leech, *Hirudo medicinalis*, to determine some physiological and morphological properties of neurons.

Leech nervous systems

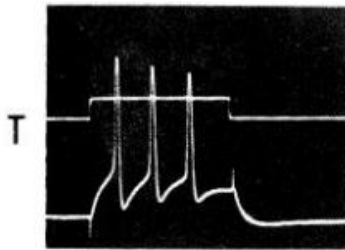
Leeches are **annelids** (segmented worms), just like our earthworm friends. They have a ventral nerve cord containing a total of 32 ganglia: 4 fused head ganglia, 21 segmental body ganglia, and 7 fused tail ganglia. We'll be recording from the body ganglia.

Each ganglion has about 400 neurons in it, most of which are paired — in other words, most of them are connected to one other neuron. The ganglia are all connected via connectives that run the length of the leech. In addition, each segmental ganglion has lateral roots that reach out to the skin. The neurons in each ganglion are laid out in stereotypical positions, and have well-defined electrophysiological properties. We'll be looking at the ganglion from the ventral side, where many neurons are accessible:



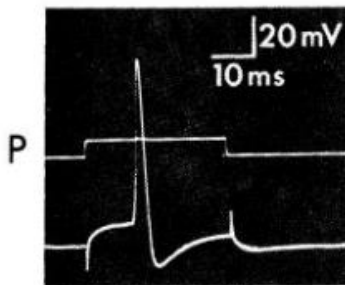
Types of cells in the leech

Rz (Retzius): the biggest cells of the leech ganglion, located in the middle of the ventral side of the ganglion. These cells often show a second component in their action potential due to the close electrical coupling with the other Rz cell. These cells do *not* overshoot.



Sensory neurons: Each of these neurons project its axon from the ganglion to a particular territory of the skin, where they have specialized mechanoreceptors. Unlike the Rz cell, these sensory neurons have overshooting, very high amplitude action potentials.

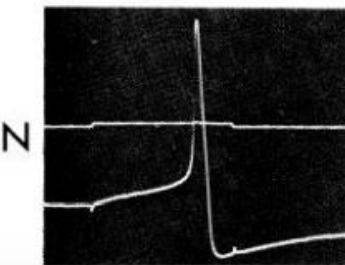
T (touch): three of these in the anterolateral packets on each side of the ganglion. Typically very short, relatively low amplitude waveform (compared to other sensory neurons; sample waveform at left)



P (pressure): two of these in each posterolateral packet (sample waveform at left)

N (nociception): two in each of the anterolateral packets. These cells have very long undershoots (sample waveform at left).

AP (anterior pagoda): large cells near the center of the anterolateral packets



AE (annulus erector): motor neurons to the AE muscles in body walls; medium-sized cells in the posterior ventromedial packet

S: unpaired cell, unlike the cell types listed above, much smaller than the other neurons listed. Record from this neuron and you shall gain infinite respect from the Gods of leech neurobiology.

About the microelectrode & amplifier

Instead of using a wire, we'll be recording from these cells with a glass pipette that has been pulled to have a very tiny and sharp tip and is filled with conductive fluid. These are typically called **microelectrodes**. Recording from nervous systems in this way was a huge leap forward in 1949 — before then, we weren't able to record from *inside* of cells (Ling & Gerard, 1949).

Back in the day, physiologists would make these small pipettes by pulling the glass over a flame. Now, we've got a dedicated machine to pull pipettes, allowing us to reproduce many microelectrodes that are the same size.

And about the size — these micropipettes are really small. This small cross-sectional area of the microelectrode greatly reduces the flow of current through the electrode. In other words, it has a **really high resistance** (often imprecisely termed "impedance" in electrophysiology circuits). This is the quality that allows us to record very tiny changes in voltage in our cells.

This high resistance is necessary, but also brings up some other problems. With really high resistance across our electrode, membrane, and PowerLab, we would almost record no voltage from a 60 mV cell.

So, for this lab, we're using a different amplifier. Because the impedance of our microelectrodes is much higher than the electrodes we used for the earthworm, we need an amplifier that can "impedance match" the microelectrode with the PowerLab. It also means that the intracellular recording system must be shielded from electrical noise much more than the extracellular system.

But with all this in place, you'll get a chance to see some *real* action potentials from a strange, slightly terrifying creature.

In these labs you will:

- Learn how to operate and calibrate an intracellular amplifier
- Record action potentials from Retzius cells, both with and without current injection
- Fill a single cell in the leech with a fluorescent dye to visualize its structure

Resources

- HHMI Virtual Leech Lab <https://www.hhmi.org/biointeractive/neurophysiology-virtual-lab>
- JOVE Video <https://www.jove.com/video/50631/intracellular-recording-sensory-field-mapping-culturing-identified>

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Intracellular Equipment

The goal of this first experiment is to troubleshoot the noise in your rig, learn how to check the electrode resistance, and balance the bridge on your amplifier. You'll also get familiar with the microscope we're using and finding tiny things in a dish.

In subsequent labs, we'll be recording from tiny cells in the leech ganglia, as well as filling these cells with a fluorescent dye. Those experiments will require that you can quickly identify and manipulate a microelectrode underneath the microscope in order to find your cell of choice. Time is of the essence here — we're keeping cells alive in a dish, and they won't last forever. Becoming proficient at these steps will save you time during the recording experiments, and therefore increase your likelihood of success.

Supplies

- Clear plastic holder for petri dish
- PowerLab 26T, connected to computer
- A-M Systems 3100 Intracellular amplifier & silver head stage probe
- Microelectrode holder
- Microelectrode
- Faraday Cage
- Metal baseplate & foam pads
- Micromanipulator & stand
- Stereomicroscope on boom stand
- Light source
- Syringe & needle
- 4 cm diameter petri dish with microbeads
- Ruler

Cables & connectors

- BNC-DIN8 Adapter
- BNC Cable
- Banana plug cords (5)
- Alligator clip adapters (5)
- Ground wire & clay or wax
- Fake neuron

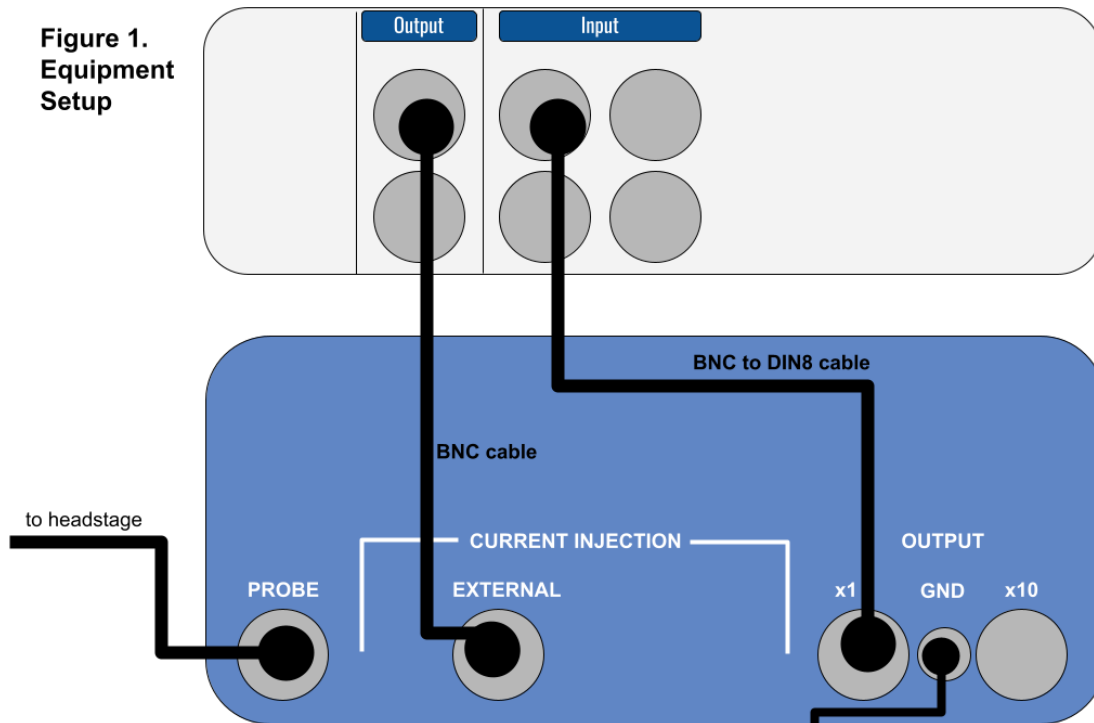
Solutions

- 3M KCl
- Saline

I. Preparing your recording setup

Setting up PowerLab & the Amplifier

1. Make sure that your PowerLab is off.
2. Configure your PowerLab & amplifier as in Figure 1:



3. The amplifier settings should be as follows:

DC Offset knob: Counterclockwise

DC Offset (+/Off/-): Off

Transients: Counterclockwise

DC Balance: Counterclockwise

Low Pass: 5 kHz

Ω TEST: Off

Notch: Off

Capacity Compensation Knobs: Fully counterclockwise

Current Injection: Fully counterclockwise (zero) * **VERY IMPORTANT!**

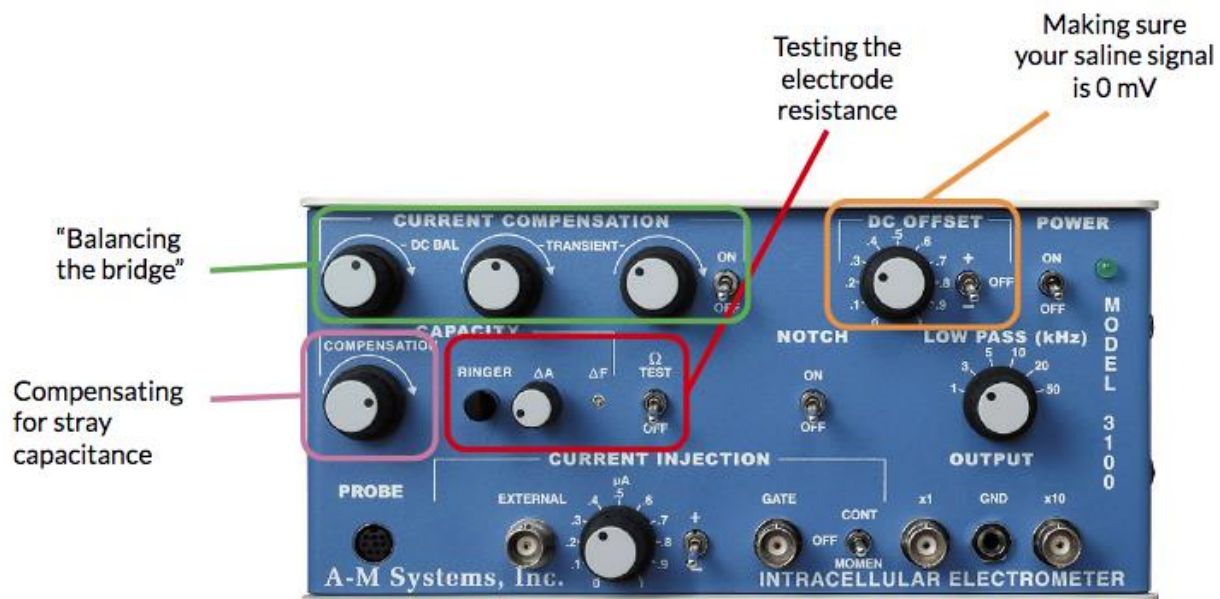
Current Injection Knob (CONT/Off/MOMEN): CONT

4. Once everything is connected, turn on your PowerLab and the Amplifier.
Note: The AM 3100 amplifier needs to warm up for 5 minutes before doing any recording. You can proceed to set up your electrode though.

II. Fake Neuron

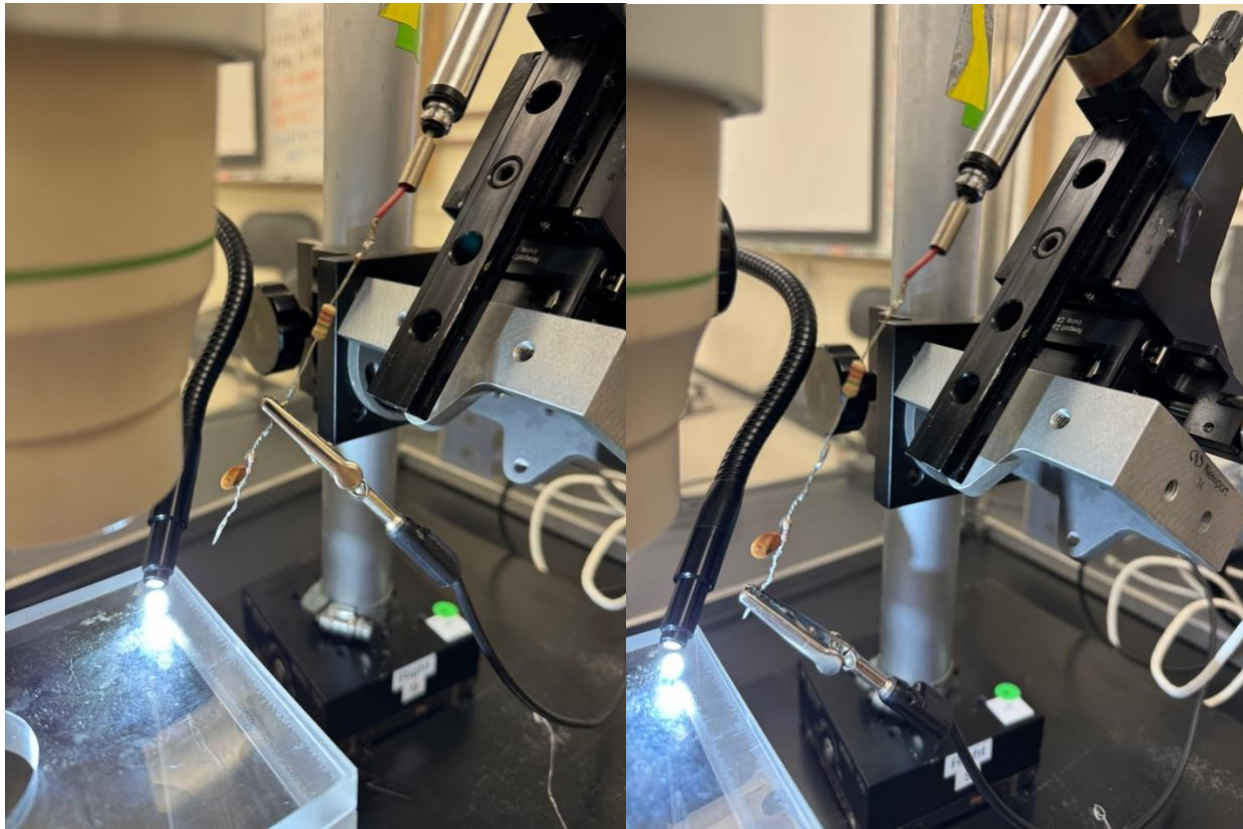
In this experiment, you'll use a "fake neuron" to model what happens when we inject current across the electrode and across the cell membrane.

1. Plug the end of your fake neuron into the silver headstage (picture on next page).
2. Plug a banana wire into the "GROUND" on your intracellular amplifier.
3. Put an alligator clip onto the other end of the ground banana wire and clip it onto the fake neuron, **in between the resistors** (as you see on left, on next page).
4. Open LabChart with the Leech Settings file (<http://bit.ly/labchart>).
5. Open the **Input Amplifier** for your recording ("Action Potential") channel.
6. Change the zoom on your signal and the range (if necessary) so that your signal fills about $\frac{2}{3}$ of the screen.
7. Chances are you'll see 60 Hz noise. In order to get rid of this, ground (one-by-one, watching to see how they change the signal), your: Faraday cage, metal plate, microscope, and light (clip on to the metal parts) to the ground pin on the back of the PowerLab. It will help to piggy-back banana plug cords. You also have a big metal grounding block to help organize your ground wires.
8. Once you've reduced your noise as much as possible, you can move on.



For the next steps, you'll need to use these knobs on your amplifier.

9. In LabChart, program a pulse with the following characteristics:
 - Pulse width: 1 second
 - Pulse height: 0.01 V. This corresponds to a 1 nA injection into the circuit.
 - Delay: 0.1 seconds.
 - Marker channel: Channel 2.
 - Number of repeats = 1
10. In Scope view, inject your current and observe the change in voltage. Note: The Duration of your Scope window needs to be longer than your stimulus!
11. Use the change in voltage to determine the value of the first resistor: _____.
12. Move the ground alligator clip so that you're recording across both the resistors and the capacitor (as you see on right, below).



13. Inject your current and observe what happens across the electrode.
14. Use the curve that is generated to estimate the time constant, just as you did in the RC Circuit lab.

15. Note that you may want to balance the bridge, but regardless keep in mind that the initial jump in voltage is because of the “electrode” resistance!
16. Note that you may also want to lengthen your pulse width to be able to see the circuit charge completely.
17. Using your estimated time constant and $\tau = RC$, estimate the capacitance: _____.

Challenge question: If this “neuron” had 1 uF specific capacitance, how big would you estimate this neuron to be?

III. Tiny beads, optics, & manipulators

This next activity will prepare you for recording from a leech ganglion. The steps you take to set up your microscope, light source, and manipulator will be almost identical in the leech lab. You might have already used a microscope before, but it’s less likely that you’ve had to do something very precise while doing so!

Determining the diameter of your viewing field

1. Compute the total magnification of your microscope at 1.8X by multiplying 1.8X times the objective magnification (typically 10x, check your oculars). Write this value in Table 1.
2. Similarly, compute the total magnification for ~5X and 11X and write these values in Table 1.
3. Place the ruler underneath the microscope at 1.8X and look through the oculars. Determine the diameter of your viewing field and record this in Table 1.
4. It’ll be tough to do the same at high magnification, so instead we can use this formula:

$$\text{high power field diameter} = \text{low power field diameter} \times \frac{\text{low power magnification}}{\text{high power magnification}}$$

5. Using this equation, determine the diameter of the viewing field for 5x and 11x.

Table 1.

Objective	Total Magnification	Diameter of viewing field (mm)
1.8x		
5x		
11x		

Setting up your lights and microscope

1. First, set your dish with microbeads into the clear acrylic holder. Be careful placing these into the holder — it's designed to fit on the ledge inside.
2. Fill your petri dish with saline.
3. Arrange your lights so that they're shining onto the dish.
4. Bring the beads into focus at a low magnification (1.8-3x).
5. Move to the highest magnification and bring the beads into focus. Adjust your light source so that the beads are clearly visible.
6. Use your calculated viewing field diameter to estimate the size of the **biggest** bead in your dish.

Given the diameter of your viewing field, what is your estimated diameter of the *smallest* microbead in your dish?

diameter = _____

Using your manipulator

1. Get a micropipette from the front and use the syringe to fill it with KCl.
2. Gently put the micropipette onto the wire of the microelectrode holder and screw it in.
3. Put the microelectrode holder into the silver headstage by pushing the gold pin into the bottom.
4. Make sure that your manipulator is set up next to your dish and plastic holder at a right angle. In other words, when you turn the “y” knob of the manipulator, it should move directly up and down in your dish, and turning the “x” knob should move it left and right. Any angles that are not perpendicular will make it difficult to maneuver your micropipette.
5. Move the manipulator base (you need to turn the magnetic switch to OFF) close to the dish so that the tip of the “electrode” is within 7.5 cm of your dish.
6. Use the knobs on the manipulator to bring the tip of your pipette about ½ inch above your beads.
7. Bring the focus above the dish so that the tip of the pipette is now in focus.
8. Lower the focus and bring the tip of the pipette into that field of view.
9. Repeat this until the tip of the pipette is at the same focus as the glass beads.

10. At a high magnification (11X), you should see your electrode bending once you poke the beads (or the bottom).
11. Once you have the tip of your pipette at the same focus as your beads, raise your electrode.

Determine how long it takes each of your group members to lower the electrode (without breaking it!) to the beads and record it in the table below:

Group Member	Time (in minutes)

Troubleshooting

Observation	Likely issue(s)	Possible solution
The view through the scope is blurry	You're out of focus	Adjust the focus
There are weird light artifacts and reflections that make it difficult to see	The light is not set up well	Change the angle, intensity, and distance of the light source. Try 1 light source vs. 2.

Prepare the saline, manipulator, electrode, and ground wire

1. Your manipulator and electrode should already be set up from the previous step. However, if you broke your electrode during Step I, you should prepare a new one.
2. Using the clay, position the **ground wire** (the wire that has a banana plug on one end and an exposed wire on the other) so that it is in the saline but at the edge of your petri dish. Your ground wire should be plugged into "GND" on the amplifier. You may need to piggy-back it onto another banana plug cord to do so.

IV. Noise, resistance, and balancing the bridge

Check for noise in your signal

First things first — with this recording configuration, we'll probably pick up much more noise than we need to troubleshoot.

3. Maneuver the tip of the microelectrode into the saline.
Note: Be very careful while moving your microelectrode around! If you break the tip of it, you'll need to get a new electrode.
4. LabChart should be open with the Leech Settings file (<http://bit.ly/labchart>).
5. Open the **Input Amplifier** for your recording ("Action Potential") channel.
6. Change the zoom on your signal and the range (if necessary) so that your signal fills about $\frac{2}{3}$ of the screen.
7. If needed, adjust the de-noising cords around your rig to reduce your noise as much as possible.

Note: As you're moving things around your rig, make sure that your lights can reach your petri dish and that you can easily manipulate the electrode & knobs on your amplifier.

Check the resistance of your electrode

1. With the Input Amplifier window open, note the voltage of the trace. It will likely not be exactly at zero. This is due to the slight differences in the potentials of the ground and recording electrodes (**the offset potential**) and will change when you change electrodes.
2. Switch the **DC OFFSET** to positive (+). Using the **DC OFFSET** knob on the amplifier, position the tracing so that it is very close to 0 volts. You may need to set the switch next to **DC OFFSET** to negative (-) depending on what direction you need to move your signal.



Note: If the voltage is drifting wildly, make sure that your ground wire is in place in the saline and that everything is still grounded properly. Restarting LabChart & turning on/off the PowerLab is also a good troubleshooting step.

3. With the tip of your electrode in the saline, press **Start** in LabChart. You should be viewing in Chart mode.

4. Very quickly, turn on and off the **Ω TEST** on the Amplifier. This **Ω TEST** sends 10 nA of current into the electrode.
5. Press **Stop** in LabChart.
6. For the data you just recorded, autoscale the channel in time and voltage so that you can clearly see a square wave pulse.
7. Measure the height of this square wave in volts.

Calculate the resistance of your electrode according to Ohm's Law ($V = IR$). In the units we're working with here, $1 \text{ M}\Omega = 1 \text{ mV} / 1 \text{ nA}$. Since the **Ω TEST sends 10 nA, with 1X gain a 10 mV shift in the output voltage corresponds to 1 MΩ resistance.**

Electrode resistance = _____

The resistance of your electrode should *generally* be between 20-60 MΩ. If it's too high (>60 MΩ), it might be clogged. Use the **Ringer** button and try again. If it's too low (<20 MΩ), the tip of your electrode probably broke — get a new electrode.

Please note that what we consider an acceptable resistance depends on how the electrodes look each quarter and how the lab is progressing. We will let you know if you should refer to a different resistance range than the one given above.

Balancing the bridge for current injection

We need to balance the bridge circuit of the amplifier if we want to inject current into a cell. The cell and the electrode each have their own capacitance and resistance. You dial out the resistance of your electrode before you impale the cell; otherwise your recording will reflect the voltage change across the electrode, instead of the actual voltage of the cell.

1. Ensure that the **CURRENT INJECTION (CONT-OFF-MOMEN)** is on **CONT** and the **μA** knob rotated fully counterclockwise.

Note: If this switch is not on **CONT** (continuous) *no current will be injected.*

2. Go to the **Setup** menu and select **Stimulator**. Set up the Stimulator to deliver a 10 mV (1 nA current) as a train of 100 pulses:

Start Delay	0 seconds
Repeats	100
Max Repeat Rate	0.66
Pulse Height	0.01 V (10 mV)
Pulse Width	1 s

3. Set Channel 2 as your stimulus Marker Channel.
4. Close the Stimulator dialog and open the Stimulator Panel. **Start** recording, and if necessary zero the electrode again using the **DC OFFSET** knob.
5. Turn the stimulator on to inject a train of 1 nA of current pulses.
6. Adjust the **DC BALANCE** knob as necessary to remove the electrode voltage drop from the output signal.

7. There will still be “rabbit ears” or transients – these are large spikes when you turn the current on and off – but very little change in standing voltage (Figure 2).
8. Adjust the Current Compensation **TRANSIENT** controls to minimize transients occurring when the current is gated on and off. The left knob controls the slope of the transient, the right knob controls the peaks.
9. Stop recording.
10. Set the **CURRENT INJECTION** knob back to **OFF**.
11. Save your data as “Leech electrode setup”. If you made any changes to the settings file (e.g., Channel names, range, etc.) you’ll also want to save it as a new settings file by going to **File > Save As LabChart Settings File**. Save this in a safe, clouded place.

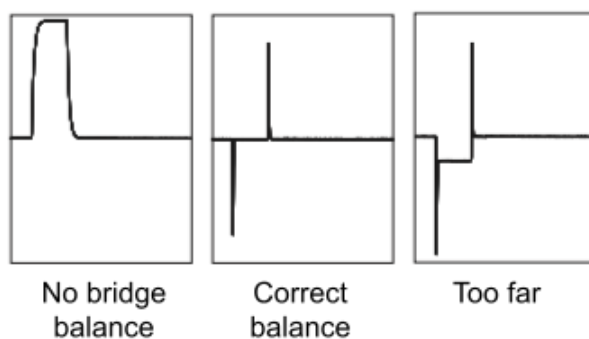


Figure 2

Recording from Retzius cells in the leech

I. Preparing for recording

First, you need to fill and arrange your electrode, troubleshoot noise, and configure your amplifier in the same way that you did in the intracellular equipment lab.

Once you've confirmed that your manipulator and recording equipment is ready to go, get a leech ganglion.

1. Maneuver your electrode into the saline bath. At the moment, there isn't any need to get it close to the ganglion — as long as it's in the saline, you can measure the resistance and troubleshoot noise.
2. Make sure the peak-to-peak noise level of your recording is < 5 mV. There shouldn't be any obvious sinusoidal noise in your data. If there is, find the source and ground it.
3. Test the resistance of the electrode (follow the steps from the previous lab):

Electrode resistance: _____

4. Quickly balance the bridge.

Getting your leech & microelectrode in focus

Before we can do any recording, we need to ensure that your light is set up well so that you can see your ganglion.

1. Gently rest your petri dish in the plastic holder.
2. Bring the leech ganglia into focus and adjust your light and manipulator positioning as necessary.
3. When you can clearly see your ganglia, see if you can spot the two Retzius cells in the middle (Figure 3).
Note: If you cannot see the two Retzius cells, it might be because your lighting needs to be adjusted, or because your ganglia is mistakenly set with the ventral side up.

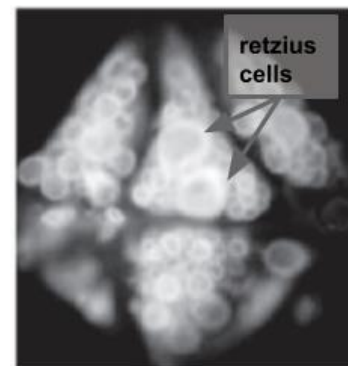


Figure 3. Ventral view of leech ganglion.

4. Using the clay, position the ground wire so that it is in the saline but at the edge of your petri dish. Your ground wire should be plugged into "GND" on the amplifier.

II. Recording from a Retzius cell

Finding and impaling a Retzius cell

1. Slowly and carefully move the tip of your electrode down towards the ganglia, until it is very close to the Retzius cells.

Note: It helps to do this at medium magnification (5-8X) but in the end, you should be looking at your cells and electrode at the highest magnification possible (11X).

Note #2: The tip of your electrode will probably be very hard to see. You can move it slightly from side to side if you lose track of it. You may need to adjust your light source so that you can see both the ganglion and the tip of the electrode.

2. While hovering above the cell but before getting too close, **check your resistance again**. Be sure to also keep the recording voltage close to zero while you're outside of the cell.
3. Once you think you're as close as possible to the cell, even dimpling the cell under the pressure from the electrode tip, start recording.
4. Firmly tap the electrode to encourage it to enter the cell. While tapping, watch the recording screen.
5. If you entered the cell, you should see your membrane potential rapidly drop to about -50 mV and stay there.

Note: If your membrane potential goes below -65 mV, you're likely in a glia cell.

6. If all goes well, you should see your Retzius cell firing very clear spontaneous action potentials!
7. If your first attempt is unsuccessful, slightly advance the electrode towards the cell and try again. If this doesn't work, reposition the electrode to a different part of the cell and try again. Frequently check the resistance of your electrode to ensure that it is not broken or clogged.
8. **SAVE YOUR DATA** as soon as you have anything to save. Periodically save your data throughout the entire lab.

HAVING TROUBLE? Make sure the following is true:

- Both the recording electrode & ground wire are in the saline
- Electrode is *fully inserted* into electrode holder
- Amplifier is ON
- The amplifier CURRENT INJECTION knob is at 0 μ A (fully counterclockwise)
- If trying to zero signal: DC OFFSET switch is on + or -
- See TROUBLESHOOTING at the end of this protocol for additional tips.

Once you have successfully impaled a Retzius cell, take note of the resting membrane potential of the cell:

Resting membrane potential: _____

Note: The resting membrane potential will change slightly over time. Your most accurate measurement will be from when you first pierce the cell.

Also take note of the shape of this action potential. Does it overshoot (past zero)? Does it undershoot?

Recording spontaneous action potentials

We'll collect 10 different 10 second windows of spontaneous action potentials from your cell, in order to estimate the **spontaneous firing rate** and **measure the action potential height, spike width, and after-hyperpolarization** of your cell.

Note: You can also do this with the data you recorded directly after you impaled the cell. But having 10 separate windows will make it a bit easier to calculate the spontaneous firing rate.

1. Configure **Scope View** to collect data with a 10 second duration.
2. Set repeats to 10, and press Start.
3. Confirm that you have 10, 10 second recordings with spontaneous action potentials by looking through your pages.

You will ultimately use this data to fill in Table 1, but **don't take the time to do this now**. Move on to stimulating your Retzius cell while you have one.

III. Stimulating the Retzius cell

Once you have characterized the spontaneous activity of your cell, we can also inject current to see how the cell responds to inputs.

1. Set up your Stimulator as a pulse with the following parameters:

Delay	5 ms
Max Repeat Rate	0.5 Hz
Repeats	1
Pulse Width	500 ms
Pulse Height	10 mV (PowerLab will tell the amp to send 1 nA for each 10 mV)

2. Make sure that your **CURRENT INJECTION** toggles are on positive (+) and **CONT**.
3. Configure **Scope View** in LabChart so that you can see one, 1 second sweep when a stimulus pulse is sent. In other words, set the Duration to one second.

4. Press **Start** to send current into your cell. Observe whether or not this elicits action potentials from the cell, and whether the cell adapts to the stimulus.
5. Repeat the stimulation 10 times (or more, if it seems highly variable). You'll need these repeats for your spike train metrics in your lab report.
Note: For this step, **do not** use averaging. Averaging will repeatedly send the same stimulus with only a short break. Give the cell a few seconds to recover in between stimuli.
6. **Repeat the stimulation** with the following current levels (with 10 repeats per current level): **-0.5 nA, 1.5 nA, 2.0 nA**. You'll ultimately need to count the number of action potentials for each of these (you should have **four** different current levels total) for a figure in your lab report.

Sample data Notebook for Retzius cell recording

Table 1. Spontaneous action potential characteristics.

Action Potential	AP Height	AP Spike Width	After-hyperpolarization Amplitude
1			
...			
Mean			
SD			

Note: This table is a suggestion for how to ultimately collect data for this portion of the report. Set up a similar table *after* you have collected your data.

Table 2. Characteristics of spike trains elicited by a 500 ms, 1 nA square wave stimulus.

Trial	Latency to spike	First interspike interval (ISI)
1		
...		
Mean		
SD		

Note: This table is a suggestion for how to ultimately collect data for this portion of the report. Set up a similar table *after* you have collected your data.

Filling Leech Ganglia

Additional Supplies

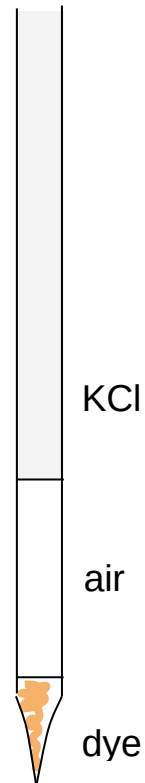
- Glass slide & coverslip
- Glycerol
- Bulb & glass pipette
- Small vial of either **Alexa Fluor 488 (green)** or **Rhodamine (red) conjugated dextran dye** (5% in dH₂O)
- Phosphate buffer (PB)
- 2% Paraformaldehyde (PFA) in 0.1 M PB
- Waste containers (2) Preparing the electrode

I. Preparing Your Experiment

First, we have to prepare an electrode that has dye in the tip. These dyes diffuse well through cells and will latch onto proteins when we fix the ganglia.

We'll use one of two dyes: an Alexa Fluor 488 dextran (green), or a Alexa Fluor 594 dextran (red). If you have time today, you are welcome to fill more than one cell to compare two neurons.

1. You'll be given a small vial with dye in it.
2. Place your glass pipette with the **back** end (the *non*-sharp end) into the small vial. The dye will be wicked along the fiber inside of the pipette.
3. Hold the pipette there for ~10-30 seconds until you can see traces of dye in the sharp tip of the glass pipette. As soon as you can see any trace of dye in the tip of the pipette, it is ready.
4. Remove the pipette from the small vial of dye.
5. Using the syringe, fill the pipette with 3M KCL solution. Leave a small gap (3-5 mm) of air between the dye and the KCL.
6. Carefully attach your glass pipette into the gold microelectrode holder by sliding the metal wire into the pipette, pressing firmly, and screwing it tight. Make sure that the pipette is not going to fall out of the holder.
7. Attach the pipette holder to the head stage.



Checking the resistance of your electrode

Note: For this experiment, you need to check the resistance, but *you do not need to balance the bridge*.

Measure the resistance of your electrode by following the steps in the previous lab protocol. In the same way, use Ohm's law to calculate the electrode resistance. The resistance with the dye **will be much higher** than before.

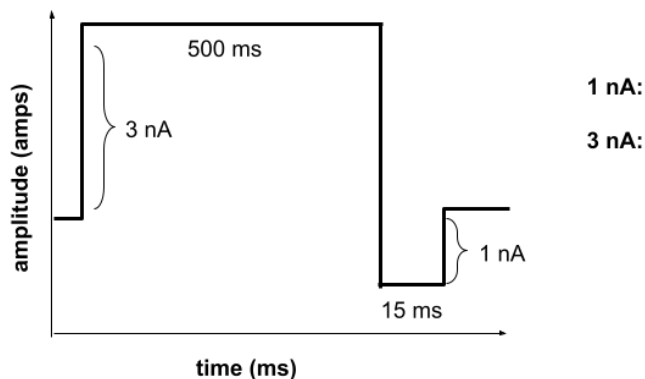
Electrode resistance = _____

If your resistance is higher than 1000 MΩ, try the Ringer button to clear it. If this doesn't work, consult the teaching staff about how to proceed.

Configuring LabChart & stimulus pulses

Note: While someone in your group is configuring the stimulus pulses, another group member should move onto Part II: Impaling a neuron.

Your configuration for LabChart will look similar to our leech recording session. Before we impale a cell, we'll also configure our pulses to look like this:

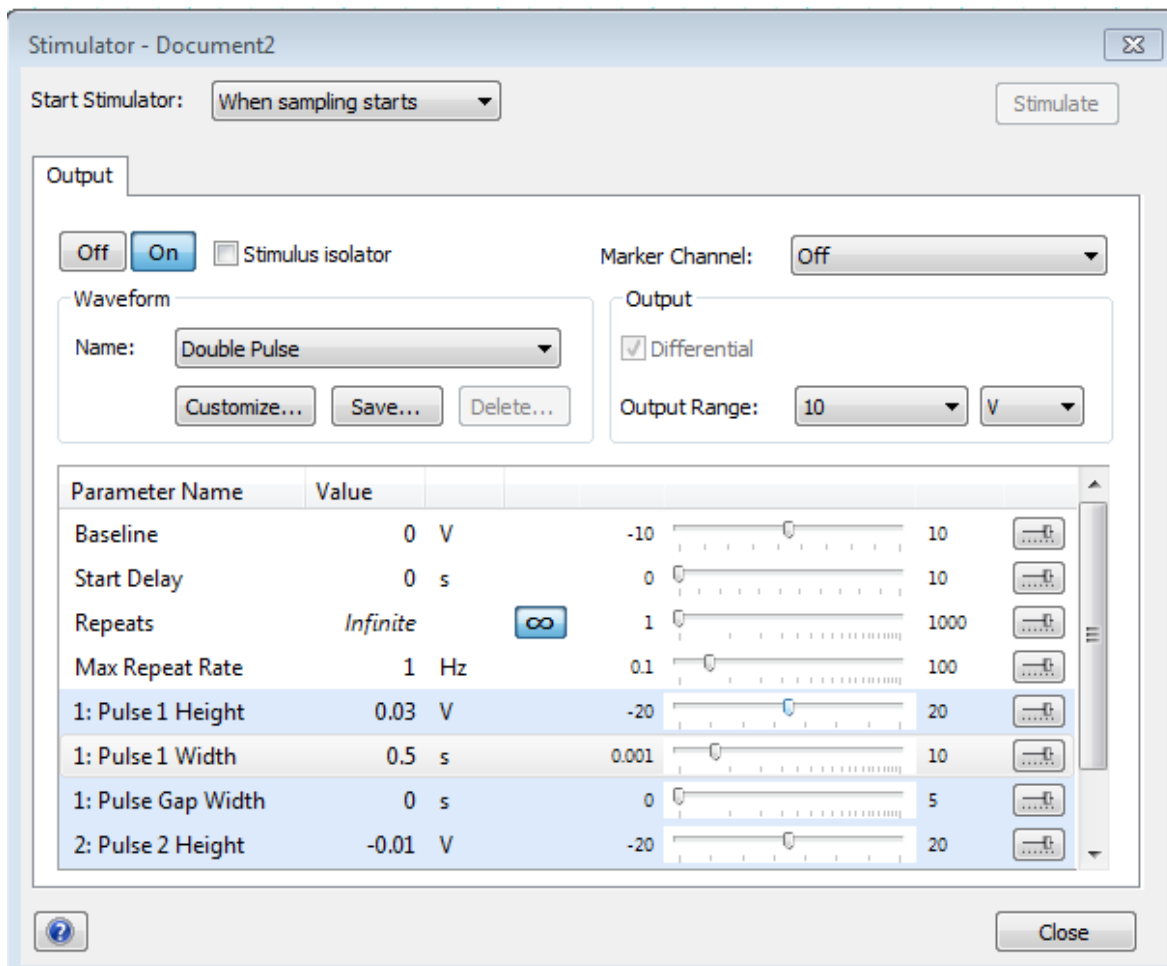


LabChart will tell the amplifier to send 1 nA for each 10 mV of stimulus given. Calculate the stimulator voltage we need for 1 and 3 nA and record above.

1. Open a new experiment in LabChart with the Leech Experiment settings.
2. Open the Stimulator window (**Setup > Stimulator**).
3. Change the wave type to "Double Pulse" setting in the Stimulator.
4. Set the parameters of your stimulus using the diagram below:

Hint: The waveform above is one positive pulse immediately followed by a negative pulse. You may need to change the possible *range* of these parameters () before you can modify them to your needs.

Note: Change Pulse Width of the second stimulus based on the diagram above. You can click on "Customize" to reassure yourself that you've configured the stimulus correctly.



5. Close the window when you're done.

Optional step: Directly monitor your stimulus waveform

To confirm that you've configured your stimulus correctly, you might want to actually monitor the stimulus waveform. But if you're running low on time, you should skip this step.

6. Unplug the **EXTERNAL** input to the amplifier and plug it into PowerLab **Input 2** (using the BNC-DIN8 adapter). This will allow us to directly monitor the stimulus.
7. Change to Scope view, and change the Duration to **1 s**.
8. Press **>Start** and observe the shape of your waveform on your Stimulator channel (Channel 2). You may want to remove the Stimulus Marker from this channel if it's impeding your ability to see the waveform.
9. Plug the External cord back in. It should be as in the Setup diagram, with PowerLab Output (+) going into the EXTERNAL input on the amplifier.
10. Return to Chart view.

II. Impaling a neuron & filling your cell (1 hour)

Impaling a neuron

1. Impale a neuron of your choice, just as you did in our leech ganglion recording experiment. You should try to target the cell you recorded from previously.
2. *Briefly* confirm its identity electrophysiologically. In other words, observe its firing rate and the shape of the action potential. If you're still not sure, inject current as we did in the recording experiment.

Filling your cell

Eject dye from the electrode by passing pulses of current into the cell. You should have the pulses configured already.

1. Make sure that the Current knob on your amplifier is on **CONT** (Continuous).
2. Turn the Stimulator ON and Press **Start**.
3. Carefully turn off your surgery lights to avoid bleaching the dye.
3. Monitor the resistance of your electrode by observing the voltage change during each current pulse.
 - a. If the electrode starts to clog, your resistance (and change in voltage) will go up dramatically.
 - b. If your electrode looks like it's clogging, slowly remove it from the cell. Test the resistance (see above if you need a refresher on how to do this). Press the Ringer button on the amplifier, and test again. Repeat as necessary. If this does not solve your problem, make a new electrode (if time allows).
4. **Every five minutes**, turn off the Stimulator. Observe the resting potential and action potentials of the cell to ensure that you're still in it. Turn on your lights and check to see if you can see any dye in the cell.

Note: Even if you can't see dye, that doesn't mean it's not there! We'll know for sure after looking under the filters with the microscope.
5. Once you are content (or there is one hour left in class), slowly remove the electrode from the cell and stop recording in LabChart.
6. Save your data, just in case.

III. Fixing & mounting your cells (30 minutes)

In order to view the filled cells, we need to rinse and mount the ganglion on a microscope slide.

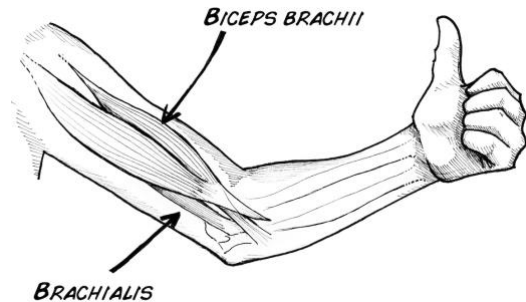
1. Pour the leech saline from the dish into the **buffer waste container**.
2. Get a glass microscope slide and a coverslip.
3. Under the microscope, **carefully remove the pins (4) from the leech ganglion dish with a pair of forceps**. Leave the pins in the dish. The leech ganglion should be resting there.
4. Using the dropper, put a tiny drop of glycerol solution on the slide.
5. Gently pick up your ganglion and place it in the glycerol. Ensure that the ganglion is spread out properly, ventral side up, and the roots are not overlapping the ganglion.
6. Add a coverslip. If you've never coverslipped before, ask for a demo.
7. Bring your slide to the fluorescent microscope to see what it looks like and take a photo!

Troubleshooting

Observation	Likely issue(s)	Possible solution
Signal has a lot of 60 Hz noise in it	Something needs to be grounded	Make sure that everything metal is grounded (microscope AND manipulator stands) and that the Faraday cage itself is grounded
Signal is oscillating wildly, but not at a particular frequency	Electrode	Change your electrode
Signal cannot be zeroed even with DC Offset turned all the way up	Amplifier power source is bad	Change amp power source
	Amplifier itself is bad	Change amplifier
Resistance is really low (<10 mOhms)	Electrode tip broke	Replace your electrode with a new one
Resistance is too high (>50 mOhms)	It is clogged	Remove the electrode from the cell (if applicable) and press the "Ringer" button to try to unclog it. If this doesn't help after several tries, get a new electrode.
There are no longer any spikes	Your electrode is no longer in the cell; your cell is very quiet; your cell died	Tap your electrode holder or gently move it closer to re-insert it into the cell
When I inject voltage (or check resistance), the reading says "Out of Range"	Your range is set too low	Increase your range (1 or 2 V makes sense for resistance testing)
	Your resistance is really high	Remove the electrode from the cell, and use the Ringer button. Test your resistance again.
Turning the manipulator knob doesn't move the electrode	Manipulator is at the end of its range	Turn the manipulator knob clockwise to tighten it, then reposition electrode using coarse movement to be closer to the ganglion
Ohm test doesn't show any change in signal	Amplifier bad	Change amplifier

Experiment #5: Electromyography

In this experiment, you will explore the electrical activity of skeletal muscle for both voluntary and evoked muscle actions and will learn how to record an electromyogram, or EMG. From the data, you will also be able to measure nerve conduction velocity.



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Background

Nerve and muscle disorders cause the muscles to react in abnormal ways. Measuring the electrical activity in muscles and nerves can help detect the presence, location and extent of diseases that damage muscle tissue (such as muscular dystrophy) or nerves (such as amyotrophic lateral sclerosis: Lou Gehrig's disease). In the case of nerve injury, the actual site of nerve damage can often be located.

Skeletal muscles do the majority of the work for locomotion and support of the animal skeleton. The skeletal muscles produce movement, maintain posture, and assist with body temperature maintenance, among other actions. Skeletal (striated) muscles attach to the skeleton by tendons (strong bundles of collagen fibers), aponeuroses (flat, sheet like tendons) and fascia (bands of connective tissue). Each muscle is made up of individual muscle fibers organized in fascicles (Figure 1).

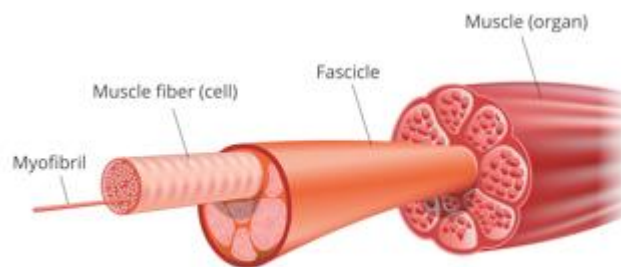


Figure 1. Skeletal muscle structure. Image courtesy of ADInstruments ©

A typical skeletal muscle contains thousands of muscle fibers. Each individual fiber is innervated by a branch of a motor axon. Under normal circumstances, a neuronal action potential activates all of the muscle fibers innervated by the motor neuron and its axonal branches. The motor neuron, together with all of the individual muscle fibers that it innervates, is termed a **motor unit** (Figure 2). This activation process involves:

1. the initiation of an action potential, either voluntarily or as a result of electrical stimulation of a peripheral nerve
2. conduction of the action potential along the nerve fiber
3. release of neurotransmitter at the neuromuscular junction
4. depolarization of the muscle membrane with resultant contraction of the muscle fibers.

During a contraction, there is synchronous activity in a number of fibers in the same muscle. The electrical signal recorded from a contracting muscle is called an electromyogram, or EMG. The EMG provides a depiction of the timing and pattern of muscle activity during complex movements. The raw surface EMG signal reflects the electrical activity of the muscle fibers active at that time.

Motor units fire asynchronously and it is sometimes possible, with exceedingly weak contractions, to detect the contributions of individual motor units to the EMG signal. As the strength of the muscular contraction increases, however, the density of action potentials increases and the raw signal at any time may represent the electrical activity of perhaps thousands of individual fibers.

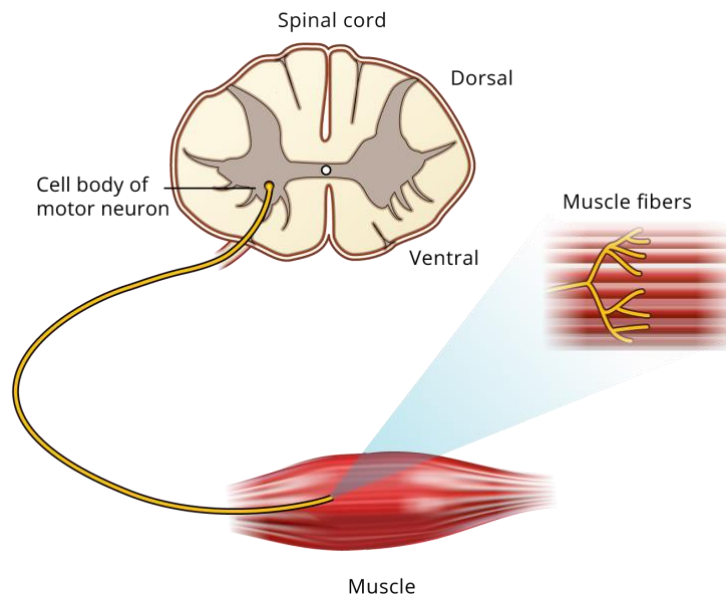


Figure 2. The components of a motor unit. Image courtesy of ADInstruments ©

The raw EMG signal during voluntary contractions may be processed in various ways to indicate the intensity of EMG activity. In the method used here, known as the **root mean square (RMS)**, the negative-going portions of the EMG are inverted by squaring the whole signal, and then the whole signal is averaged (by dividing the signal over a certain time window) and the square root calculated. This process smoothes out individual spikes and makes the time course of changing activity much clearer.

In this experiment you will examine **coactivation**, a phenomenon in which contraction of a muscle leads to more minor activity in the antagonist muscle. The physiological significance of this is not entirely clear, but it has been suggested that it helps to stabilize the joint.

You will also record EMG signals produced by electrical stimulation of a motor nerve supplying a muscle. The abductor pollicis brevis muscle is one of the intrinsic muscles of the hand, more specifically an intrinsic muscle of the thumb, on the palmar surface of the hand. The motor

nerve to this muscle, which is the median nerve, is simple to stimulate at the wrist and elbow. When brief electrical pulses are administered through the skin to the nerve, the time it takes for the muscle to contract in response to the electrical pulse is recorded. The speed of the response is dependent on the conduction velocity. In general, the range of normal conduction velocities will be ~50 to 60 meters per second, however, the normal conduction velocity may vary from one individual to another and from one nerve to another. In a clinical setting, EMG and nerve conduction studies are usually conducted together.

Written by staff of ADInstruments, edited by A. Juavinett, M. Marino, and I. Maita
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Lab Protocol

Supplies

- LabChart & PowerLab
- BioAmp Cable
- Shielded Lead Wires (5 Snap-on)
- Disposable electrodes (6)
- Stimulating Bar Electrode
- Electrode Cream or Paste
- Nuprep Abrasive Gel
- Alcohol Swabs (2)
- Cotton Swabs (2)
- Marker
- Resistance bands (4)
- Scissors

Equipment Setup and Electrode Attachment

1. Make sure the PowerLab is turned **off** and the USB cable is connected to the computer.
2. Connect the 5 Lead Shielded Bio Amp Cable to the Bio Amp Connector on the front panel of the PowerLab (Figure 3). **The hardware needs to be connected before you open the settings file.**

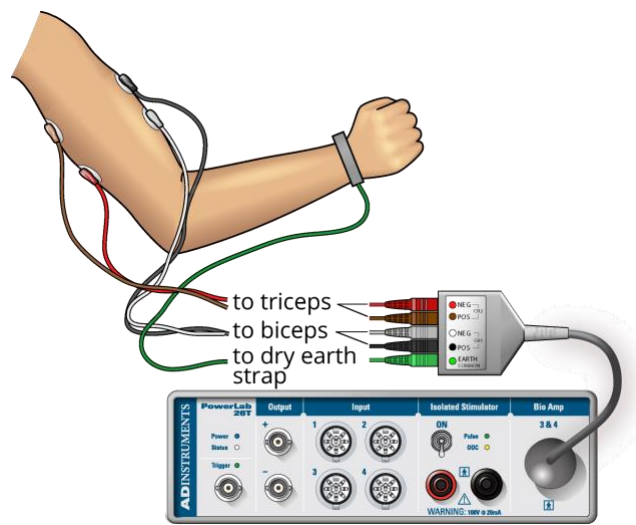


Figure 3. Equipment Setup for PowerLab 26T. Image courtesy of ADInstruments ©

3. Remove any jewelry from the volunteer's hand and arm. Use the ballpoint pen to mark two small crosses 2-3 cm apart on the skin above the biceps muscle and triceps muscle. Use Figures 3 and 4 as a guide. Abrade the skin with abrasive gel or alcohol pad. This is important as abrasion helps reduce the skin's resistance. After abrasion, clean the area with an alcohol pad to remove the dead skin cells.
4. While the skin is drying, attach the Shielded Lead Wires to the Bio Amp Cable. Attach the Disposable Electrodes to the end of the Channel 1, 2, 3, and 4 wires. Follow the color scheme on the Bio Amp Cable.
5. Attach the four electrodes and velcro grounding strap (Earth lead) to the volunteer. Channel 1 will lead to the biceps (place the negative [white] wire 2-3 cm above the positive [black] wire), Channel 2 will lead to the triceps (place the negative [red] wire 2-3 cm above the positive [brown] wire), and the Earth (green) will be connected to the velcro grounding strap via the alligator clip. Refer to Figures 3 and 4 for proper placement.

Note: If you have trouble getting a proper ground connection with the wrist strap, you can also directly attach a Disposable Electrode and place the ground at the wrist.

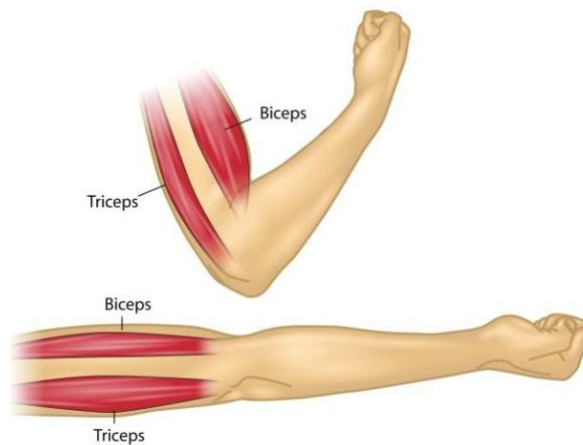


Figure 4. Skeletal muscle structure. Image courtesy of ADInstruments ©

6. Check that all four electrodes and the wrist ground strap are properly connected to the volunteer and the Bio Amp Cable before proceeding. Turn on the PowerLab.

Exercise 1: Voluntary Change in Contractile Force

In this exercise, you will examine changes in voluntary muscle contraction and how contractile force changes with increasing demand.

1. Launch LabChart and open the settings file “Voluntary Change Settings” from the EMG folder in <http://bit.ly/labchart>.

Note: Channels 1 and 2 are the RMS activity of the biceps and triceps muscles. RMS activity is commonly used in the assessment of muscle function because it is easier to quantify. Use these two channels when completing your analysis.

2. Have the volunteer sit in a relaxed position with his/her elbow bent 90° and palm facing upward. Make sure the volunteer’s elbow is not on the table. The volunteer’s other hand should grasp the wrist of the recorded arm. Make sure the volunteer is facing away from the monitor.
3. Select **Bio Amp** from the Channel 3 Channel Function pop-up menu. Have the volunteer make a strong contraction of the biceps muscle. This is done by bending the recorded arm further while resisting this movement with the other arm. Observe the signal and adjust the range in the dialog so that the maximal electrical response occupies about one half to two-thirds of the full scale. Click OK.
4. Repeat step 3 for the triceps signal in Channel 4. A strong contraction of the triceps muscle is made by trying to straighten the recorded arm while resisting this movement with the other arm.
5. Set up the volunteer with the resistance band. They can hold the middle of the band down with their foot, while gripping the end of the band.

6. **Start** recording. Add a **comment** with the volunteer's name. Have the volunteer make a strong contraction of the biceps and then the triceps. Add a **comment** at the start of each contraction. **Stop** recording.
7. Have the volunteer return to his/her original relaxed position. **Start** recording. The blue line in Chart View will help you indicate the change in procedure.
8. Prepare a comment with "first band." After a few seconds, add the **comment** and have the volunteer hold the band up so that there is some resistance. Hold for three seconds and relax. Repeat this process with each band of increasing resistance. Add a **comment** each time you use a different band. Note: You may need to have your volunteer use multiple bands simultaneously to see a difference in the EMG signal.
9. **Save your data.** Leave the disposable electrodes in place for Exercise 2.

Exercise 1 analysis

1. Examine the data in the Chart View. **Autoscale**, if necessary. Note the changes in activity in the "Biceps" channel. Note also that placing weights on the hand gives rise to little or no activity in the triceps muscle.
2. Select a small part of the "Biceps" activity and examine it in **Zoom View**. The raw EMG signal is composed of many up-and-down spikes (Figure 5).



Figure 5 Raw EMG Signal

3. Note the relationship between the "Biceps" channel and the "RMS Biceps" channel. The height of the RMS trace reflects the overall activity of the raw EMG signal and gives a simpler view of the muscle's electrical activity. Note the changes in the RMS trace as bands were added and removed.
4. Select the peak data point "RMS Biceps" when the first band was used. Enter these values in Table 1 of the Data Notebook (be sure to take your peak *relative* to your baseline). The height of the trace correlates with the force produced by the muscle.
5. Do the same for each band and record in Table 1.

Exercise 2: Alternating Activity and Coactivation

In this exercise, you will examine the activity of antagonist muscles and the phenomenon of coactivation.

1. Open the settings file “Coactivation Settings” from <http://bit.ly/labchart>.
2. Have the volunteer sit in a relaxed position with his/her elbow bent 90° and palm facing upward. Make sure the elbow is not on the table. The volunteer’s other hand should grasp the wrist of the recorded arm. Make sure the volunteer is facing away from the monitor.
3. Have the volunteer practice activating the biceps and triceps immediately after one another. The volunteer should practice this until it feels like both muscles are being equally activated in turn. Pause shortly after each activation; this makes the data clearer.
4. **Start** recording. Add a **comment** with the volunteer’s name, and record baseline EMG for 30 seconds.
5. Add a **comment** with “activation,” and have the volunteer use the alternating pattern of activation for 30 seconds. **Save your data when you are finished recording.**

Exercise 2 analysis

1. Examine the data in the Chart View for both the biceps and triceps, and **Autoscale**, if necessary. Note the large-scale alternation of activity.
2. Note when the biceps muscle is activated forcefully; there is a minor increase in the activity of the triceps. Correspondingly, there is a minor increase of activity in the biceps trace when the triceps are activated. This phenomenon is coactivation. Its physiological meaning is not well understood, but it is thought to stabilize the elbow joint.
3. Select data points from the RMS EMG peaks for both muscles during contraction of the biceps and contraction of the triceps. Enter these values in **Table 2** of the Data Notebook.

Exercise 3: Stimulating the median nerve

In these exercises, you will electrically stimulate a volunteer’s median nerve at the wrist and elbow to measure nerve conduction velocity.

Equipment Setup

 **Exercises 3 and 4 involves application of electrical shocks to muscle through electrodes placed on the skin. Students who have cardiac pacemakers or who suffer from neurological or cardiac disorders should not volunteer for these exercises. If the volunteer feels major discomfort, discontinue the exercise and consult your instructor.**

1. Leave the Shielded Bio Amp Cable attached to the PowerLab. Remove the Channel 2 Lead Wires from the Cable and detach the Channel 1 Lead Wires from the Disposable Electrodes, leaving the wires connected to the Bio Amp Cable.
2. Use the ballpoint pen to mark two small crosses 2-3 cm apart on the skin above the abductor pollicis brevis muscle. Use Figures 6 and 7 as a guide. Abrade the skin with abrasive gel or pad. After abrasion, clean the area with an alcohol swab to remove the dead skin cells. Trim the Disposable Electrodes (If the volunteer has small hands) to match those in Figure 7 and stick them to the skin, ensuring the negative (white) electrode is closer to the

wrist. Secure the ground wrist strap around the volunteers wrist. Alternately you may use another disposable electrode attached to the volunteer at the wrist or back of the hand.

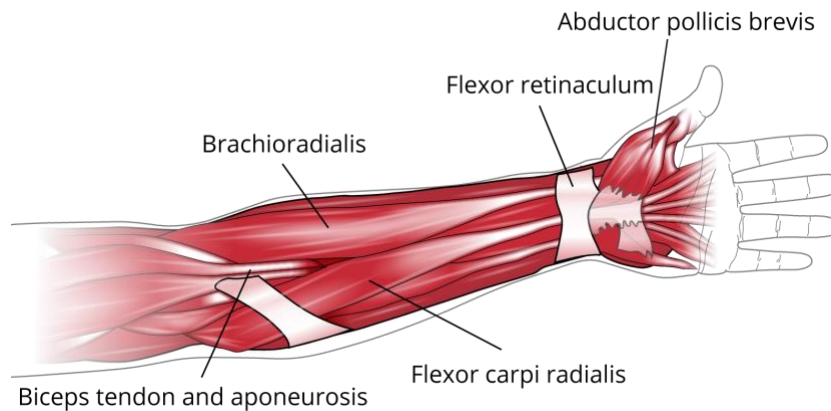


Figure 6. Muscles of the Forearm and Hand. Image courtesy of ADInstruments ©

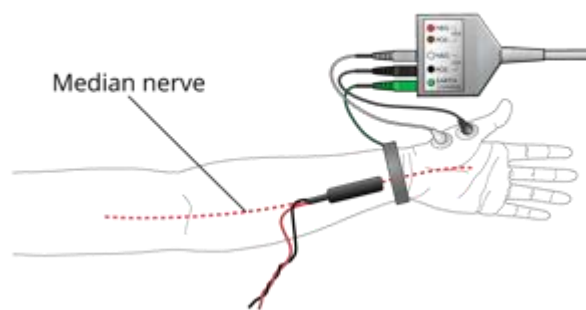


Figure 7. Evoked EMG Setup at the Wrist. Image courtesy of ADInstruments ©

3. Connect the Stimulating Electrode to the Isolated Stimulator output of the PowerLab. Make sure the red (positive) connector is in the red output and the black (negative) connector is in the black output.
4. Place a small amount of Electrode Cream or Electrode paste on the two silver pads of the Stimulating Bar Electrode and place it over the volunteer's median nerve at the wrist (Figure 7). The Stimulating Bar Electrode should lie along the long axis of the arm, with the leads pointing toward the hand; the wires pointing toward the inner elbow – a red (positive) dot on the back of the bar should be placed away from the hand.
5. Have the volunteer hold the electrode in place with the opposite hand.

Tips on finding the Median nerve

Due to anatomical differences, the exact spot for the most efficient stimulation of the Median nerve will vary between volunteers. In the first part of this exercise, you will find the “sweet spot” for the volunteer before recording the EMG activity.

Muscular responses will be observed by watching the hand of the volunteer. Some motor effects that may be observed include:

- Movement of the thumb towards the fingers (due to stimulation of adductor pollicis and flexor muscles of the thumb)
 - Bending of the wrist (due to the flexor carpi radialis and flexor carpi ulnaris muscles)
 - Bending of the last segments of the fingers (due to the long finger flexor muscles)
 - Movement of all fingers, combined with the pulling of the thumb towards the index finger (due to the intrinsic muscles of the hand innervated by the ulnar nerve)
 - Lifting of the thumb (due to stimulation of abductor pollicis at the base of the thumb innervated by the median nerve)
-

Finding the Median nerve

1. Launch LabChart and open "Nerve Effect Settings."
 - **Note:** No data will be recorded in this file. Its purpose is to control the Isolated Stimulator.
2. Open the Stimulator settings. Change the **Pulse Width** to 50 μ s (if it is not set to that already). The current should be set to 6 mA.
3. Have the volunteer sit in a relaxed position. Make sure the volunteer is still holding the Stimulating Bar Electrode in place over the median nerve.
4. Turn on the Isolated Stimulator by flipping the switch on the PowerLab.
5. Open the **Stimulator Panel**. The Isolated Stimulator only becomes active when the On button in the Stimulator Panel is selected.
6. Select the **On** button in the Stimulator Panel. Observe the volunteer's hand. Look for the twitch contractions affecting the thumb and fingers. Have the volunteer describe the effects he/she is experiencing.
7. Make small adjustments by moving the bar around the wrist and examine the effect. Locate the position giving the largest twitches. You can turn the isolated stimulator off at any time by selecting the **off** button in the Stimulator Panel. When the volunteer feels a slight tingle in the thumb, they are in the right area for the next part of the exercise.

Note: If nothing happens, you may need to increase the stimulus amplitude (current) to observe a twitch. Increase the amplitude in the Stimulator Panel Start at 6 mA and increase to 12 mA in 1 mA increments if necessary.

Alternately: No response may be due to a bad connection between the bar electrodes and the volunteer. If the yellow OOC (Out of Compliance) LED is blinking with each stimulus, this is likely the case. Turn off the stimulator and clean the stimulating bar and the volunteer's arm with alcohol wipes. Reapply conductive cream to the electrodes and continue with the exercise.

8. Explore the results of stimulating at other places in the forearm. Have the volunteer describe the effects he/she is experiencing. Each time you move the electrode to another location wipe away the residual Electrode Cream from the skin with an alcohol wipe to prevent short-circuiting. Remember the two pads need to be aligned along the arm's length.
 - **Note:** Some volunteers may fail to show any thumb twitch response stimulating the median nerve. In some people, the abductor pollicis brevis muscle is innervated by the ulnar nerve instead of the median nerve. This is an example of anatomical variation. Try moving the Stimulating Bar Electrode to the ulnar nerve (see Figure 8).

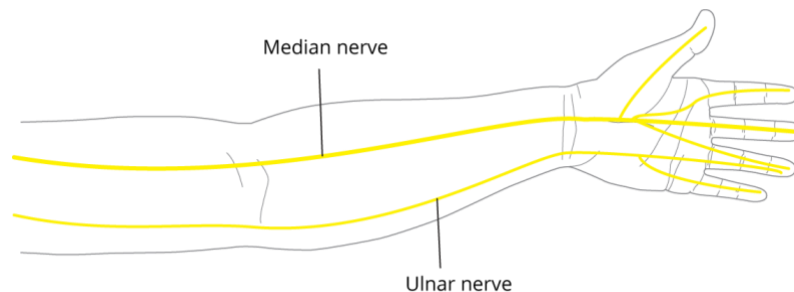


Figure 8. Position of the Median and Ulnar Nerves. Image courtesy of ADInstruments ©

8. Stimulate the ulnar nerve at the level of the elbow. The nerve passes behind a bony prominence called the medial epicondyle on the humerus. At this location, the nerve is exposed to minor mechanical injury and is known to children as the “funny bone.” Stimulation at this site gives large and obvious motor effects.

Note: Stimulation in most places should give minimal discomfort. In some locations on some volunteers, there is substantial sensory effect. There may be painful sensation in the forearm or hand away from the site of stimulation toward the fingers. At these locations, a cutaneous sensory nerve is being stimulated.
9. When you have observed a good thumb twitch reaction, leave the Stimulating Bar there and select the **Off** button in the Stimulator Panel to stop the stimulator. You do not need to save your data as nothing was recorded.

Stimulating the Median nerve

1. Open the settings file “Evoked EMG Settings” from the EMG folder in <http://bit.ly/labchart>.
2. Have the volunteer return the stimulating bar electrode to the spot on the wrist that gave a good thumb twitch response. Have the volunteer apply pressure to the Bar Electrode to make sure the nerve is stimulated and the electrode does not move. The volunteer must keep holding the electrode throughout this exercise.
3. Make sure the volunteer is facing away from the monitor, and **Start**. LabChart will record a single block of 0.05 seconds. The stimulator panel will be preset to Current to 6.0 mA. Add a **comment** with the volunteer’s name and “wrist” to denote the area being stimulated.
4. Increase the Current to 8.0 mA. Click **Start** to apply the stimulus. If a response is not recorded, have the volunteer move the electrode to a different spot on the nerve (Figure 9)

and apply pressure. If the volunteer does not feel the stimulus, increase the stimulus amplitude by 2 mA each time (up to 20 mA). The lights next to the switch on the PowerLab will light up when a stimulus is made.

5. Once you record an evoked EMG response, increase the amplitude by 2 mA and click **Start** again. Continue increasing the amplitude by 2 mA until you reach 20 mA or the volunteer requests you to stop. Add a **comment** with the amplitude each time you begin recording.
 - **Note:** *The evoked response should increase with increasing stimuli until a maximal response is reached, after which increasing the stimulus does not further increase the response amplitude.*
6. Move the Stimulating Bar Electrode over the median nerve at the elbow (Figure 9). Mark the spot in between the pressure imprints on the skin. Everything else will remain the same.

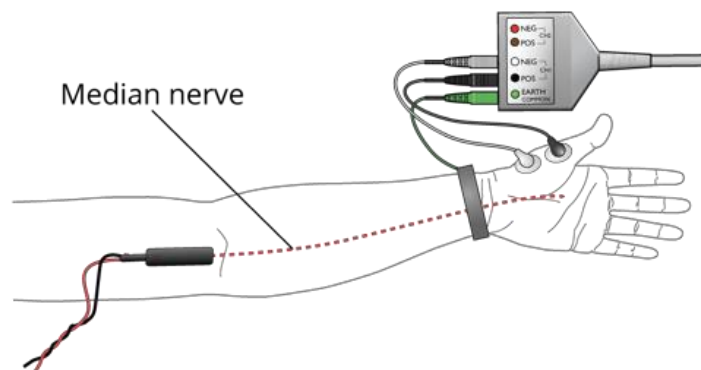


Figure 9. Evoked EMG Setup at the Elbow. Image courtesy of ADInstruments ©

7. Turn on the Isolated Stimulator and return the amplitude to 6 mA. Repeat steps 3-4.
8. Once you have found a response, increase the amplitude to maximum amplitude used at the wrist. This should be 18 to 20 mA. Add a **comment** with the amplitude used when you start recording. Record three responses at this amplitude.
9. Turn off the Isolated Stimulator. Mark the spot in between the pressure imprints on the skin and disconnect the equipment from the volunteer. **Save your data.**

Exercise 3 analysis

1. Examine the data in the Scope View for the evoked response of the wrist and elbow at maximum amplitude.
2. Use **Zoom View** to measure the latency of a single waveform for each type of response (wrist and elbow). Latency is the time elapsed from the start of the stimulus (the start of each record) to the start of the evoked response. Record these values in Table 3 of the Data Notebook under "Latency." In addition, calculate the difference between the two latencies and enter the value in Table 3.

Note: *You may see a very early deflection in response. This is a stimulus artifact and must be ignored when calculating the latency (Figure 10).*

3. Measure and record the distance between the marks at the wrist and elbow and record it in **Table 3** of the Data Notebook. This is the distance between stimulation sites.
4. Using your recorded distances & latencies, calculate the nerve conduction velocity of the volunteer. Enter the velocity under **Table 3** of the Data Notebook.

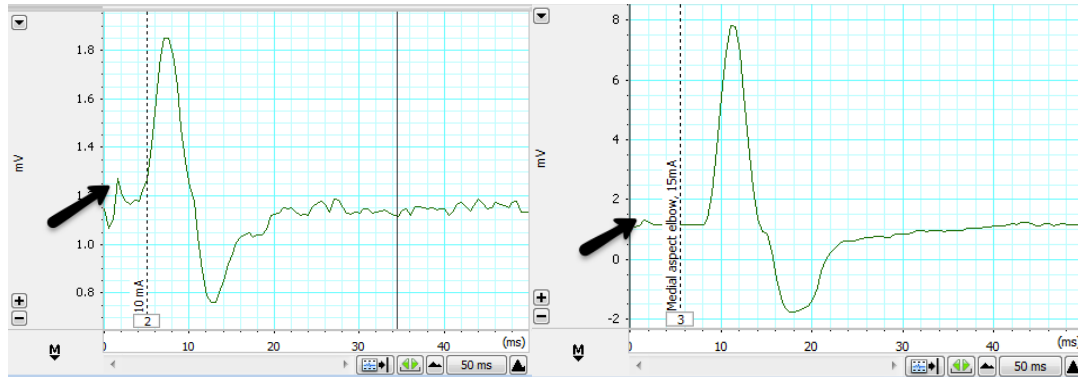


Figure 10. Evoked EMG from wrist (right) and elbow (left). Arrow denotes a stimulus artifact.
Image courtesy of ADInstruments ©

5. Record the nerve conduction velocity from at least *five* other groups in **Table 4** of the Data Notebook. You'll need this to submit the work for this lab.

Data Notebook

Table 1. Force Produced by Different Bands

	Band Resistance (Kg)	RMS Biceps Amplitude (mV_{rms})
First Band		
Second Band		
Third Band		
Fourth Band		

Table 2. Coactivation of Biceps & Triceps

	RMS Biceps Peak Amplitude (mV_{rms})	RMS Triceps Peak Amplitude (mV_{rms})
Biceps Contracting		
Triceps Contracting		

Table 3. Evoked EMG

Stimulus Location	Latency (ms)
Wrist	
Elbow	
Difference (ms)	
Distance Between Stimulation Sites (mm)	

Table 4. Nerve Conduction Velocity

Group	Nerve Conduction Velocity (m/s)
(your group)	
...	

Questions

Unlike the discrete waveform from an electrocardiogram, the electromyogram waveform is irregular. Why do you suppose this is?

Describe the root mean square analysis and why it is useful for EMG data.

What happened to the triceps EMG trace when the biceps was activated? Does the data support the phenomenon of coactivation?

Experiment #6: Visualizing the nervous system

Background & Goals

How do we visualize different parts of the nervous system?

Brain parts are tiny. We need a microscope to be able to target specific cells to record from them or to see them after we fill them with a dye or a tracer. We also need other types of chemical stains or antibodies if we want to visualize specific structures in the tissue.

If you continue doing research beyond our lab class, you'll almost definitely find yourself using a microscope for one reason or another. The skills you build in this lab will help you be much more efficient while peering through the oculars, and will also help you interpret what you see.

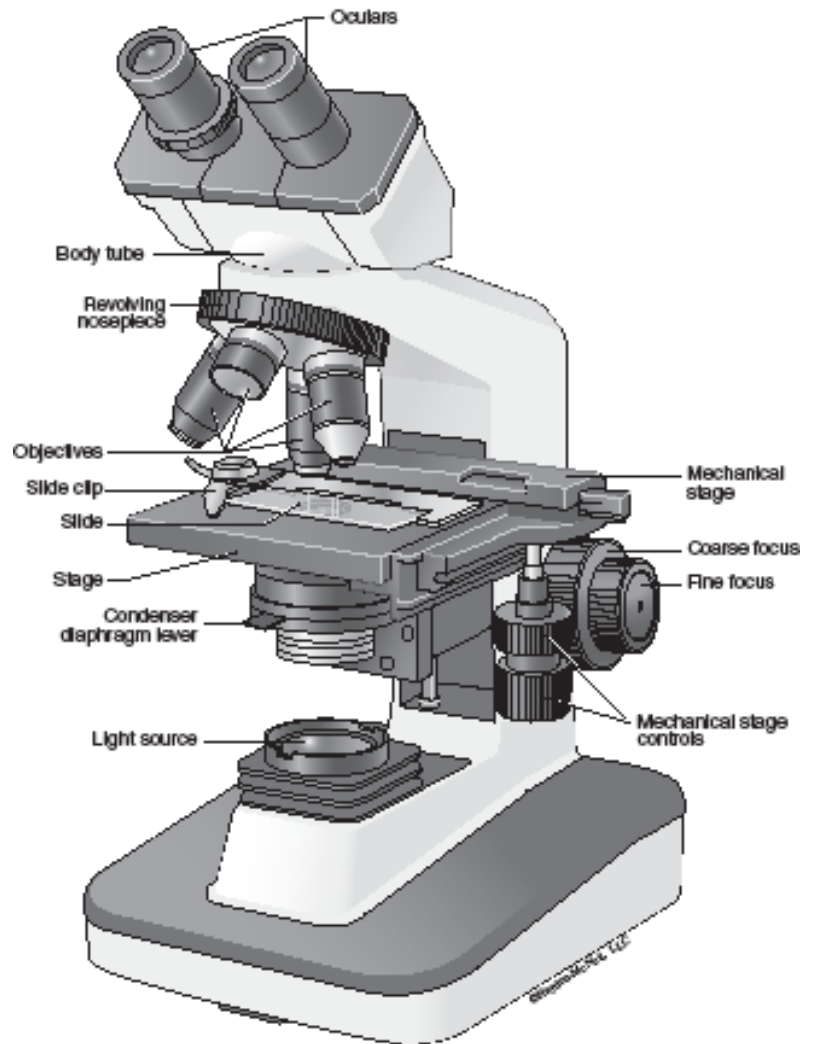
The two kinds of microscopes that we'll use in our course are the **compound microscope** and the **stereoscopic dissecting microscope**. You will learn the use and care of these microscopes in this laboratory exercise. Any time you are using a microscope, it is important to remember that a microscope is a delicate instrument and should be handled with care.

Compound Microscope Basics

Learning and practicing proper focusing technique is important to prevent you from carelessly damaging your microscope or the microscope slides. Refer to the figure below for the following discussion of the parts of the compound microscope. The goal is to focus as closely as possible on an object. The compound microscope consists of a set of lenses; the higher the magnification, the closer the object in the slide will appear. There are two lenses that magnify an object: the ocular and the objective. At the top of the microscope is the **ocular** (eyepiece); a microscope can have one ocular (monocular) or two oculars (binocular). The power of the ocular lenses in this binocular microscope is 10×; however, these lenses are replaceable by lenses of different magnifications. Next are the **objectives**, which are separated at the proper working distance from the oculars by the **body tube**. The microscope in lab is equipped with four objective lenses. The objectives differ in length depending on their magnification power; the smaller the length of the objective, the lower is its power. The revolving nosepiece of your microscope includes a **4× scanning objective**, a **10× low-power objective**, and a **40× high-power objective**. Some microscopes may have an **oil immersion lens**. Oil prevents the loss of light rays due to defraction. Because the brightness of the field decreases as the magnification power increases, oil is needed to collect sufficient light under the highest-power objective.

The **total magnification** of a microscope lens system is the product of the powers of magnification of the ocular lens and the objective lens. The oculars on most microscopes are 10×. So, if you are looking at a specimen under the 40× high-power objective lens, the total magnification of that specimen is 400× ($10 \times 40 = 400$). This means that the specimen appears through the microscope to be 400 times larger than it actually is.

The object to be viewed is usually affixed to a slide and protected by a coverslip. The slide is then placed under the objective on the **stage** and held there firmly by a spring-loaded **stage clip**. On your microscope, the slide can be manipulated under the objective by means of a movable stage. Two **stage control knobs** located under the stage regulate this movement. You are able to bring your specimen into focus by moving the stage up or down. Two knobs located on the arm of the microscope control this movement: the **coarse-adjustment knob**, which moves the stage large distances, and the **fine-adjustment knob**, which moves the stage very slight distances.



When first looking at a new slide, always start with a short objective lens, either the 4× scanning lens or the 10× low-power objective. Place the slide in the clip on the stage. While watching from the side of the microscope, use the course-adjustment knob to carefully bring the stage up to its highest position (it will still be a safe distance from the specimen). Now look through the ocular lenses and slowly lower the stage until the specimen comes into focus.

This microscope is designed to be **parfocal**, that is, once you have focused on a specimen, the specimen should remain relatively in focus when you move to a different objective lens. Once you have a specimen in focus, you need to make adjustments of focus only with the fine-adjustment knob. Never use the coarse-adjustment knob when the higher-power objectives are in position.

The distance between the objective lens and the slide is called the **working distance**. It is quite large with the lowest-power objective lens but decreases with each increase in objective power. The working distance for your highest-power lenses is very small. The 40× and 100× objective lenses are spring-loaded to prevent damage when rotating them into place.

The **light source** is located in the **base** of your microscope. You can turn it on and adjust the intensity of light with a rheostatic knob also on the base. Look immediately beneath the stage. A **condenser** gathers light from the light source and directs it through the opening in the stage.

The **iris diaphragm**, which can be opened various degrees, regulates the amount of light passing through the specimen. Proper control of the amount of light passing through a specimen can greatly facilitate your studies. When looking at slides today, experiment with the iris diaphragm and the light source and note its effect on the detail you are able to see. Any change in objective or specimen may require lighting adjustment.

Using a compound microscope

Before each use, the ocular and objective lenses of the microscope should be cleaned with lens paper. Only lens paper should be used to clean the lenses—never tissues, paper towels, or any other material that might scratch the lenses. If the lenses are very dirty, a liquid lens cleaner may be used with the lens paper.

1. Rotate the nosepiece so that the lowest-power objective (4×) is in position.
2. Prepare the slide and place it on the stage with the coverslip facing upward.
3. Make sure to secure the slide to the stage with the spring-loaded stage clip.
4. Look through the ocular lens. The first goal is to get the slide in focus. To do this, raise the coarse adjustment knob while looking through the ocular until you see the slide. The image that you see is called the **field of view**. The lens system of the microscope causes the image in the field to appear to be upside down and backwards.
5. Slowly rotate from the 4× objective to the next-highest objective (10×). You may not see your specimen and will need to refocus the lens. To do this, use the fine-adjustment knob. The field also appears darker; you may need to increase the amount of light passing through the lens system. Thus, as the power increases, the brightness and area of the field on the slide decreases.
6. Now view the slide under the 40× objective, and note the change in the area and brightness of the field. Again use the fine-adjustment knob to bring it into focus and be aware that the slide is close to the lens. The slide should not touch the lens.

Before changing slides, always rotate the lowest-power objective into position. Always view the slide under the lowest-power objective first and increase the power progressively. The thickness of the area in focus is termed the **depth of focus**.

Today you will:

- Learn the basic components of a microscope and how they work
- View brain tissues and identify the type of staining and brain regions
- Participate in citizen science by annotating EyeWire images

Lab Protocol

You may work on the following activities individually *or* as a group. We have one compound microscope for each setup.

Supplies (for all three activities)

- Compound microscope (with 4x, 10x and 40x objectives)
- Microscope slides
- Small ruler
- Colored pencils

I. Viewing nervous system slides

First, identify the different parts of your compound microscope (see Background).

Determining the diameter of your viewing field

In order to know how big our brain sections and neurons are, we need to first determine the diameter of our viewing field.

1. Place the ruler underneath the microscope at 4X and look through the oculars.
2. With your 4X objective, use your ruler to estimate the diameter of your viewing field in micrometers and enter it in Table 1.
3. It'll be tough to do the same at high magnification, so instead we can use this formula:

$$\text{high power field diameter} = \text{low power field diameter} \times \frac{\text{low power magnification}}{\text{high power magnification}}$$

4. Determine the diameter of the viewing field for 40x and 100x, and record in Table 2.

Investigating your slides

We've collected various types of brain tissue thanks to generous donations from Drs. David Linden (Johns Hopkins) & Edward Callaway (Salk). These are rare tissues that you'll have a chance to look at, so **please demonstrate the utmost care and respect in working with these slides.**

1. Get **one** of the following slides from the front (you can complete these in any order):
 - A **spinal cord** section from the brown box (any species)
 - One slide from the **myelin** stained box (mouse sections)
 - One slide with a filled **biocytin** cell (in mouse cortex)
2. Put the slide under the microscope. With the 4X objective and your coarse focus knob, bring one of the sections into focus.

Note: Do *not* use the coarse focus knob at magnifications higher than 4X. You could damage the objective and the slide.

3. Using clues from the slide (including any descriptions on the slides themselves), fill out the information for your slides on your worksheet.

For the **myelin** mouse slide, you may need to use the Allen Brain Atlas as a reference to figure out where your sections & cells are.

For your **biocytin** filled cell, you should first look for a very dark mark on one of your sections at 4X. Once you've found this, change to the 10X objective. At this magnification, you should be able to tell whether it is a cell body or not. Change to the 40X objective to draw an image of the cell.. You should label the cell body, dendritic arbor, and axon (if visible).

Note: Most of our slides have filled neurons in them, but not all. If you have trouble finding a neuron on your slide, grab another one. If you're still having trouble, ask the teaching staff.

II. Looking through electron microscopy data on EyeWire

1. Go to <http://eyewire.org>. You'll need to use your email or Facebook to create an account to get started.
2. After signing in, you'll be brought to a screen with a Starburst cell. You can change the cell in your demo by clicking on **change cell**.
Note: Cells take a few seconds to load. They'll just look like a bunch of cubes while loading.
3. After picking a cell, click **Start Playing**.
Note: There's some very nice relaxing music and useful sound effects, if you'd like to plug in a pair of headphones.
4. **Work through all five tutorials and the five sample cubes.**
5. When you're done, post your points on our EyeWire scoresheet.
Hint: You can see your points for the day by clicking on your username in the top right corner.



Experiment #7: *Drosophila* Behavior & Optogenetics

Background & Goals

Drosophila as a model in neuroscience

With just over 100,000 neurons but incredible genetic tractability, the fruit fly is an important model in neuroscience. About 65% of human disease-associated genes have a homolog in *Drosophila melanogaster* (Ugur et al., 2016). Like many model organisms, they're easy to reproduce, cheap, and easy to take care of in a lab. But on top of that, they've been studied for more than a century. *Drosophila* have been used to study everything from learning, memory, vision, sleep, addiction, courtship, and aggression.

In these labs, we'll aim to characterize ethological, olfactory, and locomotor behavior in transgenic fruit flies, solving a few mysteries along the way.

References

Ugur et al. (2016) *Drosophila* tools and assays for the study of human diseases. *Dis Model Mech* 9(3):235-44.

In these labs you will:

- Operationalize and quantify the behavior of a fruit fly
- Use a computer vision program to track locomotor behaviors
- Identify different types of transgenic flies used for neuroscience research

Day 1: The Case of the Mislabeled Fly Vials

Something terrible has happened in the lab. The flies you once knew, that were once perfectly organized and labeled, are now completely jumbled. You're pretty sure the flies are transgenic, of some sort or another. It's up to you to figure out what type of flies these are, based on three behavioral assays: an **olfactory y-maze**, an **ethological survey**, and a **locomotion test**.

Everyone will receive one batch of wildtype flies, and one batch of an experimental group. Ultimately, **you will present your findings to the class** (see Canvas for details) so that we can determine the phenotype of our mutant flies.



Supplies

- Two vials of flies (~10-15 WT, ~10-15 Mutant)
- Fly Y-maze chambers (2)
- Filter papers (2)
- Petri Dishes with sylgard bottom (6)
- Brush
- Magnetic stand & camera

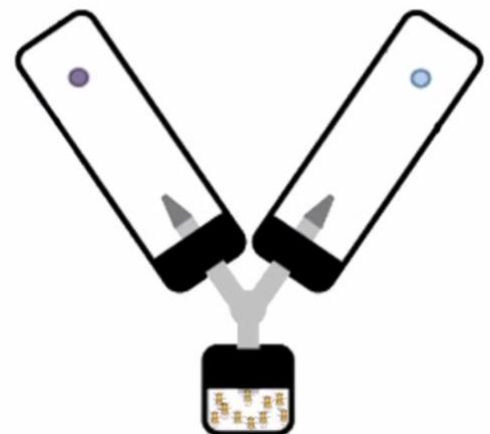
Solutions

- Solvent (H₂O)
- Apple Cider Vinegar (10% in H₂O)

Test I. Olfactory Y-Maze

For the wildtype and mutant flies, you will set up **two** different Y-mazes. For **each** of your Y-mazes, do the following:

1. Assemble your Y-mazes from the components.
2. Soak one filter paper in the solvent and one paper in the apple cider vinegar.
3. Place filter paper soaked your solvent in one end of the Y-maze, and the other filter paper at the other end. Label each end of the Y-maze with tape.
4. Put the flies on ice for **2-3 minutes** until they are immobile.



Note: Keep a close eye on your flies and do not leave them on ice too long! As soon as they're not moving, you can move them.

5. Using a brush, **very** gently isolate 6-12 flies (depending on how many flies you have in total) into your loading vial. The flies are very fragile.
6. Cover the Y-maze so that the flies don't get any visual cues (or disturbances).
7. Leave the Y-maze in the dark, undisrupted. We'll check on them in our next lab meeting, when you can calculate the olfactory index (if most flies left the loading vial):

Calculating the Olfactory Index

The **olfactory index** is equal to $(O-C)/T$, where O = number of flies on experimental solution side, C = number of flies on control (solvent) side, and T = total number of flies that moved from holding vial.

Test II. Ethological approach to behavior

It's possible your flies do something different that's *not* captured by a standard behavioral assay. The best way to figure this out is to watch the flies, choose several behaviors to quantify, and see if those behaviors differ between your experimental and wildtype flies.

Choosing behaviors to quantify

Your next goal for today will be to quantify **three** different behaviors that your flies perform. It is up to you to define and operationalize these behaviors.

1. Freeze both groups of flies for 2-3 minutes until they stop moving.
2. Isolate three flies of each type in three separate petri dishes.
Note: Make sure they're completely awake and moving before moving on to the next step. This could take about 10 minutes.
3. Take a few minutes to observe both batches of flies.
4. Choose three different behaviors that the flies do, and record them below. You also need to provide a definition for the behavior.

Table 1.

Behavior name	Definition	AVERAGE # Instances in WT	AVERAGE # Instances in Mutant

5. For 5 minutes, each person in your group should observe one behavior and record the number of instances those behaviors happen in the WT flies. Record the number of instances in the table above.
6. Do the same for the Mutant group.

Test III. Recording locomotor behavior

For the final test, you'll record the movement of your two different types of flies in a petri dish and use an automated tracking program to track velocity and heading of your flies. Your mutants might have altered locomotor behavior — this will help you test that.

Set up your camera

First, we need to arrange the camera and petri dish so that you can get everything in frame, in focus, and well lit.

1. Ensure that your camera is connected via the USB cord to the computer. The camera should be plugged into a USB3.0 slot (the blue ones).
2. Set up your camera and petri dish as in Figure 1. You want your camera to be about 6" from the petri dish.

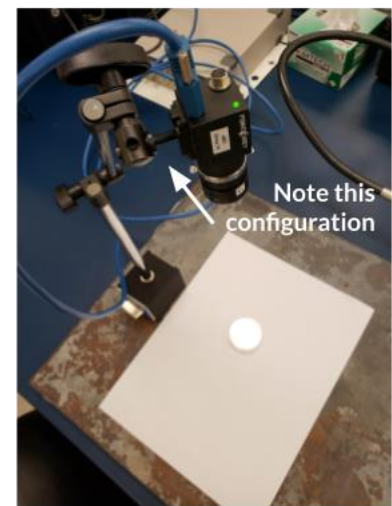


Figure 1. Fly locomotor recording setup.

3. On the computer, open **Point Grey FlyCap2**.
4. Make sure that it recognizes a "Flea3" camera, and press OK.
5. Adjust the **focal length** knob on your camera (closest to end, ranges from infinity to 0.1) in order to get the image in focus.
6. Adjust the image on your screen using the toolbar at the top:



Figure 2. FlyCap menu bar.

7. Isolate one fly in your petri dish.
8. Click on the button to open up the Camera settings (e.g. exposure).

- a. Uncheck auto for brightness, exposure, and shutter.
- b. Increase the exposure almost to the point where your image is overexposed. You want the flies to stand out very clearly against the white background, and the edges of the petri dish to be invisible:



Figure 3. Sample frame of fly tracking video.
Notice that the fly is in focus and very dark, while the background (including the edges of the petri dish) are very

9. Ensure that you can easily see your fly throughout all areas of the arena.
Note: Optimizing the way your fly recording looks will make the tracking *much* easier. Make sure there are no glares from lights in the room, no dark spots on the petri dish, and clean off any fingerprints on the glass. **Every light and dark spot matters, and anything can throw off the tracking.**

Set up your flies & record

1. When you're ready to record, go to **File > Capture Image or Video**
2. A dialogue will pop up. Set up a 60 second recording with the following settings:

Filename: browse to create your own (make sure you label it as WT or Knockout). **SAVE YOUR FILE LOCALLY**, not on your username server. **Your video file will not save correctly if you do not do this.**

- ☐ Capture for 60000 ms
- ☐ Buffered

(Click on **Videos** tab)

Video Recording Type: M-JPEG

Frame Rate: 15.0

AVI Split Size: 0

JPEG Compression Quality: 75

3. Press **Start Recording**, and wait while the program records the video.
Note: this window will not close, but the message at the bottom will change.
4. When it is done, **check that the video recorded correctly by finding the file and opening it. Check that the video is one minute long. If not, change the Frame Rate to 60, or click "Use Camera Frame Rate."**
5. For the first fly you record, **move on to the next step to ensure that your video works well for tracking. If the tracking doesn't work well, you'll need to adjust your lighting and/or image acquisition.**
6. When you're confident the tracking works well with your videos, obtain videos for three flies from each group (you've already done one).
7. When you're done, save your videos in a clouded location for safekeeping.
8. Close FlyCap.

Recording Settings

Save Filename
 Filename: C:\Users\ajuavine\Videos\fly_test
 A timestamp, sequence number and file extension based on output type will be appended
 Browse

Saving Options
☐ Capture 0 frames*
☐ Capture 1 out of every frames from a run of frames*
☒ Capture for 60000 ms
☐ Capture once out of every ms from a run of ms*
 * Specify zero to capture until manually stopped

Recording Mode
☒ **Buffered:** Frames are buffered in physical memory while recording. Recording will stop if physical memory usage hits 95%
☐ **Streaming:** Frames are not buffered in system memory while recording. Dropped frames may occur depending on the speed of the PC and the current camera/recording settings.

☐ Only save corrupt frames when capturing

Images Videos

Video Recording Type: M-JPEG
Frame Rate: Use Camera Frame Rate
AVI Split Size: 0 MB (0 indicates no file split)
M-JPEG Options
JPEG Compression Quality (1-100): 75

Good Frames:	1198	Available Physical Memory:	12522.50 MB
Corrupt Frames:	0	Total Physical Memory:	16304.59 MB
Buffered Frames:	0	Usage:	23%
Saved Frames:	1198		

Hide Recording is stopped Start Recording

Quantify your locomotor behavior

Now that you have videos of your specimen, you need to quantify the locomotor behavior. We'll use a MATLAB program (that someone wrote, and I've edited) to do so.

Download the code we need

1. Download the code for **fly_tracker** from Github: https://github.com/BIPN145/fly_tracker.
To do so, click on the green "Code" button on Github and download as a .zip file.
2. Extract all of the contents of the zipped folder into your **local** Documents folder (*not* a folder on your username).
3. Open MATLAB.
4. On the top bar of MATLAB, find the "Set Path" button:



use this button to tell
MATLAB where your code
is

5. In this window, click "Add with Subfolders."
6. Navigate to the folder where you extracted fly_tracker. Hit **Save**.
Note: You might get an error about pathdef.m. Select **Yes** and save this file in your **local** Documents folder.

Test the tracking for a sample video

7. In the MATLAB window, type

```
[mean_velocity SD_velocity all_velocity] = BIPN145_flytrack2(4,15)
```

and press enter. If you used a different frame rate (e.g., 60), change 15 to your frame rate.

8. Navigate to where your movie is and select it.
9. In the next window, draw a rectangle (click and drag) around your entire petri dish.
10. Double-click inside of the rectangle to initiate tracking.
11. Wait for the tracking to finish. A window will pop up that shows the trajectory of your fly and its velocity across time.
Note: If you'd like to edit these figures in any way, you can go to **Tools > Edit Plot**.

Run your analysis for a group of flies

Once you've tested your tracking & recorded all of your movies, you can run this script for **several movies** at a time (e.g., one group of flies). You should record videos for 3 WT flies and 3 mutant flies. Do this by re-doing steps 7-11 above, but **select all of the videos for one group of flies** in the window where you are asked to pick a movie. After you've done the analysis for your wildtype flies, run the BIPN145_flytrack script again with your mutant fly videos.

The script outputs a variable called "mean_velocity". This will show the mean velocity (in mm/s) for *each* of your videos. SEM velocity is the standard error of the mean of the velocity for these videos.

To generate a mean value for the video(s) you just tracked, type:

`mean(mean_velocity)` and press enter.

Record these values in the table below (you'll ultimately want to plot these).

Table 2. Comparison of mean velocities for control and mutant flies.

	Control	Mutant
Mean		
SEM		

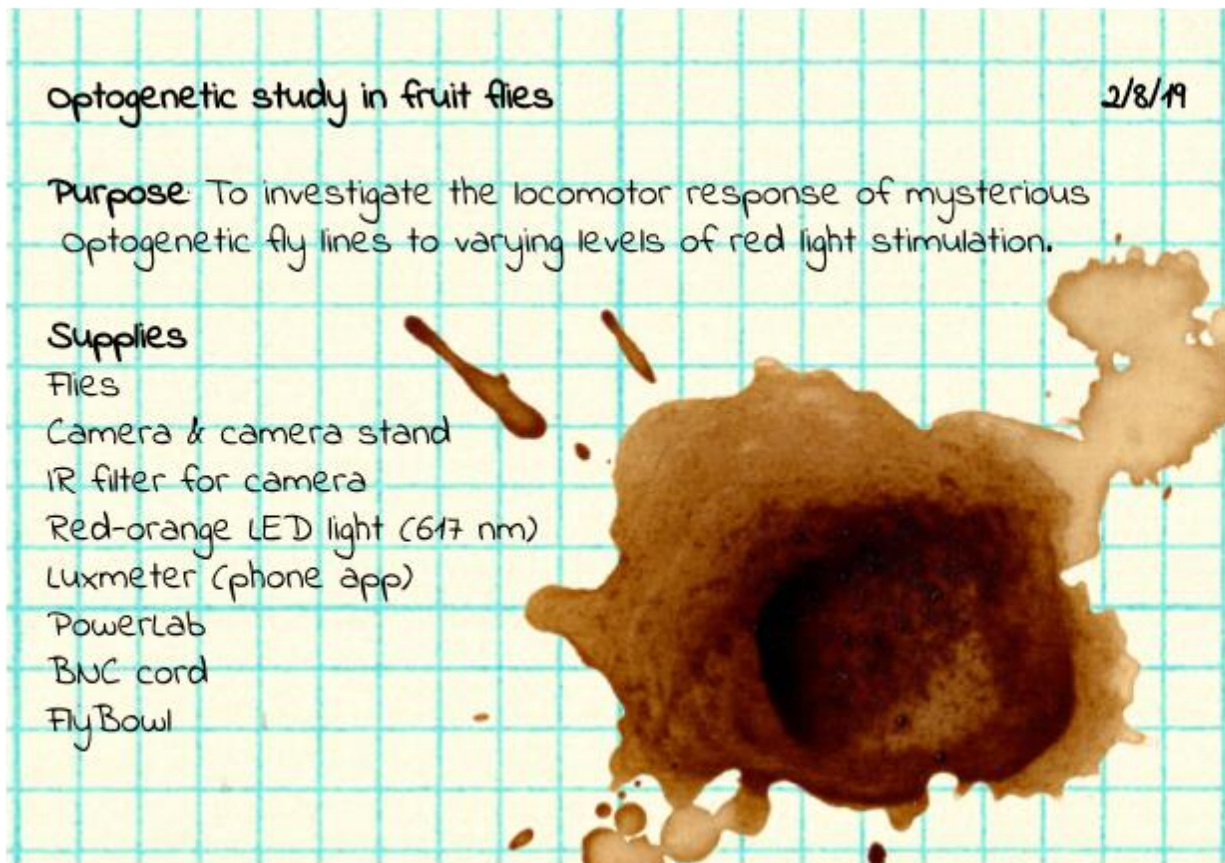
Troubleshooting

Observation	Likely issue(s)	Possible solution
The camera output is black	Lens cap is on	Remove the lens cap
Camera not recognized	Camera not plugged in	Plug in the camera
Video file is not recognized in MATLAB or the movie is the wrong length/size	You wrote the video file to the server	You need to re-record your video but with the file saved directly to the computer locally

Day 2: The Case of the Missing Methods

Just when you thought everything was back to normal in the lab, a remarkably inconveniently placed coffee spill completely drenches several pages of the notes given to you from your mentor.

Unfortunately, those pages detailed all of the methods that you needed in order to perform your experiment. You've managed to save the supplies list, at least:



Hope is not lost. You've got some supplies, and you suspect that this study had something to do with the role of the genetically-modified neurons. Certain neurons have optogenetic channels, but what do those neurons actually control?

It might take a little bit of research to refresh your memories about those transgenic flies, but that's easy enough. And you've done at least some fly behavior before — you're pretty much the fly whisperer.

You can re-write the protocol and get some interesting results, can't you?

Tips for designing your experiment

You can control the intensity of the light source through LabChart by following these steps:

1. Plug the BNC input on the light into the OUTPUT of the PowerLab using a BNC cable.
2. Program the stimulator (**Setup > Stimulator**) to send different pulses with heights from 2V - 10V. You should notice that the intensity of the light changes with more voltage.
Do not stimulate your flies at higher than 1 Hz frequency or with stimulus pulses longer than 0.5 seconds.
3. Measure the intensity of the light using a phone app luxmeter (I recommend Lux Light Meter Free). You should measure the light intensity **from the same distance** as you are using it in the experiment.

Don't forget about control groups! It may help to compare these flies to each other, as well as to wildtype flies. We have wildtype flies if you need them.

You will need to put the red light **within several centimeters** of your flies in order to see an effect of the stimulation.

Experiment #8: Measuring Alpha Wave Activity Using Electroencephalography

Background

The cerebral cortex contains huge numbers of neurons. Activity of these neurons is—to some extent—synchronized in regular firing rhythms. These are referred to as brain waves. Electrodes placed in pairs on the scalp can pick up variations in electrical potential that derive from this underlying cortical activity. This recording of the electrical activity in the cortex is called an electroencephalogram (EEG). EEG signals are strongly affected by different states of arousal and show characteristic changes in different stages of sleep. EEG signals are also affected by stimulation from the external environment and brain waves can become entrained to external stimuli. Electroencephalography is used, among other things, in the diagnosis of epilepsy and the diagnosis of brain death.

Recording EEG signals

EEG recording is technically difficult, mainly because of the small size of the voltage signals, which are typically 50 μV **peak-to-peak**. The signals are small because the recording electrodes are separated from the brain's surface by the scalp, the skull, and a layer of cerebrospinal fluid. A specially designed amplifier, such as the Bio Amp built into the PowerLab, is essential to record EEGs. It is also important to use electrodes made of the right material and to connect them properly. Even with these precautions, recordings may be spoiled by a range of unwanted interfering influences, known as artifacts.

In this lab, you will record EEG activity with an occipital electrode on the scalp at the back of the head. A third (ground or earth) electrode is also attached, to reduce electrical interference. In clinical EEG, it is usual to record many channels of activity from multiple recording electrodes placed in an array over the head.

Origins of the EEG signals

The EEG results from slow changes in the membrane potentials of cortical neurons, especially the excitatory and inhibitory post-synaptic potentials (EPSPs and IPSPs). Very little contribution normally comes from action potentials propagated along nerve axons. The EEG reflects the sum of the electrical potential changes occurring from large populations of cells. Therefore, large amplitude waves require the synchronous activity of a large number of neurons. The rhythmic events that these waves reflect often arise in the thalamus whose activity is in turn affected by a variety of inputs including structures in the brainstem reticular formation.

Components of the EEG waveform

The EEG waveform contains component waves of different frequencies. These can be extracted and provide information about different brain activities. The types of brain waves are:

- **alpha** (between 8 to 13 Hz; average amplitudes of 30 to 50 μV peak-to-peak) which will be studied in this experiment. Alpha rhythm is seen when the eyes are closed and the volunteer relaxed. It is abolished by eye opening and by mental effort such as doing calculations or concentrating on an idea. It is thus thought to indicate the degree of cortical activation. The greater the activation, the lower the alpha activity. Alpha waves are strongest over the occipital (back of the head) cortex and also over the frontal cortex.

- **beta** (13 to 30 Hz; <20 μ V peak-to-peak) which are prominent in alert individuals with their eyes open. The beta rhythm may be absent or reduced in areas of cortical damage and can be accentuated by sedative-hypnotic drugs such as benzodiazepines and barbiturates.
- **theta** (4 and 8 Hz; <30 μ V peak-to-peak) which are seen in awake children but not adults. The theta rhythm is normal during sleep at all ages. However, some researchers separate this frequency band into two components, low theta (4 - 5.45 Hz) activity that they correlate with decreased arousal and increased drowsiness, and high theta (6 - 7.45 Hz) activity that it is claimed is enhanced during tasks involving working memory.
- **delta** (0.5 and 4 Hz; up to 100 - 200 μ V peak-to-peak) which is the dominant rhythm in sleep stages three and four but not seen in conscious adults. The delta rhythm tends to have the highest amplitude of any of the component EEG waves. EEG artifacts caused by movements of jaw and neck muscles can produce waves in the same frequency band.
- **gamma** (30 and 50 Hz). Most people recognize gamma rhythm, but its importance is controversial. It may be associated with higher mental activity, including perception and consciousness and it disappears under general anesthesia. One suggestion is that the gamma rhythm reflects the mental activity involved in integrating various aspects of an object (color, shape, movement, etc) to form a coherent picture.

Written by staff of ADInstruments & A. Juavinett

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www.ADInstruments.com

Lab Protocol

Supplies

- LabChart software & computer
- PowerLab 26T
- 5 Lead Shielded Bio Amp Cable
- EEG Flat Electrodes (gold contacts)
- Electrode Paste
- Abrasive Gel (NuPrep)
- Alcohol Swabs
- Elastic bandage
- Cotton swab

Part I. Spectral Analysis Tutorial

A spectrum is a representation of data based on the frequency distribution of its component sine waves. Spectra indicate the strength of the various frequencies in a time-varying waveform. Spectrum View allows you to observe the frequency distribution of data that might not otherwise be easily seen. For example, it could be used to break down an EEG waveform into its various components: beta waves, alpha waves, theta waves and delta waves. A mathematical technique known as the Fast Fourier Transform is applied to the raw data. The results of this analysis can be displayed as a plot of the power (vertical axis) of different frequencies (horizontal axis) relative to each other in the input signal. This is called a **Power Spectrum Density (PSD) plot**. The data can also be displayed as 3-dimensional color plot of spectral power, frequency, and time called a **Spectrogram**.

Note: The questions here are written to ensure that you are understanding the steps of the tutorial. You do not need to answer them, and they do not directly relate to your lab report. You *should* understand how to interpret a spectrogram, however.

1. Open the EEG Spectral Analysis Tutorial settings file (<http://bit.ly/labchart>).
2. Examine the Chart View. Use the **View Buttons** to view each block. You should see five blocks of data. The first record is a slowly oscillating sine wave
3. Open Spectrum view by clicking on the Spectrum View button in the Toolbar:

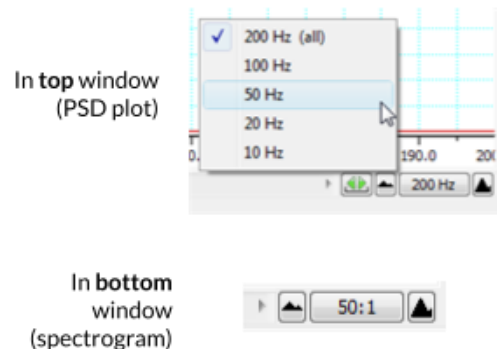


4. Click the **Smart Tile** button in the LabChart Toolbar to display both windows in full screen mode.
5. In Chart View **Select** the first record by double clicking in the Time axis. This will perform a spectral analysis for this record and displays the result in the Spectrum view.
6. Adjust the horizontal scaling of plots to view the results:

- Set the **horizontal scaling** for the **Power Spectrum Density (PSD)** plot to 50 Hz
- Use the horizontal scroll bar to display the 0 Hz to 50 Hz region of the plot.
- You may need to Autoscale the axes.

- Set the **horizontal scaling** for the Spectrogram to 50:1.

7. Examine the PSD plot and then the first section of the Spectrogram. Expand the vertical axes if necessary. Use the waveform cursor to identify the frequency in Hertz (Hz) of the peak in the PSD plot and the band in the Spectrogram. Values are displayed at the top of each plot.



What is the frequency in Hertz (Hz) of this sine wave?

8. Select the second record and again view the result in the Spectrum view.

What is the frequency in Hertz (Hz) of this second sine wave?

9. Select the third record and again view the result in the Spectrum view. You should now see two prominent peaks (PSD plot) and bands (Spectrogram) in the result.

Are these two peaks/bands the same as for the first two records? How is this possible?

10. Select the fourth record and again view the result in the Spectrum view.

Is there any regular signal within this record?

10. In Chart View compare the signal amplitudes of the fourth and fifth records. Note that the fifth record has lower amplitude compared with the fourth record (you may need to Autoscale to see a peak at all).

11. Select the fourth record again. In the Spectrum view examine the PSD plot. Move the **Waveform Cursor** to the prominent peak.

What is the frequency (Hz) of this signal?

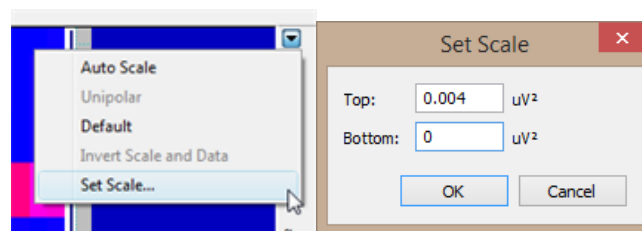
What is the power (mV^2) of this signal?

12. Select the fifth record and examine the PSD plot. Move the **Waveform Cursor** to the peak.

What is the frequency (Hz) of this signal?

What is the power (mV^2) of this signal?

13. Examine the Spectrogram. Note that the band corresponding to the signal's frequency appears to be missing. This is because the power of the signal is small compared with the previous four records. Expand the scale on the right hand side of the Spectrogram by setting the scale closer to the power of the signal (e.g., right click **Set Scale** to 0 to 0.004 V^2). Note that the band is now visible at the expected frequency.



14. The fifth record is the same signal as the fourth record, except that the quality of the raw signal has been affected. Compare your features (amplitude, power, frequency) of the fourth and fifth record.

How has the quality of the signal affected the wave features?

Part II. Detecting alpha waves

Equipment Setup and Electrode Attachment

1. Make sure the PowerLab is turned **off** and the USB cable is connected to the computer.
2. Connect the 5 Lead Shielded Bio Amp Cable to the Bio Amp Connector on the front panel of the PowerLab (Figure 1). The hardware needs to be connected **before** you open the settings file.

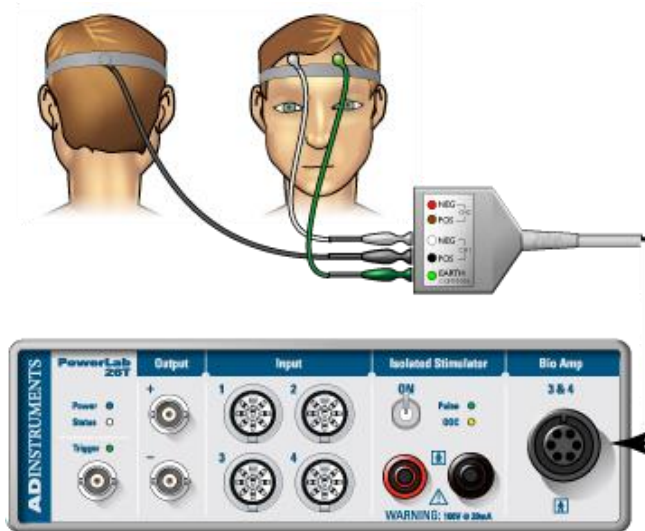


Figure 1. Equipment Setup for PowerLab 26T. Image courtesy of ADInstruments ©

3. Attach the leads of the EEG Flat Electrodes to the Earth, CH1 NEG and POS pins closest to the labeled side on the Bio Amp Cable (the side with the stickers labeling the connections). Channel 1 “**positive**” will lead to the inion (the bump on the back of the head above the neck) and Channel 1 “**negative**” will lead to the forehead. Channel 2 will be empty and the Earth will lead to the temple. Refer to Figure 1 for proper placement, but do not attach them to the volunteer yet. Follow the color scheme on the Bio Amp Cable.
4. Remove any jewelry from the volunteer’s face, ears, and neck.
5. With Figure 1 as a guide, locate where you will place the electrodes. Abrade the skin with the NuPrep Abrasive Gel.
Note: This is important as abrasion helps reduce the skin’s resistance. Your recordings will be much better if you have abraded the skin well (but not too much — that will cause irritation).
6. After abrasion, clean the area with an alcohol swab to remove the dead skin cells.
7. While the skin is drying, scoop Electrode Paste into the EEG Flat Electrodes. When the skin is dry stick the electrodes to the skin (Figure 1). Immediately hold the electrodes and wires in place with the medical tape. Make sure the tape firmly holds the electrodes against the head. Use the elastic bandage to wrap tightly around the head. This will help the electrodes maintain good contact with the skin.
8. Have the volunteer lie in a comfortable position on his/her back (or sitting comfortably), with the head turned so that none of the electrodes are disturbed or compressed.
9. Check that all three electrodes are properly connected to the volunteer and the Bio Amp Cable before proceeding. Turn on the PowerLab.

Exercise 1: Recognizing Artifacts

In this exercise, you will learn to recognize some of the artifacts that can appear on an EEG.

1. Launch LabChart with the EEG Settings file (<http://bit.ly/labchart>).

2. Select **Bio Amp** from the EEG Channel Function pop-up menu. Make sure the settings are as follows: **Range 200 μ V, High Pass 0.5 Hz, and Low Pass 50 Hz.**

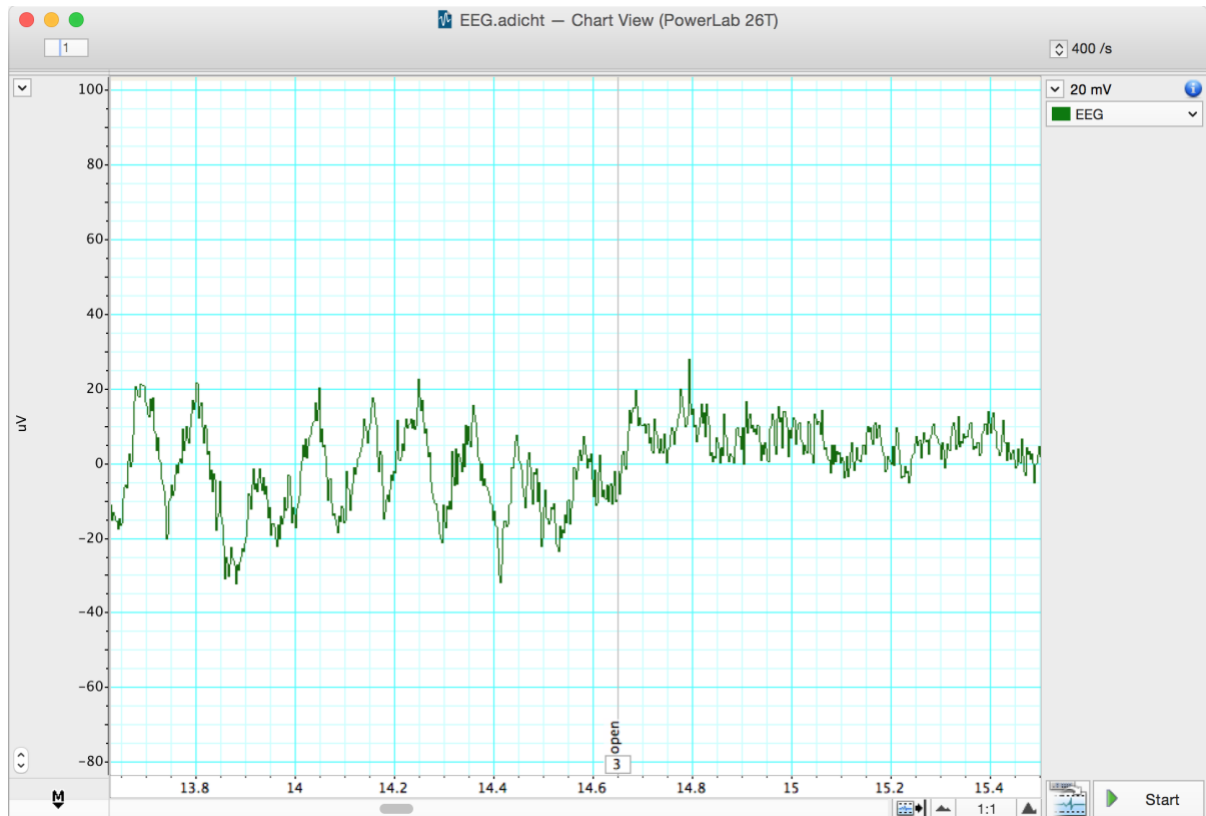
💡 **Why are we applying high and low-pass filters to our data?**

3. **Start** recording. Add a **comment** “blinking,” and have the volunteer blink repeatedly. **Stop** recording after 10 seconds.
4. Repeat step 3, this time, have the volunteer make eye movements. Add a **comment** “eye movements.” Have the volunteer gaze up-and-down and left-and-right in a repeated pattern. Make sure the volunteer’s head is still and only the eyes move.
5. Repeat step 3, this time, have the volunteer make head movements. Add a **comment** “head movements.” Have the volunteer gently move his/her head in a repeated pattern.
6. Examine the vertical scale at the left of the Chart View, and note the positions corresponding to +50 μ V and –50 μ V. True EEG signals rarely exceed these limits.
7. Examine the entire data trace and **Autoscale**, if necessary. There may be some large signals outside the ± 75 μ V range. **Such large signals are artifacts.**
8. **Save your data**, and open a new file with the same settings.

Exercise 2: Alpha Waves in the EEG

In this exercise, you will examine the effects of relaxation and eye movement on alpha waves in the EEG. Do not stop and start the recording during these trials.

1. Make sure the volunteer is relaxed and comfortable. Have the volunteer close his/her eyes and remain quiet. Keep noise to a minimum and keep all distractions away from the volunteer.
2. **Start** recording. Prepare a comment with “open;” do not enter it yet.
3. Tell the volunteer to open both eyes. Immediately press Return/Enter to add the **comment**. Record with the volunteer’s eyes open for 30 seconds. Do not stop recording.
4. Prepare a comment with “shut.” When the 30 seconds of eyes open are complete, tell the volunteer to close both eyes. Immediately press Return/Enter to add the “shut” **comment**. Record with the volunteer’s eyes closed for 30 seconds.
5. Repeat steps 3 and 4, oscillating between eyes closed and open, to give you 10 sets of results. **Save your data.**



EEG signal with eyes shut and open. Note alpha waves during eyes shut (left).

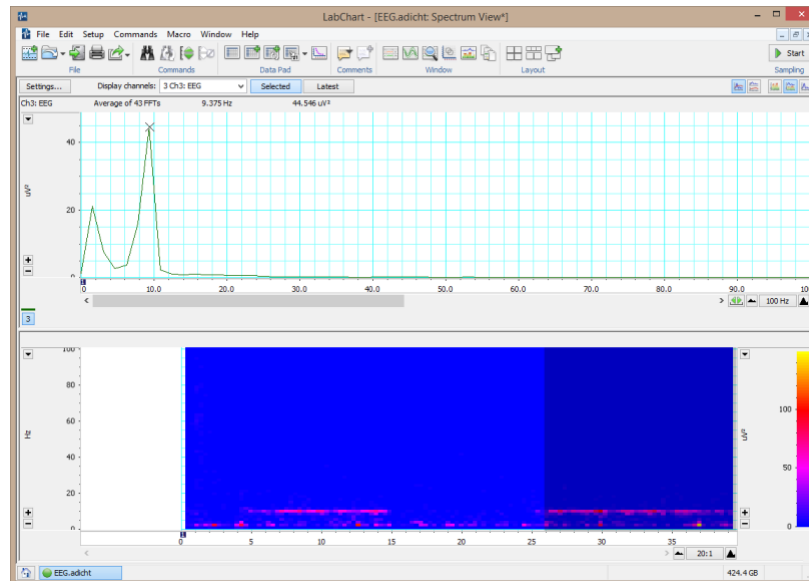
Wave Amplitude Analysis

1. Use the **View Buttons** to change the horizontal compression to see data with eyes open and shut. Make a data selection that includes some data from both eyes open and eyes shut conditions. View this selection in **Zoom View**. This should make it easier to see the alpha wave activity. **Autoscale**, if necessary.
2. In Chart View scroll through the parts of the recording that were made with the volunteer's eyes shut to look for alpha waves. Use the **View Buttons** to change the horizontal compression if necessary – you should zoom into about 1-2 seconds of data to be able to see the alpha waves. The alpha waves can be recognized by their amplitude (usually 30 to 50 μV peak-to-peak) and their frequency. Each cycle of an alpha wave – in other words, the width from trough to trough – lasts ~ 0.1 s (100 ms).
3. Use the **Marker** and **Waveform Cursor** to measure the amplitude of the alpha waves. Place the **Marker** at the lowest point of the wave and move the **Waveform Cursor** to the peak of the wave. Measure the amplitudes of 10 waves from when the volunteer's eyes were closed. Record the values in Table 1 of the Data Notebook.
4. Now measure 10 wave amplitudes when the volunteer's eyes were open. Note that it may be more difficult to find these, but you can use the spectral analysis to guide where you look in your recording. Record these values in Table 2 of the Data Notebook.

Spectral Analysis

Now you will use Spectral Analysis to examine the EEG you recorded.

1. Find the part of the recording when the volunteer had their eyes **shut**. Click-and-drag across this part of the data trace to **select** it.
2. From the **Window** menu, select **Spectrum**. In the Spectrum View choose **Selected** (underneath the main toolbar at the top).



Spectrum of an EEG

3. Next to Display Channels, choose Ch 3: EEG.
4. Alpha activity should show up in the PSD plot as a clear peak in the 8-12 Hz range.
5. Alpha activity shows up in the Spectrogram as a band of color in the 8-12 Hz range.

Note: If you cannot see any alpha activity (indicated by brighter colors in the 8-12 Hz range in the spectrogram), manually scale the colorbar. You can do so by right clicking on the color bar and choosing "Set Scale". The scale will likely be a very tiny number, with an exponent. Change the *exponent* of this number and see how it changes your spectrogram. For example, if it originally plots with 10^{-12} , try 10^{-13} or 10^{-11} .

Note #2: Spectrogram displays all the recorded data and that the selection you have made is highlighted in a darker blue color.

6. Make a data selection of several seconds from when the volunteer had their eyes **open**. Select **Spectrum**. Note that in the PSD plot the peak in the alpha activity range of 8-12 Hz is small or absent and in the Spectrogram the band of color in the alpha activity range of 8-12 Hz is weak or absent.

7. In the Spectrogram scale the horizontal axis so that all the data is visible. Note the presence and absence of the band of color in the alpha activity range of 8-12 Hz which correspond with the eyes shut and eyes open conditions.
8. If you're happy with your spectrograms (e.g. you can clearly see alpha activity in closed conditions but not open conditions), you've completed the task for this lab. Save your data in a safe location (we'll deal with plotting it later).

Troubleshooting

Observation	Likely issue(s)	Possible solution
There is a lot of 60 Hz noise in your recording.	Your electrodes are not making good electrical contact with the scalp.	Remove electrodes, clean both the scalp and the electrode, and attached again. If a less noisy recording cannot be obtained by better electrode placement, a Notch filter can be applied in LabChart. To do so, click on the recording channel and choose "Digital Filter." Then, change it to "Notch" and 60 Hz.

Data Notebook

Table 1. Alpha Waves with Eyes Open

	Amplitude (μ V)
Wave #1	
Wave #2	
Wave #3	
Wave #4	
Wave #5	
Wave #6	
Wave #7	
Wave #8	
Wave #9	
Wave #10	

Table 2. Alpha Waves with Eyes Shut

	Amplitude (μ V)
Wave #1	
Wave #2	
Wave #3	
Wave #4	
Wave #5	
Wave #6	
Wave #7	
Wave #8	
Wave #9	
Wave #10	

Questions for consideration

Under what conditions did you see alpha waves more clearly?

What are alpha waves thought to indicate?

Look at your Spectral Analysis from when the volunteer had their eyes closed. At what range do you have the greatest peak? What type of brain waves does the Spectral Analysis suggest is active?

Analyzing your EEG data

Exporting PSD data from LabChart

You will export data from this lab in the same way you've exported data for previous labs.

Follow the directions in Appendix A to export 30 seconds of your eyes closed data **and** 30 seconds of your eyes open data as separate LabChart Text Files. Ensure that you are **only** exporting the EEG channel. You will import these into a computational notebook for analysis.

Appendix A: Exporting & Plotting LabChart data

Saving recordings from lab computers

After finishing your experiment, save the raw data (.adicht) file. We recommend using either Google Drive or Dropbox to save your data so that you may access it from anywhere. Your files will rarely be bigger than 5 MB.

Viewing LabChart files from your Mac or PC

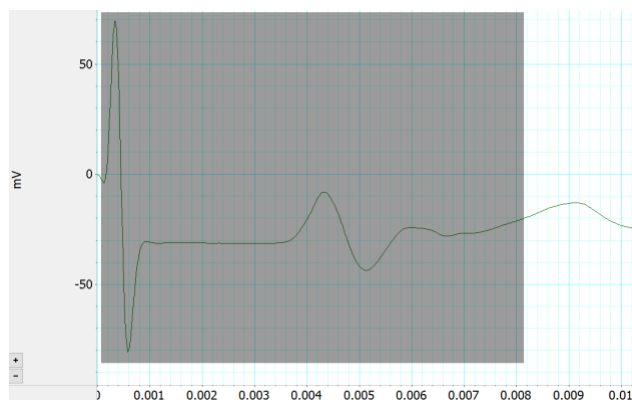
LabChart Reader

If you'd like to view your files on your personal computer without LabChart 8, you can download LabChart 8 Reader. If you're using LabChart Reader on PC, you can simply click & drag to select the data you'd like to export, copy, and then paste this into Excel. Unfortunately, LabChart Reader on Mac does not allow you to export your data.

Exporting your data from LabChart

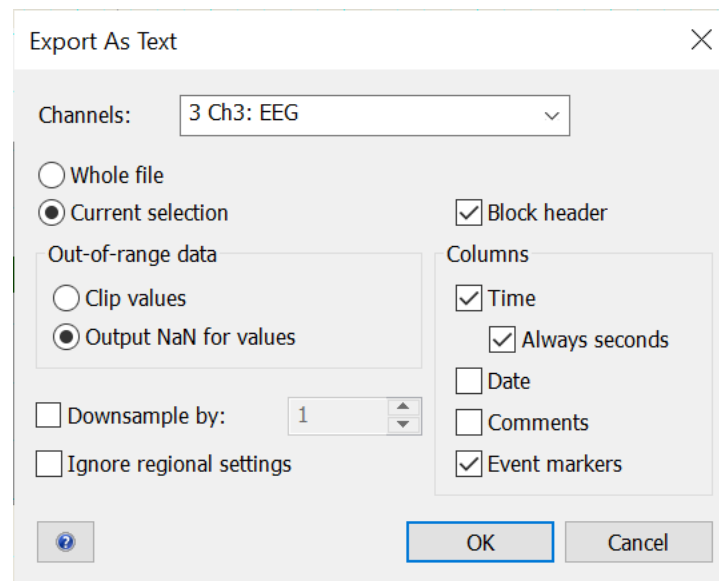
Note: This will work for LabChart8 but *not* LabChart Reader.

1. Select the data you would like to save by clicking and dragging in the window with your recording. You *can* export a large file with all of your data, but in most cases this will not be necessary.



2. In LabChart, go to **File > Export**.
3. By default, LabChart will offer to export it as a Text File. In the first window that pops up, choose a name and location for your file and press **Save**.
4. In the following window, you have options for how your exported file will look. In the drop down **Channels** window, be sure that your data channel is checked. If applicable, you may also want to save your stimulus channel, so that you can also plot when the stimulus came on (and at what amplitude).

5. Choose the bubble for “Current selection.”
6. Un-check “Comments.”
7. You may want to **downsample** your data (e.g., take fewer data points than were collected), especially if you have issues plotting it later because the dataset is too large. However, if you don’t have plotting issues, don’t bother with downsampling.
8. Here is a sample export window where we are exporting EEG data on channel 3. We’re exporting the current selection without downsampling:



Plotting in Python

If all of the steps below to use Excel looks like a lot of work to simply replot a recording trace — you’re right, it is. We have a Python script for that. Your instructor will advise you on how to import your data into Python for plotting. You are strongly encouraged to use the provided computational notebooks to plot your data.

Importing your LabChart text file into Excel (on Windows 10)

1. Open a New Sheet in Excel.
2. Go to **File > Open > Browse** and navigate to where your text file is saved.
3. Change the file type to “All files” and open your .txt file.

- In the next window(s), hit Next. In the following (screenshot below) make sure that the **Delimiter** is set to **Tab**, and click Finish. The data preview should have multiple separated columns and look something like this:

Text Import Wizard - Step 2 of 3

This screen lets you set the delimiters your data contains. You can see how your text is affected in the preview below.

Delimiters

☒ Tab
☐ Semicolon
☐ Comma
☐ Space
☐ Other:

☐ Treat consecutive delimiters as one

Text qualifier: ▼

Data preview

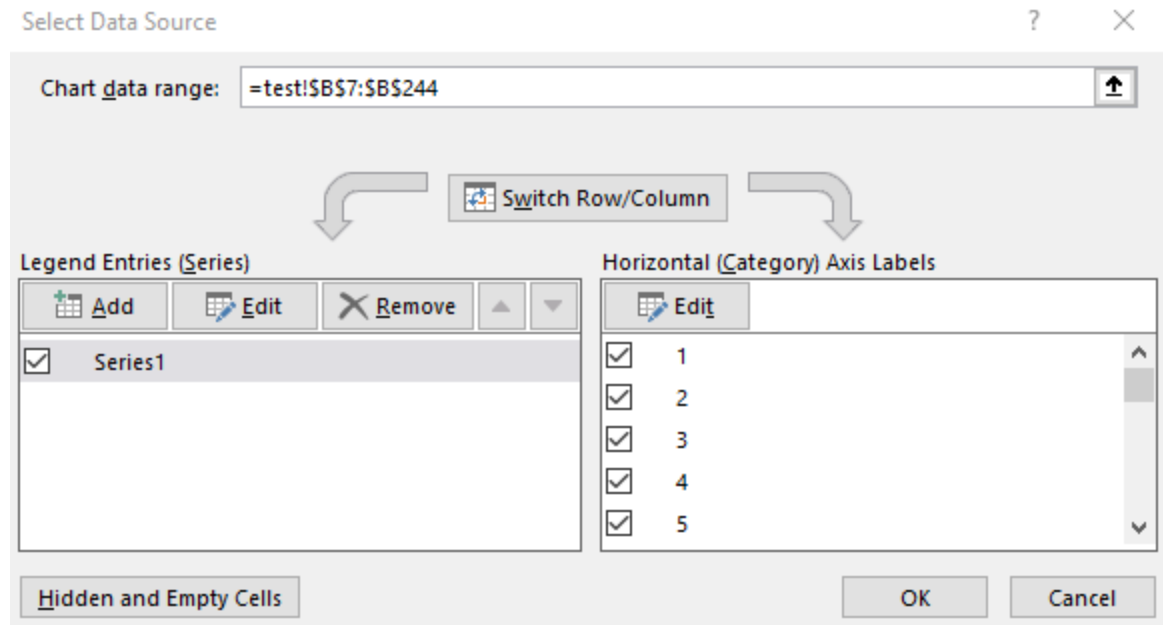
Interval=	2.5e-005 s	
ExcelDateTime=	4.3738531324010488e+004	9/30/2019 12:45:06.394507
TimeFormat=	StartOfBlock	
DateFormat=		
ChannelTitle=	BioAmp Recording	

Cancel < Back **Next >** Finish

- This file will have multiple columns. The first column will contain the timestamps for your experiment, in seconds. The second column will be the first channel of data that you've exported, in volts. In the example above, it is the stimulus channel. The third column will be the second channel of data, etc. You should always confirm that the units match what you saw on LabChart. By default, data is exported in volts and seconds.

Note: If you exported all of your data, each page of your experiment will be **vertically concatenated**. In other words, it will start at 0 and go to the end of Page 1, and then start at 0 again for the next page.

- Select the cells containing the data you would like to plot (if you exported the stimulus channel, it's the third column; if you didn't export the stimulus channel, this should be the second column of your file).
- Go to **Insert > Line Chart**.
- To add the correct time axis labels, right click on the chart and choose "Select Data."
- In the next window, under "Horizontal (Category) Axis Labels" choose "Edit." In the window that pops up, select all of the time values (the first column of your file).



10. From here, you might need to adjust the Horizontal Axis crossings in order to prevent the X axis from directly crossing your data (since your data will likely hover around zero, where the axis goes by default). For example, you might want to have your X axis cross at -100 mV, so that it doesn't overlap with your data.
11. Format your plot with axes labels and a title, and make any additional changes as necessary.

Importing your LabChart text file into Excel (Mac)

1. Open a New Sheet in Excel.
2. Go to **File > Open > Browse** and navigate to where your text file is saved.
3. Change the file type to "All files" and open your .txt file.
4. In the next window(s), hit Next. In the following (screenshot below) make sure that the **Delimiter** is set to **Tab**, and click Finish. The data preview should have multiple separated columns and look something like this:

Text Import Wizard - Step 2 of 3

This screen lets you set the delimiters your data contains.

Delimiters

☒ Tab ☐ Treat consecutive delimiters as one

☐ Semicolon Text qualifier: "

☐ Comma

☐ Space

☐ Other:

Preview of selected data:

0.00005	-2.40625
0.000075	-5.6875
0.0001	-10.625
0.000125	-15.25
0.00015	-16.34375
0.000175	-10.03125
0.0002	7.25
0.000225	37.51562
0.00025	80.28125

Cancel < Back Next > Finish

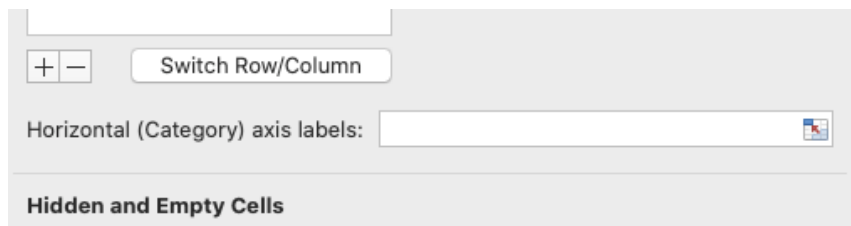
- This file will have multiple columns. The first column will contain the timestamps for your experiment, in seconds. The second column will be the first channel of data that you've exported. In the example above, it is the stimulus channel. The third column will be the second channel of data, etc.

Note: If you have exported all of your data, each page of your experiment will be **vertically concatenated**. In other words, it will start at 0 and go to the end of Page 1, and then start at 0 again for the next page.

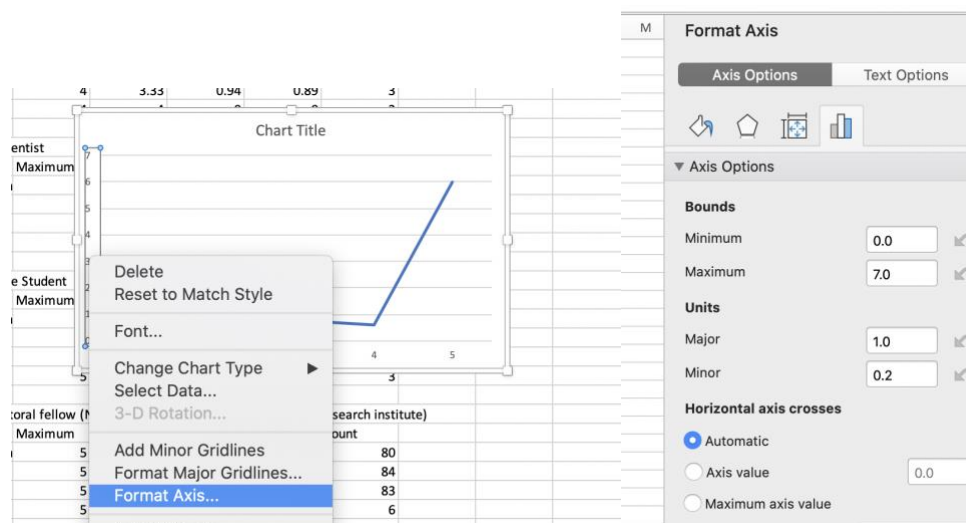
- Select the cells containing the data you would like to plot (if you exported the stimulus channel, it's the third column; if you didn't export the stimulus channel, this should be the second column of your file).
- Go to **Insert > Line Chart**.
- To add the correct time axis labels, click on Select Data under the Chart Design menu:



9. Where it says “Horizontal (Category) Axis labels, click the  button to the right:



10. Select all of your time stamps in the Excel file (the first column) and press Okay. Your x axis should now be updated with the time, instead of the samples.
11. From here, you might need to adjust the Horizontal Axis crossings in order to prevent the X axis from directly crossing your data (since your data will likely hover around zero, where the axis goes by default). For example, you might want to have your X axis cross at -100 mV, so that it doesn't overlap with your data. You can do this by clicking on an Axis, and modifying the “Horizontal axis crosses” in the “Format Axis” menu on the right (under the tab with the ). Alternatively, you can right click on the axis, and choose “Format Axis”, which will open the same menu.



12. Format your plot with axes labels and a title and make any additional changes as necessary.